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Requirements for the Application of Reliability-Centred Maintenance Techniques to HM Ships, Submarines, Royal Fleet Auxiliaries and other Naval Auxiliary Vessels

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Issue 2 of this Standard has been prepared to incorporate changes to text and presentation and to allow for declassification. The technical content has been updated in line with current practice.

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REQUIREMENTS FOR THE APPLICATION OF RELIABILITY-CENTRED MAINTENANCE TECHNIQUES TO HM SHIPS, SUBMARINES, ROYAL FLEET AUXILIARIES AND OTHER NAVAL AUXILIARY VESSELS

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Record of Amendments

Amendment	Inserted By	Date
1		
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DEFENCE STANDARD 02-45
(NES 45 ISSUE 3)

REQUIREMENTS FOR THE APPLICATION OF RELIABILITY - CENTRED MAINTENANCE TECHNIQUES TO HM SHIPS, SUBMARINES, ROYAL FLEET AUXILIARIES AND OTHER NAVAL AUXILIARY VESSELS

ISSUE 2

JULY 2000

This Naval Engineering Standard is
authorised for use in MOD contracts
by the Defence Procurement Agency and
the Naval Support Command

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SCOPE

1. The aim of Naval Engineering Standard 45 (NES 45) is to give direction on how to prepare for, conduct and utilise the results of Reliability Centred Maintenance (RCM) analyses. It is intended for use on Royal Navy ship and submarine platforms, Royal Fleet Auxiliary and other auxiliary vessels procured or supported by the Navy Department (referred to collectively in this NES as “Naval vessels”), and their associated systems and equipments. It assumes the reader has an understanding of the principles of RCM.
2. This NES is for the application of RCM techniques to:
   a. In-service platforms managed by the Naval Support Command.
   b. New naval procurements managed under the Integrated Logistics Support/Logistics Support Analysis (ILS/LSA) procedures mandated by the Chief of Defence Procurement and detailed within Defence Standard 00-60.
3. It excludes munitions and Royal Navy aircraft with their associated moveable ground support and test equipment, but includes all fixed onboard aircraft support facilities.
4. MHE on RFAs, when the responsibility of the STO(N), is also excluded.

ACKNOWLEDGEMENTS

5. The role played by Aladon Ltd and its licensees in the development of RCM, and in particular John Moubray for permission to use copyright material from his book "Reliability-centred Maintenance" in the course of preparing this NES, is acknowledged with thanks.
6. The role played by other pioneers in the development of RCM is acknowledged, particularly the originators of the process, Stanley Nowlan and Howard Heap, and the unknown authors of Mil-Std 2173 (AS) and its successor document NAVAIR 00-25-403.
7. The sponsoring section also recognises the suggestions and proposals made by users of earlier versions of this NES, both from within MOD and the RCM industry at large.

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FOREWORD

Introduction

1. This Naval Engineering Standard 45 (NES 45) is sponsored by D Tech (Sea) AD/WLS (HILS(N) RCM Group), Ministry of Defence.

2. NES 45 details the requirements for the application of Reliability Centred Maintenance (RCM) techniques to both new-build and in-service Royal Navy vessels. This NES 45 at Issue 3 supersedes NES 45 Part 1, Issue 1 and NES 45 Part 2, Issue 2.

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4. Any user of this NES, either within MOD or in industry, may propose an amendment to it. Proposals for amendments that are not directly applicable to a particular contract are to be made to the publishing authority identified on Page (i), and those directly applicable to a particular contract are to be dealt with using contract procedures.

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6. Unless otherwise stated, reference in this NES to approval, approved, authorised and similar terms, means by the MOD in writing.

7. Any significant amendments that may be made to this NES at a later date will be indicated by a vertical sideline. Deletions will be indicated by 000 appearing at the end of the line interval.

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Categories of NES

14. The Category of this NES has been determined using the following criteria:

   a. Category 1. If not applied may have a *Critical* affect on the following:

      Safety of the vessel, its complement or third parties
      Operational performance of the vessel, its systems or equipment.

   b. Category 2. If not applied may have a *Significant* affect on the following:

      Safety of the vessel, its complement or third parties.
      Operational performance of the vessel, its systems or equipment.
      Through life costs and support.

c. Category 3. If not applied may have a *Minor* affect on the following:

MOD best practice and fleet commonality.
Corporate experience and knowledge.
Current support practice.

Related Documents

In the tender and procurement processes the related documents listed in each section and Annex A can be obtained as follows:

a.	British Standards	British Standards Institution, 389 Chiswick High Road, London, W4 4AL
b.	Defence Standards and Naval Engineering Standards	Directorate of Standardisation, Stan 1, Kentigern House, 65 Brown Street, Glasgow, G2 8EX
c.	Other documents	Tender or Contract Sponsor to advise.

16. All applications to Ministry Establishments for related documents are to quote the relevant MOD Invitation to Tender or Contract number and date, together with the sponsoring Directorate and the Tender or Contract Sponsor.

17. Prime Contractors are responsible for supplying their sub-contractors with relevant documentation, including specifications, standards and drawings.

Health and Safety

18. This NES may also call for the use of processes, substances and/or procedures that could be injurious to health if adequate precautions are not taken. This NES refers only to technical suitability and in no way absolves either the supplier or the user from statutory obligations relating to health and safety at any stage of manufacture or use. Where attention is drawn to hazards, those quoted may not necessarily be exhaustive.

19. This NES has been written, and is to be used, taking into account the policy stipulated in JSP 430: MOD Ship Safety Management System Handbook.

Additional Information

20. This specification describes processes that are outwith the traditional boundaries of what is perceived to encompass “RCM”. This is intentional, since it is important for Project Managers to understand how the groundwork for RCM analyses must be prepared, and what needs to be done with the results of individual RCM analyses to enable them to be fused into a coherent maintenance strategy for a complete platform. The RCM processes proper are described in sections 9 and 10.

21. Certain ILS related statements within this NES are clearly identified as being advisory in nature. Also provided are examples taken from the Hunt Class MCMV and Type 23 Frigate RCM Studies. These sections and examples are intended to provide both Contractors and MOD personnel engaged in the conduct of RCM with advice on how particular aspects of the RCM process may be carried out, without being prescriptive. Alternative approaches to those identified within the advisory sections and examples may therefore be taken where it can be shown that the principles of RCM as defined within this NES are followed, and that the required output from the RCM process is not compromised.

22. The MOD has a legal obligation to provide a safe working environment such that the risks to MOD personnel, contractors, visitors and society in general from any hazards that may be present are As Low As Is Reasonably Practical (ALARP). Whilst an RCM study team should not set out to conduct a safety case, the group should identify those hazards that may be present, together with the associated risks to personnel and the environment, caused by any failure mode of the system/equipment under analysis.

23. ILS related statements not identified within this NES as being advisory in nature should be followed except where deviations are specified within the contract.

24. This NES provides a working level definition of the RCM process as required by the high level Logistic Support Analysis (LSA) process detailed within Defence Standard 00-60, and should therefore be regarded as a supplement to Defence Standard 00-60. To ensure that the correlation between this NES and Defence Standard 00-60 is understood, all affected areas of this NES have sub-sections detailing the correlation in LSA processes. This information is summarised at the end of each section.

25. The key players in each phase of the RCM process are indicated at the beginning of each section. Key players should ensure that they read and understand their appropriate sections.

The Regulated Environment

26. It is commonly the case that legislation governing safety makes demands on users of equipment having significant safety implications to demonstrate that they are doing whatever is prudent to ensure that their assets are safe. RCM wholly satisfies this requirement, placing increasing emphasis on an audit trail that requires users to provide documentary evidence that there is a rational, defensible basis for their maintenance programmes.

27. In the case of environmental legislation, performance requirements and constraints are usually quantified explicitly. In such cases RCM provides a maintenance strategy for achieving the performance expectation.

28. Further details of RCM in a Regulated Environment are provided at Annex E.

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CONTENTS

1. RCM IN THE ROYAL NAVY

1.1 Key Players

a. The key players in the RCM process for the Royal Navy are indicated in Table 1.1. All key players should ensure that they understand this section.

Vessel/System Status	
New Construction/Procurement	**In-Service**
Shipbuilder Platform Manager	D Ops P
Shipbuilder Equipment Managers	Warship IPT Leader
Commodity Manager	D Ops E
Warship IPT Leader	Commodity Manager
HILS(N) Management	
IPT ILS Manager	
All Study Group Members	All Study Group Members

Table 1.1 Key Players

1.2 Background

a. This section provides an introduction to the application of RCM when carried out for the RN.

b. The preventive maintenance strategy for the equipments, systems and platform structures used in Royal Naval vessels has traditionally been derived from the manufacturer's recommendations for the equipment, modified as necessary to meet statutory requirements. Occasionally, amendments have been made in response to feedback acquired from in-service experience. This approach to maintenance, which has seen service for over 30 years, has shortcomings when applied to the modern, complex systems now at sea. These shortcomings were sufficiently numerous, and of such importance, that a total review of the maintenance philosophy in the Royal Navy was deemed to be necessary. As a result of the review it was decided that a methodology was required that allowed a more reasoned and structured approach to the derivation of maintenance strategies.

c. RCM is such an approach. It has been accepted by MoD(N) as the process to be employed to derive the maintenance strategy for new procurements, managed under the principles of ILS, and for retrospective application when reviewing the maintenance requirements of platforms already in service.

d. The RCM process is essentially a formal application of common sense, and there will always be the requirement to apply experience and a measure of subjective judgement. If applied correctly, the process will result in the most cost-effective failure management strategy to ensure assets continue to function to meet required performance standards. It also allows MoD(N) to demonstrate that it is exercising the "duty of care" towards its employees and the environment as required by national and international legislation.

e. The benefits that should accrue with the introduction of RCM are:

1) A more relevant maintenance definition.

2) Use of condition monitoring where shown to be applicable and effective.

3) A clear audit trail of the maintenance management decisions taken.

4) An increased awareness of the risk being undertaken when preventive maintenance is deferred, or when consideration is given to changes to the Upkeep Cycle for the vessel.

a. An RCM analysis will produce the information needed to analyse support activities such as the ranging and scaling of spares, Level of Repair Analysis (LORA), the requirement for tools and test equipment, manpower skill levels, and the requirement, for ship and shore-side facilities necessary to support the derived maintenance strategy.

b. The RCM process generates a vast amount of maintenance data and management information. MOD(N) has therefore developed a number of software toolkits that:

1) Assist with the analysis process.

2) Derive the associated on board storing requirement to validate and support the maintenance strategy.

3) Provide an RCM based maintenance management system for the ships and their associated waterfront and headquarters support organisations.

1.3 Patterns of Failure

a. Research into failure patterns within the airline industry during the 1970s revealed that the majority of failures in modern complex equipment/systems did not follow the “bath tub” or "traditional" curves as had been assumed previously. Figure 1.1 shows the 6 dominant failure patterns that were found, with illustrative percentages of occurrence, and explains each one briefly.

b. Further research has shown that these failure patterns occur similarly within industrial plant, and also in modern merchant ships and warships with their advanced technologies.

c. It will be seen that the majority of failures are random in nature and would not, therefore, benefit from a preventive maintenance regime based solely on equipment life or operating age.

d. Only failure patterns A and B exhibit a distinct wear out zone, indicating that failure modes that conform to these failure patterns would benefit from an age related maintenance strategy. The term “age related” is used to denote a period of exposure to stress, and might be measured in hours run, miles steamed, rounds fired, number of operational cycles, etc. It is not solely a measure of calendar time.

e. The observation that the vast majority of items exhibit no perceivable wear out zone has important implications for maintenance. It follows that, unless the item has a dominant failure mode that is age related, maintenance around an assumed age limit does little or nothing to improve overall reliability.

f. Failure rate also plays a relatively unimportant role in deriving maintenance systems. Although the frequency of failure is useful in making cost decisions and in establishing appropriate intervals for maintenance tasks where a “life” has been established, it conveys no information as to what the consequences of failure might be or the tasks necessary to actually prevent the failure.

1.4 Summary of the RCM process

a. RCM is a process that takes account of patterns of failure to ensure that assets can achieve their inherent reliability in their specified operating context. It is used to determine the maintenance requirements of an asset in its specific operating environment to ensure that it continues to achieve its required performance standards. It allocates the most suitable maintenance, with the least expenditure of resources, and recognises that inherent reliability levels cannot be improved upon through maintenance if the original design is inadequate and is not subsequently modified.

b. RCM is based upon the responses to the following seven fundamental questions asked for each asset being analysed:

1) What are the functions and associated desired standards of performance of the asset group in the present operating context (functions)?

2) In what ways can the asset group fail to fulfil its functions (functional failures)?

3) What causes each functional failure (failure modes)?

4) What happens when each failure occurs (failure effects)?

5) In what way does each failure matter (failure consequences)?

6) If the failure matters, what should be done to predict or prevent each failure (proactive tasks and task intervals)?

7) What should be done if a proactive task cannot be found (default actions)?

a. Answers to questions 1 to 4 are derived and recorded during the Failure Mode, Effects and Criticality Analysis (FMECA), which is discussed in greater detail at section 9. The Marine and Combat System RCM algorithm detailed at section 10 is used to determine the answers to questions 5, 6 and 7. RCM is undertaken in the Royal Navy to derive failure management strategies and, therefore, the FMECA must be compiled using the processes detailed in this NES rather than any other approach.

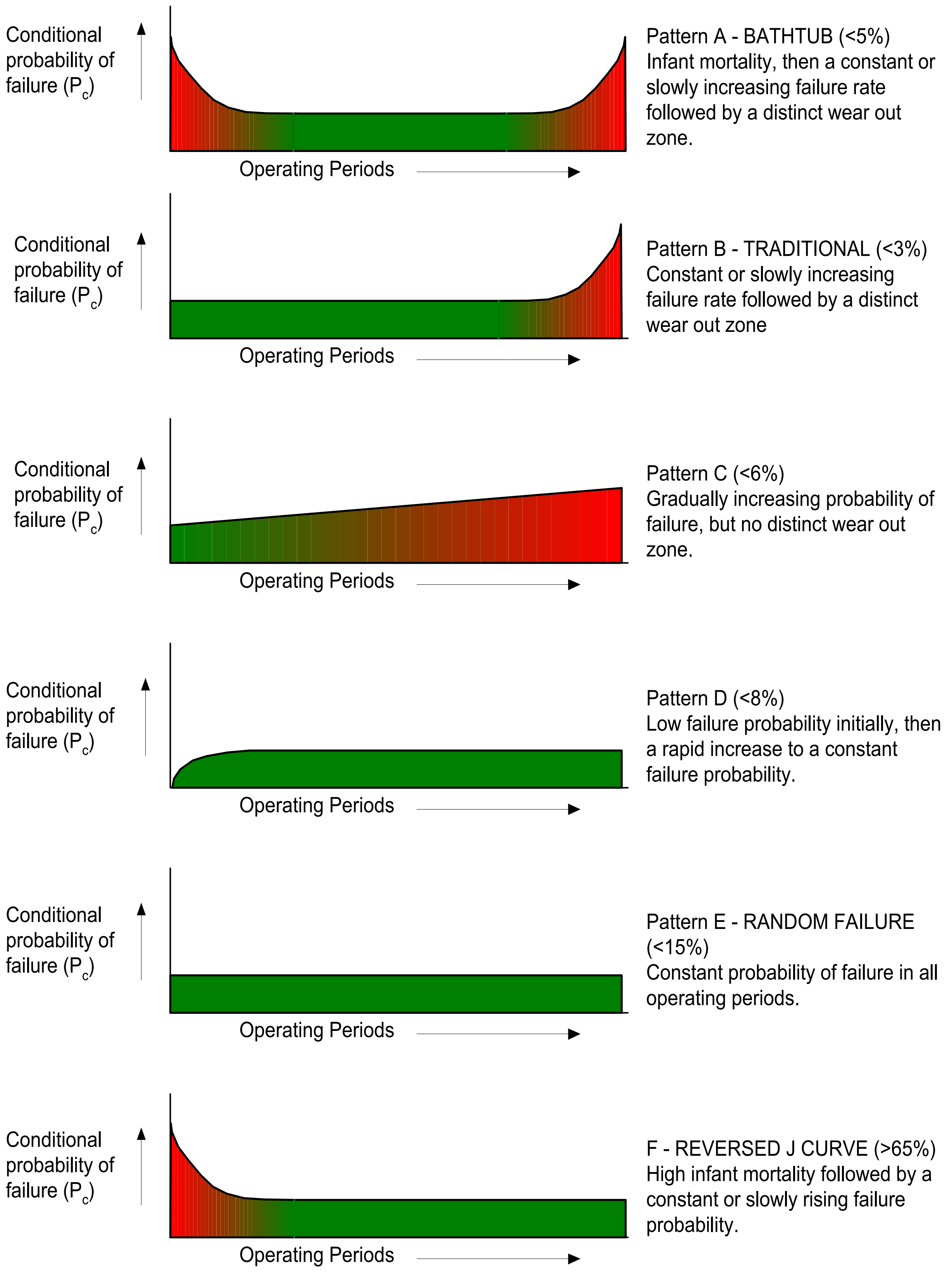

Figure 1.1 - Patterns of Failure

Note: The percentage figures quoted represent the average occurrence of the failure pattern in modern complex systems.

1.5 Training

a. A key factor in ensuring high quality RCM analyses and subsequent successful implementation is that personnel are correctly trained. MOD PE and NSC staff, contractors and sub-contractors, together with associated waterfront personnel should all be trained in RCM techniques. At a minimum, all personnel should have completed a recognised RCM Introductory course. Further detailed RCM awareness can then be provided to those with a justified requirement by attending a RCM Facilitators’ course. Facilitators, Technical Secretaries and other personnel (eg EPMs) should also be familiar with the application of the RCM (RN) and RCS (RN) software toolkits. Details of training course availability can be obtained from the sponsor of this NES.

1.6 Defence Standard 00-60 Correlation

a. LSA Task Applicability

1) The requirement to conduct RCM is identified within Task 301, subtask 2.4.2 of Defence Standard 00-60.

2) The contents of section 1 are advisory in nature within the ILS context.

2. THE RCM PROCESS WITHIN ILS

2.1 Key Players

a. The key players in the RCM ILS process are indicated in Table 2.1. All key players should ensure that they understand this section.

Vessel/System Status	
New Construction/Procurement	**In-Service**
Shipbuilder Platform Manager	Warship IPT Leader
Shipbuilder Equipment Managers	D Ops P
Warship IPT Leader	D Ops E
D Ops P	
D Ops E	
Shipbuilder ILS Manager	
MoD ILS Manager	
HILS(N) Management	

Table 2.1. Key Players

2.2 Introduction

a. All significant procurements by the MOD mandate that supportability and through life costs are given due consideration (see current CDPIs). In order to achieve this in the most cost-effective manner the principles of ILS are employed. One of the key tools used within ILS to influence supportability is the application of LSA.

2.3 Background

a. Defence Standard 00-60 has been developed by the MOD to ensure that a consistent approach is taken within ILS for all defence procurements. This standard has been derived from the amalgamation and evolution of four existing standards:

1) US Military Standard 1388-1A (for LSA).

2) US Military Standard 1388-2B (for LSA data recording and reporting).

3) AECMA 1000D (for technical publications).

4) AECMA 2000M (for spares support).

a. The use of these standards has ensured that the lessons learned from the many years of their development and practical application have been included within Defence Standard 00-60.

b. Within US Military Standard 1388-1A/2B, the RCM process was based on the standard known as MSG 3 or one of its military derivatives such as US Military Standards 1843 or 2173(AS).

c. The conduct of RCM for Naval procurement projects must therefore be regarded as much more than the traditional RCM detailed within Defence Standard 00-60 (Part 1A, subtask 301.2.4.2). Organisations involved in the planning and management of RCM and LSA programmes must ensure that the integration of RCM within the LSA process is understood and that effort is not duplicated unnecessarily. Details of the relationship between the two processes are shown within Figure 2.1.

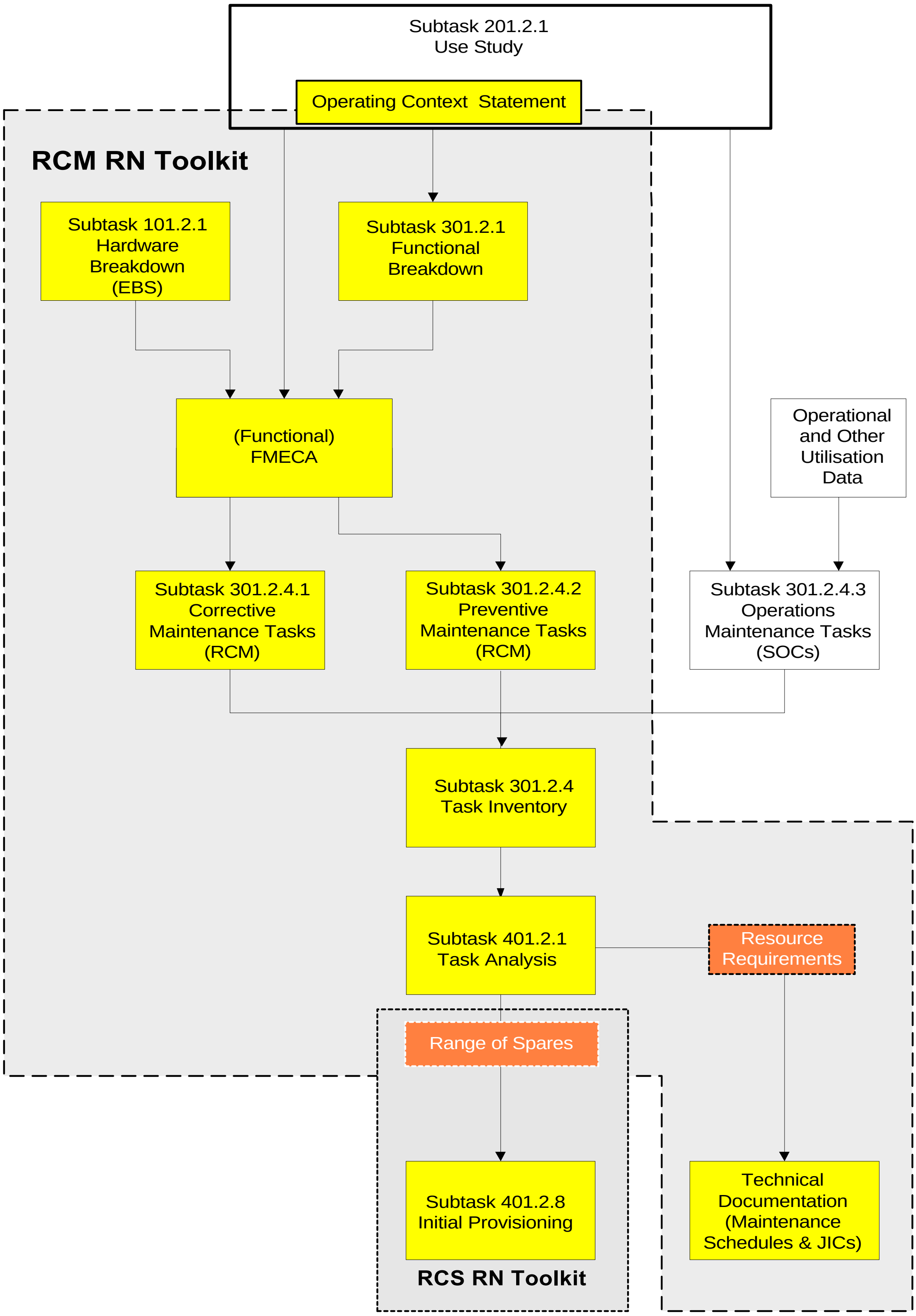

Figure 2.1 - LSA / RCM Process Map

d. The data recording requirements for the two methods also differ. This is primarily as a result of the divergent boundaries but also due to the use of different RCM decision logic matrices. Historically there has always been a shortfall within Military Standard 1388-2B in that not all aspects of a

MSG 3 based RCM study can be recorded within the defined database. This continues to be the case with studies following the NES 45 method and utilising Defence Standard 00-60.

e. With the implementation of modern safety standards and management methods it is essential that the RCM process liaise fully with the relevant safety management team. Many decisions made within the RCM process have the ability to undermine the safety justification and mitigation agreed as part of the control mechanism for a given hazard in the safety management system. It is therefore essential that the RCM team provide feedback to the safety management team to identify areas where the existing maintenance approved by a safety case may need to be modified.

2.4 Management of RCM within the LSA Process

2.4.1 Introduction

a. RCM as an ILS activity lies totally within the LSA Process as shown in Figure 2.2. A RCM Study conducted in accordance with this standard must always be regarded as an important, key activity even when applied to relatively small-scale projects. Proper planning and management of the Study is therefore essential to derive the required benefit from the Study and ensure that overall project timescales and costs are not compromised.

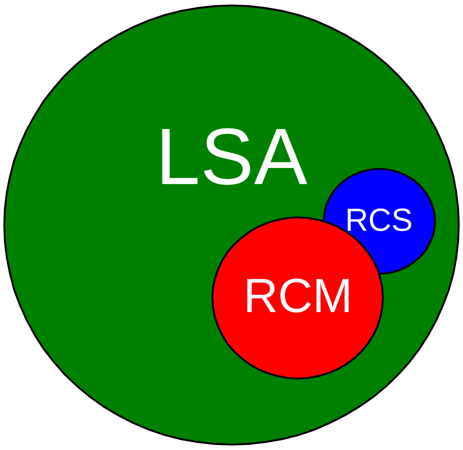

Figure 2.2 - RCM and the LSA Process

b. Due to the high level of interrelationship between the activities within this standard and those contained within Defence Standard 00-60, and also due to the need to ensure that effort is not duplicated, a single planning and management activity must be undertaken.

c. The 100 series Tasks defined within Defence Standard 00-60 provide the requirements for effective planning and management of the LSA effort. These tasks should either be applied to RCM in isolation or preferably, in order to avoid duplication of effort, RCM should be planned and managed as an integral part of the LSA programme.

d. The 100 series tasks are:

1) Task 101 - LSA Strategy

2) Task 102 - LSA Plan

3) Task 103 - Programme and Design Reviews

a. The following sections provide guidance on the issues to be considered when planning and managing an RCM Study as an integral part of a LSA programme.

2.4.2 LSA Strategy (Task 101)

a. Within the LSA programme the requirement for the conduct of RCM will be determined during the development of the LSA Strategy. It must be ensured that all Naval procurement projects requiring RCM have NES 45 specified as the detailed procedure to be employed.

b. In order to conduct RCM the following mandatory activities detailed within this standard must be undertaken:

1) Compilation of the Operating Context (section 5)

2) Functional Partitioning (section 6)

3) Assembling an appropriate Analysis Team (section 7)

4) Carrying out a FMECA (section 9)

5) RCM Analysis Using Marine and Combat Engineering Algorithm (section 10)

6) RCM Analysis of Structures and Structural Items (section 11)

7) Assembly of Maintenance Schedules and the writing of Job Information Cards (section 12)

8) Task Rationalisation and Derivation of the Platform Upkeep Cycle (section 13)

9) Spares Holding Determination (section 14)

a. RCM must be conducted at the appropriate phases of a project and the applicable phases must be identified within the LSA Strategy.

b. Commercial Off-The-Shelf (COTS) procurements of significant size[1] will require the application of RCM to determine the correct maintenance regime of equipment in its intended operating context.

2.4.3 LSA Plan (Task 102)

a. The LSA Plan must contain details not only of the RCM activities detailed above, but also details of the management aspect of RCM within the LSA process (responsibilities, resources, contractor relationships, etc.).

b. Details of the input and output data requirements for each RCM activity must also be included within the LSA Plan (functional partitioning, FMECA, etc.).

2.4.4 Programme and Design Reviews (Task 103)

a. Full consideration of RCM must be given as part of the overall LSA Programme reviews and during all design reviews.

2.5 LSA Process Map

a. Figure 2.1 identifies the major activities conducted using RCM within the LSA Plan, as defined by Defence Standard 00-60, and illustrates those that are appropriate to RCM as defined within this standard.

b. There are other LSA processes that are not covered by Figure 2.1. Some of these may have an impact on RCM while others may be impacted by the conduct of RCM. Of particular importance in this respect will be the conduct of trade-off analyses and Level of Repair Analysis (LORA).

c. Amongst the LSA Tasks not covered by Figure 2.1, Tasks 303 and 401 are the major consideration. Task 303 defines a series of trade-off studies, including LORA, which will require input details of maintenance activities, and the output may result in changes to maintenance schedules. When applied normally, LORA is an iterative process with an increasing level of detail at each iteration. Early iterations may make gross assumptions of maintenance actions but the more detailed iterations will require the level of detail provided by a task analysis as detailed within Defence Standard 00-60 Task 401.

d. Within NES 45 a task analysis is an integral RCM process. The results of the later iterations of LORA may therefore cause rework in the conduct of RCM analysis (sections 10 and 11) and all subsequent processes.

e. Figure 2.1 depicts certain activities as being within the scope of one of the two Naval RCM software toolkits, RCM RN and Reliability Centred Stockholding RCS RN. The toolkits can be

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[1] Reference should be made to current guidelines regarding cost/risk thresholds.

provided as free issue to organisations performing RCM/RCS studies on MoD(N) procurement projects, subject to detailed licensing arrangements.

2.5.1 The RCM RN Software Toolkit

a. The RCM RN toolkit shall be used to assist in the conduct of an RCM analysis, and to record the results of the analysis in the appropriate data fields.

b. RCM RN also holds the data fields necessary to plan and carry out platform, system and equipment maintenance (maintenance procedures, maintenance schedules and Job Information Cards).

c. The relevant fields within the LSAR shall be populated from the RCM RN Toolkit to ensure data consistency.

d. As the RCM RN Toolkit holds more data than can be held within the LSAR, both the RCM RN Toolkit and the LSAR must be maintained. This is illustrated in Figure 2.3.

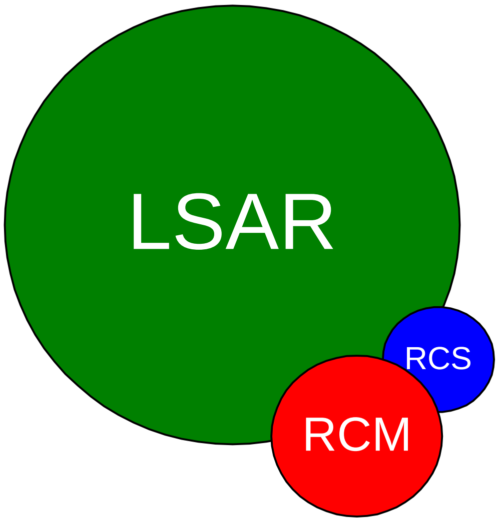

Figure 2.3 – RCM/RCS RN and LSAR Data Relationship

2.5.2 The RCS RN Software Toolkit

a. The RCS RN Toolkit can be used to record spares ranging details and to perform and record spares scaling.

The relevant fields within the LSAR shall be populated from the RCS RN Toolkit to ensure data consistency.
As the RCS RN Toolkit holds more data than can be held within the LSAR, both the RCS RN Toolkit and the LSAR must be maintained. This is illustrated in Figure 2.3.

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3. THE PLATFORM PROCESS

3.1 Key Players

a. The key players in the RCM platform process are indicated in

b. Table 3.1. All key players should ensure that they understand this section.

Vessel/System Status	
New Construction/Procurement	**In-Service**
Shipbuilder Platform Manager	D Ops P
Shipbuilder Equipment Managers	D Ops E
Shipbuilder ILS Manager	Warship IPT Leader
Commodity Manager	Commodity Manager
Warship IPT Leader	
MoD ILS Manager	
HILS(N) Management	

Table 3.1 Key Players

3.2 Preparatory Work

a. In order to optimise the benefits of RCM, the process should be applied using a top down approach starting at platform level. However, it is recognised that this may not always be appropriate, and it may at times be necessary to conduct individual studies at system or equipment level. Before any study is started, the preparatory phases detailed in sections 4 -7 of this NES must be completed, namely:

1) Identify contributing authorities.

2) Write the Operating Context.

3) Construct the functional and physical block diagrams.

4) Determine the functional and physical analysis boundaries.

5) Collect the supporting study data.

6) Interrogate the existing RCM (RN) database for templates.

7) Form the analysis team with the appropriate expertise.

8) Allocate the study identification/control number.

3.3 RCM Platform Process Map

a. The full benefits of RCM are only realised when all the failure modes of a platform have been analysed using RCM principles. Once this has been completed, the Platform Manager will be supported with the most effective maintenance package possible for sustaining the platform in its operational context.

b. The RCM process for analysing complex structure is different to that used for RCM marine and combat system analyses. Before commencing a study it will be necessary to decide whether the analysis is purely structural, purely ship system, or a hybrid of the two.

c. A flow chart, highlighting the steps required to achieve an RCM based platform Upkeep Strategy is shown in Figure 3.1. Each stage of the process annotated on the figure is then discussed in the following sections of this NES. Only the mandatory activities are shown. Those sections that are advisory in nature are omitted.

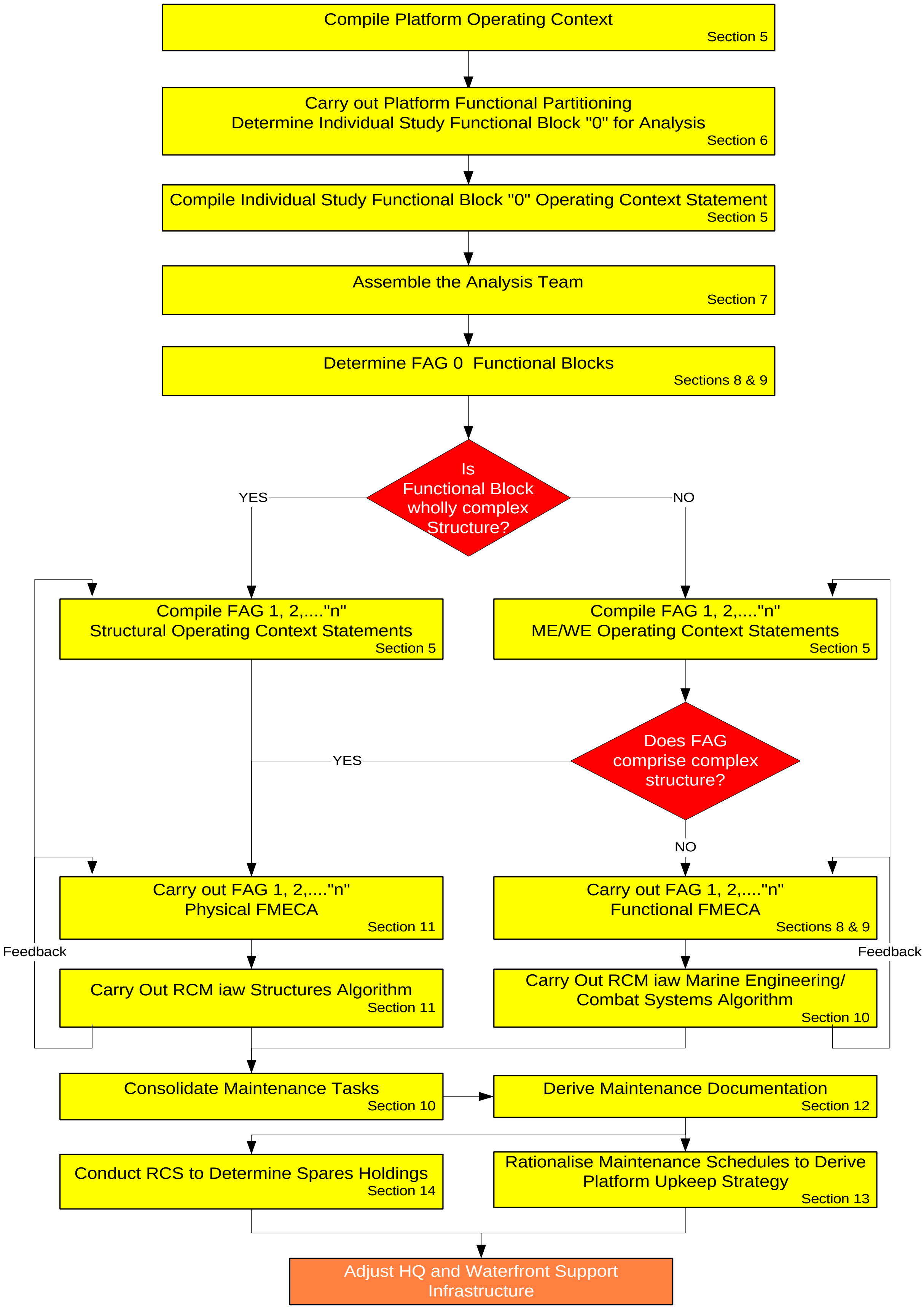

Figure 3.1 – The RCM Platform Process Map

4. ENGINEERING INFORMATION REQUIREMENTS

4.1 Key Players

a. The key players requiring RCM engineering information are indicated in Table 4.1. All key players should ensure that they understand this section.

Vessel/System Status	
New Construction/Procurement	**In-Service**
Shipbuilder Equipment Managers	D Ops P
Shipbuilder ILS Manager	D Ops E
Warship IPT Leader	Warship IPT Leader
MoD ILS Manager	
Study Facilitator	Study Facilitator
Study Technical/Design Advisor	Study Technical/Design Advisor
	Study WPM Representative
	Study EPM Representative
	Study Operator/Maintainer

Table 4.1. Key Players

4.2 Failure Data

a. The basis of RCM is an understanding of the failure modes of the assets under consideration. The first stage of any detailed analysis of a function should be to collate all obtainable failure data to help ensure that all failures that have occurred previously are covered. Refit survey records should also be obtained to give an indication of the condition of equipment after use. However, extensive historical failure data is not essential to the process as the subjective judgement of an analysis team with a deep knowledge of the equipment can be used where hard data is not available.

b. Failure data taken in isolation has limited value without a full understanding of the failure mechanisms it is revealing. Hence, failure statistics should be treated with caution because:

1) Data relating to failures with safety or environmental consequences is likely to be sparse or even non-existent, since any existing maintenance programme should have prevented the failures on which such data would be based (Resnikoff's Conundrum).

2) Conversely, frequent failures of minor consequence will generate a large amount of failure data. This will be of limited value since the failures themselves do not matter.

3) Stores usage data will reflect the existing maintenance regime.

4) Many failure modes could be suppressed by the existing maintenance strategy.

a. There are many engineering and maintenance data sources available within MOD(N) and the RN. Where applicable to a particular platform or equipment undergoing an RCM study some or all of the data from these sources may be released to contractors. However, this data is gathered in-service and relates to specific equipment operated within a specific RN operating environment. Care must therefore be taken when utilising such data for new design systems and equipment, and specialist reliability engineering advice should be sought.

b. Many contractors will have long-established data collection systems for in- service equipments similar to those used within the RN. These can record the occurrence of failures and the failure modes observed during development, production, repair and overhaul activities. Such systems should be used where they can provide useful input data to a RCM Study.

More generic failure rate and failure mode data is available within industry that could also be used for new design systems and equipment. Some of the more common commercial sources for such data are listed at Table 4.2.

Type of Data	Data Source
MTBF & failure rate (but see also Def Stan 00-41)	Non-electronic Parts Reliability Data (NPRD) Military Handbook 217F Stress/Strength Interference (SSI) calculations Commercial databases including : RELDAT SRD FARADIP HARIS OREDA GADS HRD De facto industry standard publications including : Green & Bourne D J Smith Manufacturers' product literature, equipment specifications or data sheets
Failure modes and associated distributions for failure mode MTBF	Failure Modes Distribution (FMD) Military Handbook 338 De facto industry standard publications including : D J Smith

Table 4.2. - Data Sources in the Commercial Sector

4.3 Useful Documentation

a. Some documents that may assist an RCM analysis team during the various stages of a study are given below. It is stressed that these lists are not exhaustive.

1) Writing the Platform, Engineering Discipline and Functional Block Operating Contexts:

(a) Staff Requirements (SRs) – define operational performance parameters for the function.

(b) Warship Policy Document (WPD) – define operating parameters.

(c) Agreed Characteristics (ACs)- refine performance parameters.

(d) Reliability Block Diagrams - identify survival probabilities.

(e) Books of Reference (BRs) and Charge Books (CBs) – provide technical performance parameters.

(f) Safety Case or Safety Assessments – to identify failure modes and consequences.

(g) Manufacturers' Handbooks – provide performance parameters.

(h) Operating Procedures.

(i) HATs and SATs data, Acceptance and Performance Trials test forms. – for definition of acceptance parameters.

(j) Requirements database – gives a more detailed quantification of functional and physical requirements in contractual language leading to Project Definition.

2) Determining Functional Block reliability and maintenance characteristics:

(a) S2022s, SUDS and OPDEFS - give reliability and dominant failure patterns of in service equipments.

(b) Manufacturers' Handbooks - give equipment design parameters.

(c) ACs - give required performance parameters.

(d) Current Planned Maintenance Schedules (PMSs) - give the current maintenance load.

(e) Predicted Alterations and Additions (As & As) - give estimated equipment service life.

3) Setting the Functional and Physical Boundaries for the analysis:

(a) Functional Block Diagram for the platform - map the functional boundaries and record the dependencies between functions.

(b) System Diagrams - identify the physical assets involved in delivering the function.

(c) BRs - provide technical and operational information about equipment.

(d) Drawings - give details about the physical construction of systems and equipment.

4) Conducting FMECA/RCM Analyses:

(a) S2022s, SUDS and OPDEFS - can be used to help identify failure modes and give an indication of failure rates, noting that only reported instances of failure will be visible. However, these methods of reporting do not record defects identified at overhaul, and generally report effects rather than root causes of failures.

(b) Current Planned Maintenance Schedules - can be used as a checklist after completing the FMECA to check all failure modes have been covered.

(c) The RCM (RN) toolkit - can be interrogated to assist templating of existing data, but only if applicable for a specific study. This information must be reviewed for accuracy in the new study.

5) Determining patterns of failure & Mean Time Between Failure (MTBF) values:

(a) Survey reports at refit - may indicate the condition of components at specific running hours, or after a specific period of installed operational life.

(b) Manufacturers' strip-down reports.

(c) Generic failure data from the North Sea Oil Industry (OREDA - and other reference databases – see Table 4.2).

(d) Stores usage rates - may give an indication of failure rate and MTBF.

(e) D171: Reports of Survey.

6) Writing of Job Information Cards:

(a) BR 1313 (see also section 12).

b. Further advice on the use of the above data, and information on the derivation of specific RCM parameters required during an RCM analysis, is provided in later sections.

4.4 Templating

4.4.1 Use of Existing RCM RN FMECA and Task Analysis Data

a. The importing of RCM records generated during previous analyses is known as "Templating". It can save considerable time and effort by indicating possible solutions, **BUT MUST NEVER BE USED WITHOUT PROPER CONSIDERATION OF THE OPERATING CONTEXT INTO WHICH THE TEMPLATE IS TO BE IMPORTED**. The criticality of each failure mode is also to be revisited. Templating takes several forms:

b. RN Standard. Certain analysis topics (such as leaks, pressure gauges, earthing straps, etc.) can be approached from different angles and will usually generate time consuming discussion. The RN standard templates will give guidance on the accepted approach.

c. Generic Equipment. Certain common task equipment (such as motor driven centrifugal pumps) will be susceptible to similar basic failure modes. An appropriate generic equipment template may be used as an aid when formulating the failure modes for a specific occurrence.

d. Whole Equipment. The results from a previous analysis of the same equipment employed in a different context can be used to focus the team's deliberations in generating failure modes and indicating possible maintenance tasks. It must not, however, be used verbatim.

e. Rubber Stamping. This is a direct "cut-and-paste" procedure, which can only be used when the team is completely satisfied that they have already considered identical equipment with an identical operating context. In this special case, the record can be copied in its entirety.

4.4.2 Available RCM (RN) Templating Data

a. RCM RN databases contain information with the potential to save considerable time and effort. An interrogation of the database held by the RCMIT should be completed by the study facilitator, the MILSM, or the appointed representative with responsibility for RCM, before an analysis is undertaken to establish whether any suitable RCM (RN) templates exist. For RCM studies that are undertaken by contractors any templates that are identified as being suitable will be provided to the contractor but it remains the responsibility of the contractor to ensure the suitability of the template to the RCM study being undertaken.

b. Contractors may also develop their own generic template during an analysis where no such template exists within the RCM (RN) Toolkit. In such circumstances the master of the new template should also be provided to the RCMIT for the RCM (RN) toolkit and to provide assurance of its generic applicability and correct use in all operating contexts.

4.5 Documentation Amendments

a. The RCM process, applied correctly, will result in recommendations for the most effective method of operating systems and equipments within their operating context in order to extract the maximum useful life from them. Recommendations for changes to BRs and technical publications are therefore likely outcomes from an RCM study.

b. It should be noted that any proposed documentation changes would be specific to a system/equipment only within the operating context of a specific RCM study. Any proposed documentation changes will not necessarily be applicable for the equipment in all its operating configurations, and must not be templated across without careful checking.

c. Where a system has an existing Safety Case, it is imperative that this is revisited before a RCM based maintenance strategy is implemented. Safety cases are based on existing approved maintenance and any changes will require the Safety Case to be revalidated.

4.6 Defence Standard 00-60 Correlation

a. LSA Process/Task Applicability

1) Data collection has no direct correlation to any process defined within Defence Standard 00-60.

2) Task 303 does, however, require the use of quantitative data as input to trade-off studies and in the development of a baseline comparison system.

3) Reliability data is normally generated under the control of a reliability programme and is used within a baseline comparison system.

4) The contents of this section are advisory in nature.

5) The conduct of this activity is a desirable prerequisite to the performance of RCM.

5. THE OPERATING CONTEXT

5.1 Key Players

a. The key players in this part of the RCM process are indicated in Table 5.1. All key players should ensure that they understand this section.

Vessel/System Status	
New Construction/Procurement	**In-Service**
DOR (Sea)	DOR (Sea)
Warship Type Commander	Warship Type Commander
Shipbuilder Platform Manager	Squadron Staff
Shipbuilder Equipment Manager	Warship IPT Leader
Shipbuilder ILS Manager	D Ops E
Warship IPT Leader	Equipment Operators/Maintainers
MoD ILS Manager	D Ops P
Study Facilitator	Study Facilitator

Table 5.1. Key Players

5.2 Requirement

a. An operating context statement describes the physical environment in which the asset is operated, gives precise details of the manner in which the asset is used and specifies the performance capabilities of assets as well as the required performance of the systems in which assets are embedded. This is the foundation for the FMECA and all subsequent RCM decisions, as RCM determines the maintenance required to preserve the function of a system or equipment within its specific operating context. It is therefore essential that all interested parties agree the operating context before starting the analysis.

b. Operating contexts, at all levels, are to be regarded as "living" documents and, as such, are to be sustained throughout a ship's life. They should be reviewed regularly by the Warship Platform Manager as platform configuration changes and by the Type Commander when usage policy and/or tactical doctrines evolve. Annex G describes the various levels of Operating Context.

c. The operating contexts for the whole platform and the individual engineering disciplines will be fundamental to the Functional Partitioning process, whilst more specific operating context statements will be required before commencing the FMECAs of individual functions. The operating contexts for the analysis of platform functions must therefore be written in a cascading hierarchy to correspond with the level under consideration as the study proceeds. The following paragraphs outline the basic requirements of each level of operating context, and are illustrated with extracts from those of the Hunt Class MCMV.

5.3 Platform Level Operating Context (Level 2)[2]

a. The platform level operating context is written first. It is generic to a class and briefly describes the vessel's main physical characteristics, as illustrated in Example 5.1.

> "... the vessel is made of glass reinforced plastic and attention has been paid to the reduction of the ship's noise and magnetic signature. The vessel has an overall length of ..."

Example 5.1 - Physical Characteristics

b. It addresses the vessel's primary role, along with any secondary roles and tasks, identifying the systems installed to achieve them, as illustrated in Example 5.2.

[2] Level 1 Operating context information is contained within the Warship Policy Document (see Annex G)

"... when dealing with mines it is essential that the ship is able to search for and approach a contact slowly and hover in a fixed position …an auxiliary diesel engine supplies power to a slow speed drive and bow thrusting system is for this purpose ...... in its secondary role the vessel carries out general patrol and fishery protection duties during which it patrols littoral waters inspecting the nets and catches of any fishing vessels ..."

Example 5.2 - Vessel Roles

c. The typical peace and wartime mission profiles are outlined, along with the appropriate support regimes, as illustrated in Example 5.3.

"... the MCMV can be deployed for periods of several weeks away from base support. It is capable of operating singularly or with a small squadron, possibly assisted by a Forward Support Unit (FSU). In its primary role the vessel expects to operate at up to A days transit from the main base for B days continuous mine sweeping or mine hunting within a maximum of C hours passage from the FSU. The maximum primary mission length without support is therefore "D" days. The vessel might expect to undertake up to E such missions per year, with passage, restore, maintenance periods, general patrol, visits, trials and training in between ..."

Example 5.3 - Roles and Support

d. The platform operating context also describes the vessel's operating environment, as illustrated in Example 5.4.

"... as the vessel is required to be capable of operating over a wide area, all equipment must give full specified performance over the range from Standard Tropical to Standard Arctic Conditions ..."

Example 5.4 - Operating Environment

5.4 Engineering Discipline Operating Contexts (Level 3)

a. The 3 main engineering discipline operating contexts are then written for the overall Marine Engineering, Combat systems and Hull, that together make up the total platform operating context. These engineering discipline operating contexts are still at a high level, but describe in more detail the way that the vessel uses its primary assets to fulfil its designated operational roles. The format for these operating contexts is at Annex G.

b. The marine engineering discipline operating context for a HUNT class MCMV might contain the statements in Example 5.5:

"…ship's staff provide first level maintenance, and undertake ships' daily husbandry duties to maintain the cleanliness, appearance and preservation of the vessel. Routine servicing, scheduled maintenance, and the diagnosis and repair of minor defects will normally be undertaken by the ship's engineering staff …..."

Example 5.5 - Engineering Discipline

c. The combat system operating context also includes the operating and maintenance capabilities of Ship's Staff, as illustrated by Example 5.6:

" The major weapon systems comprise the Type A Command System, Sonars type B&C, Radars type D&E, the following Navigational equipments, etc. The Command System interfaces with the following systems and equipments …… and provides the following facilities to the Command……." " .. on-board maintenance capabilities exist to undertake PEC diagnosis and repair, but this facility would only be used in extreme circumstances, when upkeep by exchange is not available …."

Example 5.6 - Combat System

The above example is sanitised to meet the unclassified nature of this document. Further advice on security classification requirements, particularly with regard to Information Systems, should be sought from appropriate sources.

d. The hull operating context will include details of design assumptions as illustrated by Example 5.7. For the full requirement for this discipline, see section 11.

> “ this is the main collision bulkhead and as such it has been designed to resist the forces generated by forward vessel movement of up to 10 knots ahead with the stem in the damaged condition. The next bulkhead (WTB6) has not been designed with this degree of strength. WTB3, as a consequence, needs to remain structurally intact to preserve its protective function."

Example 5.7 - Hull

5.5 Functional Block and Functional Asset Group (FAG) Operating Contexts

a. Once the engineering discipline operating contexts have been written, the operating context statements are cascaded down to functional block and functional asset group level.

b. These operating contexts are a key element in the overall specification of a functional block and functional asset groups. (The functional partitioning process and the derivation of functional blocks and functional asset groups is described in detail in section 6.) At the functional block level, the operating context is written to precisely define the operating parameters of the function under review. Each level of further functional breakdown in turn requires an operating context statement amplifying the operating assumptions for that specific FAG.

c. At the lower levels of the functional breakdown more detail is included in the operating context statement, as by then the focus is on the systems and equipment which make up the functional block, their operating parameters and all their modes of operation. <u>Specific performance parameters are necessary to clearly determine what constitutes a failure, and what effects such failures will have upon specific equipment performance, overall system operation and, ultimately, the ship’s mission</u>.

d. An example of a complete operating context statement covering the functional block “Provision of Compressed Air and Diving Support Services” is provided at Example 5.8.

> “To produce, store and supply compressed air.
> High-pressure air is required at 207 bar at a (specified) quality for use in Breathing Apparatus Self Contained Compressed Air (BASCCA), diving breathing sets and the decompression chamber. Air is required at a reduced pressure of *X* bar for pneumatic control and propulsion machinery control, the starting of propulsion and generator diesel engines and to minehunting equipment.
> Two electrically driven compressors, each nominally capable of compressing X litres of air per minute to 220 bar, are sited in each of the Generator and Engine Rooms. Each compressor set has 12 in number air reservoir bottles of 20.4 litres capacity each, divided into two groups of six bottles, for the storage of air. One or more compressor sets can be in operation at any time depending on the demand for air. Each set can be operated locally or from remote position in the Ships Control Centre (SCC). Normal operation is for the compressor(s) to be started from the remote position. When 207 bar pressure is reached the compressor(s) is automatically shut down by action of the pressure sensing switch. The compressor(s) can be restarted from remote position again after the pressure sensing switch has reset at a pressure below 207 bar. In an emergency, an emergency press-to-stop device is used to isolate the electrical supply and stop the compressor.
> The output from the compressor is passed through filtration and dryer units via flexible hoses to remove oil and water droplets/vapour before passing into the system or storage bottles via the gauge and distribution panels.
> The Generator Room reservoir group is normally used when breathing air is required for the decompression chamber and BASCCA charging. For normal supply of all other ancillary systems, the Engine Room reservoir group will be used. In emergency, reservoir bottle groups can be blown down by operation of emergency blow-down valves sited external to compartments.
> A starter and surveillance panel is used to control and monitor the motor and compressor. The parameters monitored are the motor temperature, the compressor lubricant oil pressure (1.72 bar) and the final delivery air temperature (45°C). If any of these exceeds the set limits the compressor is stopped automatically and an audible alarm and indication is given in the SCC. The cut-out can be bypassed and the plant restarted if required. A connection is provided to allow the supply of HP air from a shore side facility. Continuity of supply must be maintained in the event of any single point failure.”

Example 5.8 – Functional Block Operating Context

e. Each FAG, once it has been identified during the FMECA, will require an even more detailed operating context to ensure that the analysis team has a clear understanding of the constraints upon the asset under consideration. e.g. the "Chilled Water System", identified as a FAG within the function of the "To Provide Conditioned Air", might contain the following performance parameters, as illustrated by Example 5.9:

> "... to allow the plant to continue running under low load conditions a capacity control line valve operates to prevent the chilled water temperature dropping from 6.1°C to 4.4°C. The valve is fully closed when the chilled water temperature at the chiller outlet has fallen to 6.5°C and begins to open when the chilled water temperature falls to 6.1°C, or when the compressor suction pressure falls to 4 bar. The valve is fully open when the chilled water temperature has fallen to 4.4°C, or the compressor suction pressure has fallen to 3.7 bar. The compressor is stopped if the chilled water temperature falls to 3.5°C, if the chilled water differential pressure rises to 0.45 bar, or if the chilled water differential pressure falls to 0.075 bar ..."

Example 5.9 - Lower order FAG Operating Context

f. All assumptions made in the production of an operating context statement must be clearly identified. This will include equipment operating procedures, availability requirements for the system, protective devices and any built in redundancy or emergency / alternative arrangements, etc., as illustrated in Example 5.10:

> "... the analysis team assumed that one plant would be operated as a dedicated duty unit whilst the second unit would be operated as a standby. In the event of a loss of the duty plant there would be no operational consequences as the vessel would be capable of continuing minehunting or minesweeping operations as long as the standby plant could be started within one hour of the failure of the dedicated duty plant. In the unlikely event that both air conditioning plants failed the ship could only continue operations if an alternative emergency cooling arrangement could be supplied to operational spaces ..."

Example 5.10 - Analysis Assumptions

5.6 Mission Phase Analysis

a. The operational state of a warship can be very much a variable, ranging from peacetime cruising, through enhanced readiness to meet specific threats, to Action Stations during hostilities and war. It may therefore be necessary to generate different operating contexts at all levels to reflect these different operational states or as a tool to more closely identify failure effects at critical stages of a platform's mission.

b. Mission Profiles. Within the timeframe of a specific mission there can also be significant differing platform operational states, as illustrated in the illustrative submarine operating profile in Figure 5.1. Although in this instance the timeframes are comparatively short and maintenance variations could therefore be discounted, it would be necessary to consider the different maintenance requirements for dived operations and extended harbour periods alongside.

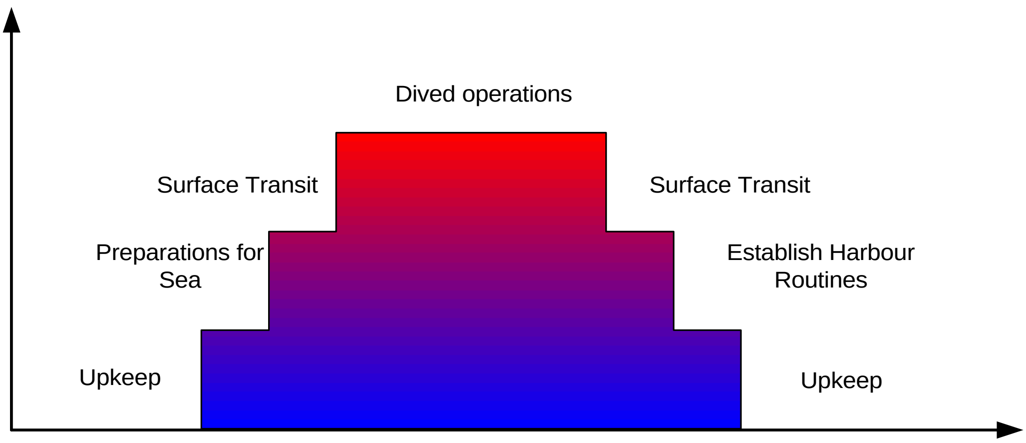

Figure 5.1. - Illustrative Platform Operating Cycle

c. System Readiness Implications. The operational demands placed on certain systems and equipments by the user may also vary to meet the changing platform requirements. This is particularly true for weapon systems, where the operational state could range from shut down, through extended readiness, operational notice, stood to, and, ultimately, engagement with the enemy. This changing demand is illustrated for a typical Close in Weapon System (CIWS) in Figure 5.2, which demonstrates a typical weapon cycle.

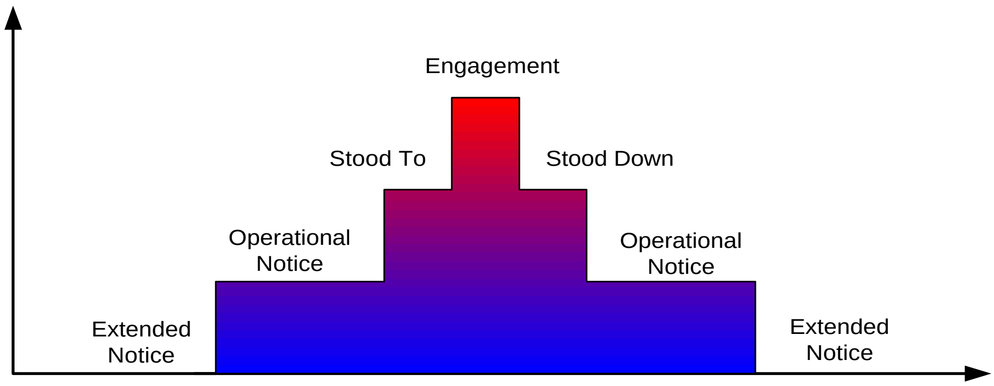

Figure 5.2. - Illustrative CIWS Readiness States

d. Readiness States. In this specific instance, the result of such changing operating context may be a set of differing maintenance requirements which may be matched to the required system readiness. These could range from an inspection based on calendar time during periods of extended notice, to those required for pre and post firing checks or based on the number of rounds fired.

e. Operating Environment Implications. A changing maintenance demand for certain systems could also be caused by different operational environmental conditions, i.e. between sub-arctic and tropical conditions. Such changes may also need to be considered in the analysis.

f. Carrying Out Mission Phase Analysis. A mission phase analysis should be carried out at the start of each study to ensure that the group understands the possible changing demands that will be placed on an asset based on the higher level operating contexts. For some equipments there will be a requirement to repeat certain sections of the actual RCM analysis to reflect the varying operating contexts called up under mission profiles and readiness states, and to derive different maintenance strategies to cater for the variations. This in turn could lead to slightly different sets of maintenance documentation.

5.7 Asset Redundancy

a. Occasionally, multiple operating contexts may need to be generated within an analysis to deal with the issue of asset redundancy, where multiple assets exist to support a single function or range of functions. When this occurs, asset redundancy should be considered under the two prime categories of Serial Redundancy and Parallel Redundancy.

b. Serial Redundancy. This is the most commonly found situation, where an identical standby machine or equipment exists to support a running machine(s), or equipment(s). The standby cuts in/is switched in the event of a failure of the operational asset. The operating context for the running and the standby machines can be very diverse and generate different failure modes. In the case of the running machine the failures modes will tend to be evident; in the case of the standby machine they are likely to be hidden. The RCM analysis for evident and hidden failures is different (see section 9), and the study group will need to clearly define the different running and standby operating contexts before commencing the FMECA process. Whenever possible, RCM analyses should address the duty/standby scenario, since the results from such analyses can be more readily adapted for differing modes of operation.

c. Parallel Redundancy. This is a less common occurrence, where two or more machines operating simultaneously are used to meet a demand, and where each machine individually has the capacity to meet the total demand. There may also be standby machines in reserve to take over in the event of failure of any of the running machines.

d. Redundancy Considerations. The operational scenario driving redundancy considerations is similar to that described in the Mission Phase Analysis. However, the analysis is now more complicated, particularly when dealing with parallel redundancy. Consider a hypothetical analysis of the main generation function. The varying operational demands are as shown in Figure 5.3, which illustrates the changes in mode of generator operation, based on the declared ship NBCD

State, and condition. This particular example is further complicated by the action damage limitation requirement to dedicate generators to specific ship sections and their associated equipments.

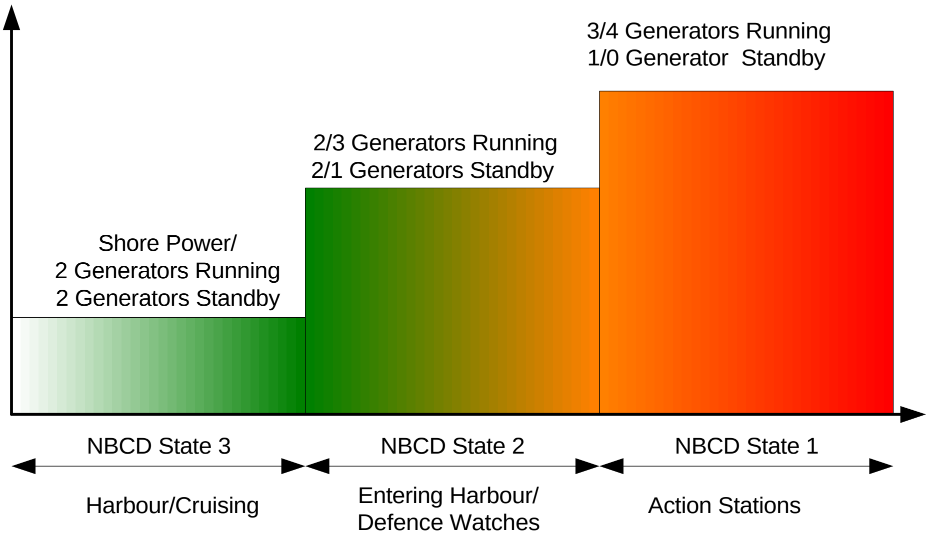

Figure 5.3 – Illustrative Generator State

e. Assuming for the purposes of this exercise that any two generators can meet the total demand, and that there are four generators in total, then any number between two and four of the generators can be running at any one time, carrying 50%, 33% or 25% of the load. Any combination of two, one or nil generators can be on standby. The potential for generating operating contexts can be daunting.

f. In all scenarios, a generator will only need to be analysed once as a standby machine.

g. Analysis of a running generator in its operational state will, however, need to consider Functional Failures for:

h. A running machine failing to carry any load at all or alternatively less than 50%, 33% and 25% of the demanded load, the failure modes will probably be identical, but the effects of the failures may vary. These functional failures will all tend to generate failure modes that are evident.

i. In the event of a failure of a running generator, a parallel running machine failing to accept its share of the total demanded load. The functional failure will generate failure modes that are hidden, and in this case will lead to a task to conduct a load trial on a periodic basis to determine whether each machine will bear full load as a worst case.

5.8 Operating Context Records

a. Ideally, the operating context statements will be held within the RCM (RN) database. However, a total platform analysis will contain classified operating context information, which could result in the whole database becoming classified. In order to avoid this, information classified “Confidential” and above should be compiled on a stand alone system authorised for the production of classified documents, and retained in hard copy in an appropriately classified Annex to the overall study report. It can then be referred to as necessary.

5.9 Defence Standard 00-60 Correlation

a. LSA Process/Task Applicability

1) Operating context statements provide a large part of the information content of a Use Study as defined within Task 201, subtask 2.1 of Defence Standard 00-60. This information set will mainly relate to the required functionality of the system/equipment and its detailed operating scenario including both qualitative and quantitative aspects.

2) The formulation of the high level operating context statements (platform and major functional blocks) as well as the initial issue of the Use Study will normally be the responsibility of the MOD ILS Manager (MILSM), or his appointed representative. Field visits and subsequent updates to the Use Study are activities that are often performed jointly by the MILSM and the Contractor. Updates to existing high level operating context statements should also be conducted jointly, with the MILSM taking ultimate responsibility.

3) The Contractor, with input and subsequent agreement from the MILSM, must normally perform operating context statements covering the lower (more detailed) levels of the functional breakdown, including the FAG level.

4) The assembly of a complete set of operating context statements covering a whole platform or system will therefore act as an invaluable input to the development of a Use Study. Due care must be taken to avoid duplication of effort when preparing a Use Study by the proper use of operating context data as an input to the study as detailed in sections 201.3.1 and 201.3.5 of Defence Standard 00-60. It must be appreciated that there are information requirements of a Use Study that will be outside the scope of an operating context statement. Operating context statements cannot therefore be used on their own as a direct replacement for a properly conducted Use Study.

5) The conduct of this activity is an essential prerequisite to the performance of RCM.

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6. FUNCTIONAL PARTITIONING

6.1 Key Players

a. The key players in the functional partitioning part of the RCM process are indicated in Table 6.1. All key players should ensure that they understand this section.

Key Roles	
New Construction/Procurement	**In-Service**
Shipbuilder Platform Manager	D Ops P
Shipbuilder Equipment Manager	Warship IPT Leader
Shipbuilder ILS Manager	D Ops E
Warship IPT Leader	Equipment Operator/Maintainer
MoD ILS Manager	
Study Facilitator	Study Facilitator

Table 6.1. Key Players

6.2 Functional Partitioning Requirement

a. Although not strictly a part of the accepted “RCM” process, to undertake a complete RCM analysis of an asset as complex as a whole warship it is invariably necessary to break down the total functionality into more manageable functional blocks prior to commencing individual analyses. This will be an iterative process in which high level functions are partitioned progressively into lower level functions that combine to form a functional model of the warship. It should be noted that there are many methodologies available that can undertake functional partitioning. The method described in this section has been found suitable for the purpose.

6.3 Functional Partitioning Process

a. The functional partitioning process is undertaken using a top down approach. The platform characteristics found in documentation such as Staff Targets, Staff Requirements and Warship Policy Documents are initially used to develop the platform and engineering discipline operating contexts (see section 5 for further details). From the platform operating context the overall platform functionality is derived. This platform functionality is then decomposed into the 3 main functions of Structure, Marine and Combat System Engineering.

1) Structural functions are those that deliver the watertight, gastight and structural integrity of the vessel or system.

2) Marine engineering functions are those that provide propulsion and hotel and utility services of the vessel.

3) Combat System functions comprise those functions that deliver the fighting capability.

b. These 3 primary engineering functions are then further decomposed until a level is reached at which functions can be seen to identify with discrete physical units, such as a single system or equipment. As each function is partitioned, the associated equipments that actually perform the function are effectively subdivided into ever-smaller groups of assets. Typically, the final level of the functional breakdown should be consistent with the lowest Functional Asset Group, (FAG), within the Functional Block. (A FAG is an asset, or group of assets, whose failure will have consequences in terms of safety, environment, mission or cost.) The functional partitioning process should be carried out using a modelling tool such as IDEF 0 or other acceptable process. On completion of the partitioning exercise, a hierarchical functional code is to be assigned to each functional block for use as the principal record identity key in the RCM (RN) toolkit.

c. Figure 6.1 presents part of the functional breakdown for the Hunt Class MCMV. The function of the Hunt Class MCMV has been decomposed to a level where functions identify FAGs that are suitable for FMECA. In this example, the Functional Block “To Provide Compressed Air Services” has eight FAGs, for which independent FMECAs have been produced. On completion of the process a check must be made to ensure that all the physical assets of the platform have been included in the functional breakdown.

d. Partitioning will identify a number of dependencies that form part of the specification of the function but which cross the selected functional boundaries. Typically, these will include inputs such as any services required by the function to meet its desired performance, but which do not fall within the physical boundary of the analysis. These interdependencies should be recorded in the overall specification for the FAG to ensure that they will be properly considered during the subsequent RCM analysis.

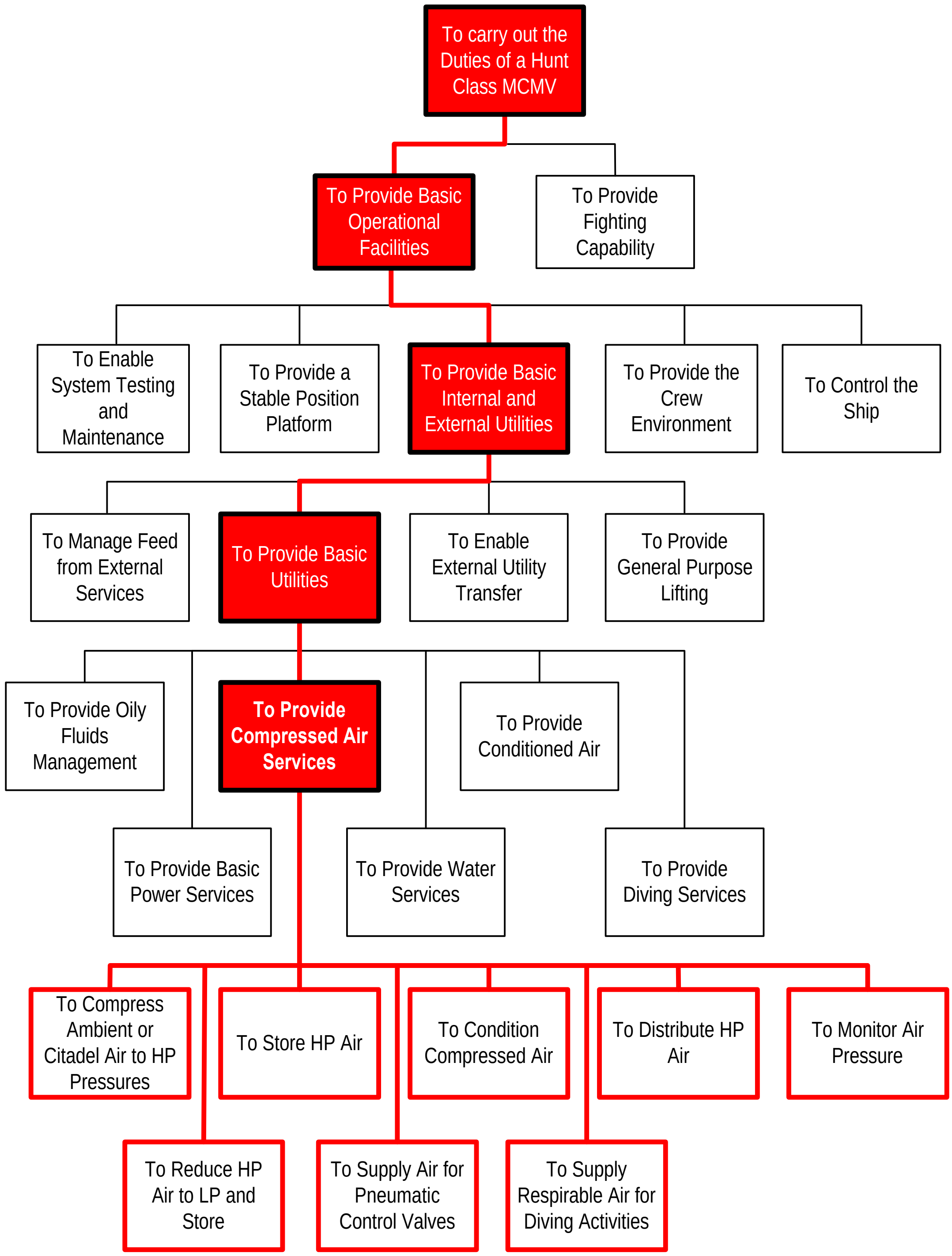


Figure 6.1 - Illustrative Functional Breakdown

e. Where a single asset performs a number of functions, it should be allocated to the functional block assessed as containing its principal function. RCM analyses of the other functions to which the asset contributes should refer to the records of the analysis of the principal function to confirm that all relevant failure modes have been satisfactorily addressed. For example, the ships main engines, as well as providing propulsion, may also, via a generator, generate electrical power. However, the principal function of the main engines is to provide propulsion, and therefore should be allocated to the functional block considering the ships propulsion rather than the generation of electricity. The RCM analysis of the electrical generation function will refer to the propulsion function analysis to ensure no failure modes are omitted.

f. Where an asset can be used to perform any number of functions depending upon its configuration, the asset itself should be considered as a utility, i.e. as though it is providing a service. The mechanism which determines the asset's configuration should be considered separately from the asset itself. The asset should only be defined once within the functional block and should be located within the part of the functional breakdown which best describes the asset in its most usual configuration. The mechanisms which determine the asset's configuration should be within functional blocks which relate to the functionality produced by each individual mechanism / asset combination. An example is the auxiliary engine fitted to the HUNT Class MCMV. This can be used either in the "Pulsing Mode" to generate electrical power for the magnetic loop, or reconfigured to the "Pumping Mode" to provide hydraulic power to the slow speed drive motors and bow thrust (or the sweep winches).

g. Functional blocks to be considered for RCM analysis are derived by inspection of the functional partitioning. These logical blocks of functionality will show a common purpose or characteristic and will usually be found at a level above the lowest layer achieved during the functional partitioning. For example, the Hunt Class MCMV functional breakdown, shown in Figure 6.1, defined "To provide compressed air services" as a functional block. However, it was necessary for the partitioning to continue down to the next level to ensure the assets to be included within the analysis could be identified.

6.4 Selection of Functional Blocks for Analysis

6.4.1. Selection Considerations

a. Priority. It will be necessary to identify an order of priority for the analysis of the functional blocks so that resources may be targeted most productively. As all functions will have to be analysed before a total platform maintenance strategy can be derived, it will not be cost effective to spend an inordinate time and effort on the process. However, consideration should be given to first undertaking studies that are likely to provide the platform upkeep cycle drivers. Experience with the HUNT class suggests that those involving underwater items, e.g. hull structure, hull preservation (and fittings to a lesser degree), all of which currently require the use of dry dock or ship lift facilities for maintenance purposes, fulfil these criteria. By conducting these studies early it is possible to note the probable changes to the current upkeep cycle at an early phase of the platform analysis and prepare the support authorities for the necessary changes to their business practices.

b. Other Factors. There are several additional factors that should be considered before choosing a functional block to be subjected to analysis:

1) Equipment characteristics within the Functional Block. Maintenance cannot improve upon:

- Design faults (Although a FMECA can help identify them).
- Design performance.
- Intrinsic equipment reliability.
- Packages which are not maintenance intensive.

It is therefore advisable to screen any failure and maintenance data pertinent to the equipment contained within each functional block to ensure that adequate knowledge is available concerning the reliability and maintenance characteristics before a study is undertaken.

2) Potential Cost Savings. The expected cost savings over the predicted remaining life of the equipment should be balanced against the cost of the study. A check should be undertaken to ensure that the equipment is not about to be replaced under A&A action. In certain

instances where the replacement programme is lengthy it may be necessary to conduct analyses for both the existing and the replacement equipment within a FAG.

3) Human resources availability. The human resources required to undertake each analysis must be identified and their availability ascertained. As the studies depend very heavily on assembling the correct mix of Operator, Maintainer and Design expertise, the programme of analyses to be undertaken will most probably be determined by the availability of specific suitable individuals.

c. The final selection of functional blocks for RCM analysis will therefore require the rationalisation of the requirement with the availability of the resources and the business need.

6.4.2. Functional Block Consolidation

a. Consideration should be given to the possibility of consolidating a number of previously identified functional blocks into a single analysis package. This may be due either to the technical nature of the assets involved or to some other common factor, such as the suitability of employing the same analysis team for all of the blocks.

6.5 Overall Specification for a Functional Block

a. The overall specification for each functional block will comprise:

1) Operating Context Statement. This defines the operational environment and performance parameters of the function undertaken. (An operating context for the "Provision of Compressed Air and Diving Support Services" is presented in section 5.)

2) Physical Description. This defines the physical assets employed by the function. An illustration of a typical physical description is presented in Figure 6.2.

3) Functional Description. This defines the scope of the function, identifies the assets or mechanisms employed to deliver the function and records the interdependencies on other functions, i.e. the inputs, services required for the asset to perform its function. An illustration of a typical functional description derived from IDEF modelling is presented in Figure 6.3.

4) Functional Logistic Control Number (LCN) Coding. On completion of the partitioning exercise, a hierarchical functional code is to be assigned to each functional block for use as the principal record identity key in the RCM (RN) toolkit**.**

6.6 Review of the Analysis Boundary Definitions

a. The analysis boundaries, both functional and physical, are to be constantly reviewed and developed to reflect decisions taken during the analysis, as a more complete understanding of the block will be acquired with experience. The boundaries of adjacent analyses must also be revised to ensure that all issues have been, or will be, addressed. The general premise should be that it is better to include an item in two analyses rather than risk a significant issue being overlooked.

6.7 Defence Standard 00-60 Correlation

a. LSA Process/Task Applicability is as follows:

1) The performance of functional partitioning correlates a part of Defence Standard 00-60 subtask 301.2.1 for the functional breakdown.

2) Where conflict exists between the guidance on performing a functional breakdown given in Defence Standard 00-60, Part 2, Annex A and this section, precedence should be given to the details given herein. In the formulation of the LSA Plan (Task 102, subtask 2.1), an Equipment Breakdown Structure (EBS) is required to identify the LSA candidate items. The interpretation of this EBS could be extended to include the functional breakdown of the system and subtask 102.2.1 for the hardware breakdown. It must be noted that, due to the required relationships between physical and functional entities within an RCM study, the correct physical to functional cross mapping cannot be achieved by use of the currently defined LSAR cross- reference table (XG).

3) The conduct of this activity is an essential prerequisite to the performance of RCM.

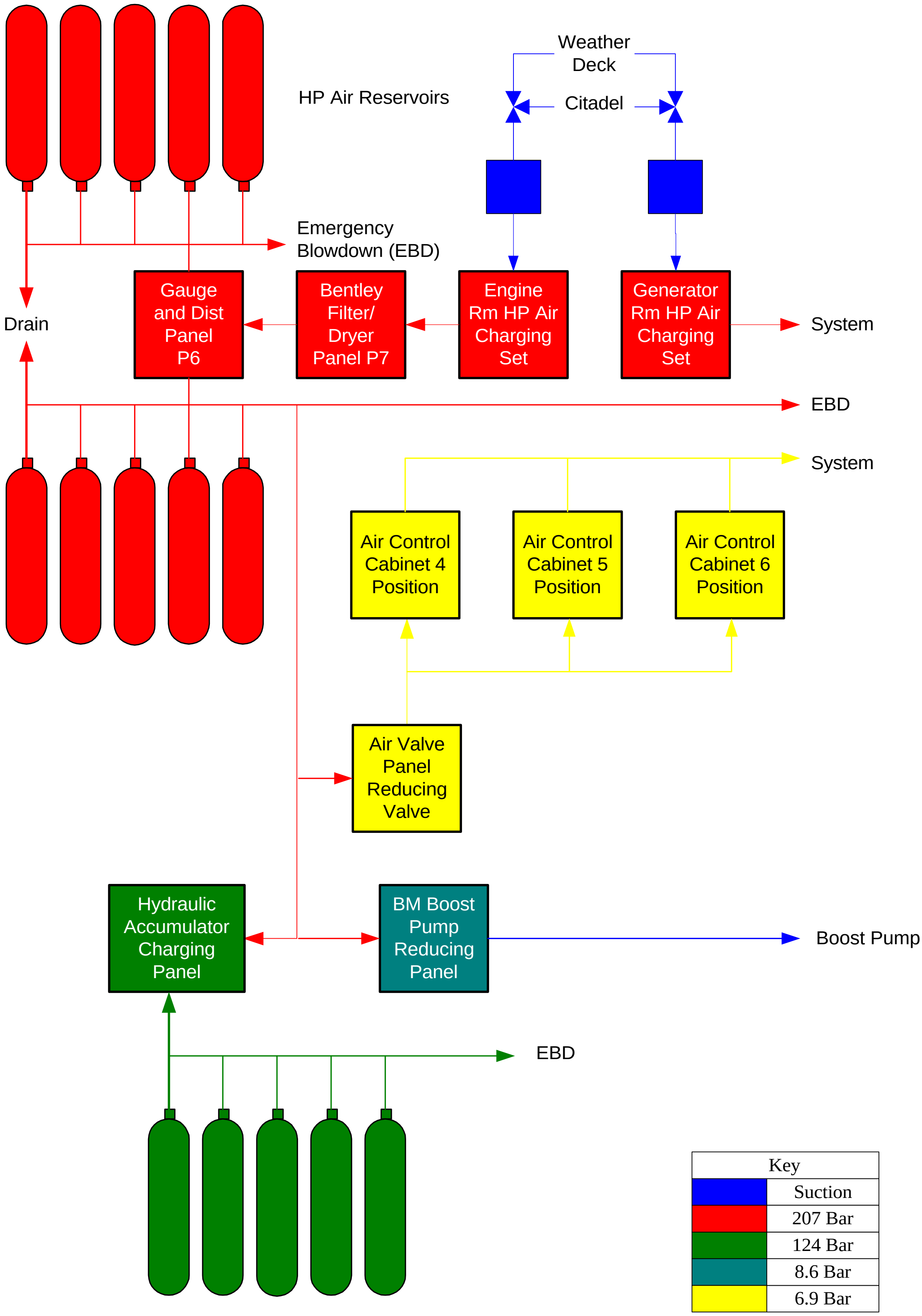


Figure 6.2 - Example of a Physical Description

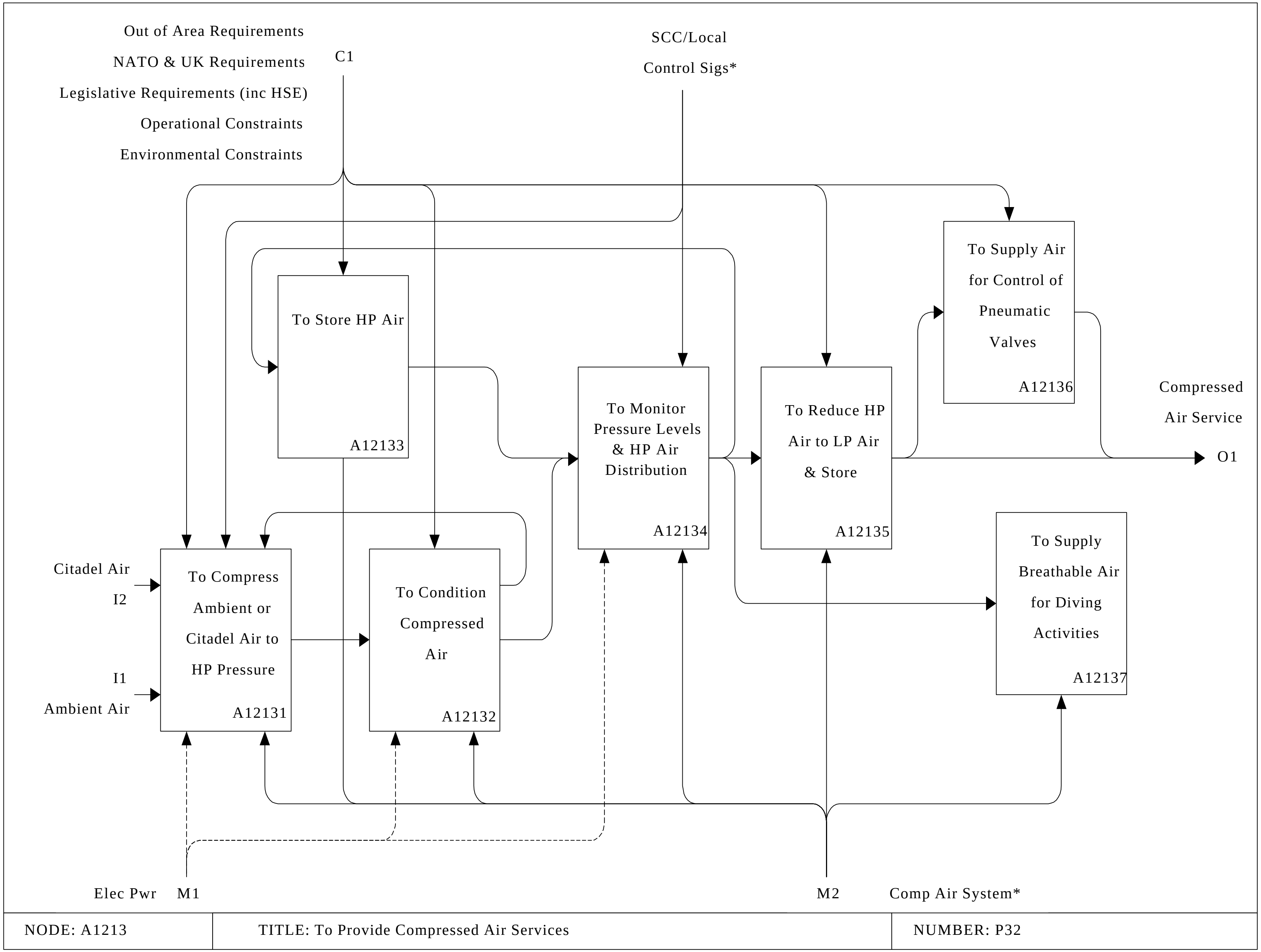


Figure 6.3 - Example of a Functional Model (IDEF0 Methodology)

7. COMPOSITION OF THE ANALYSIS TEAM

7.1 Key Players

a. The key players in this part of the RCM process are indicated in Table 7.1. All key players should ensure that they understand this section.

Platform/System Status	
New Construction/Procurement	**In-Service**
Shipbuilder Platform Manager	Warship IPT Leader
Shipbuilder Equipment Manager	D Ops P
Shipbuilder ILS Manager	D Ops E
Warship IPT Leader	
MoD ILS Manager	Study Operator/Maintainer
Study Facilitator	Study Facilitator
Study Technical/Design Advisor(s)	Study Technical/Design Advisor(s)
	Study IPT Representative
HILS(N) Management	Study D Ops E Representative
Study Technical Secretary	Study Technical Secretary

Table 7.1. Key Players

7.2 Team Considerations

a. RCM must be conducted on a team basis. The composition of the analysis team should bring together the following attributes:

1) Knowledge and experience of the RCM process and applying NES 45.

2) Detailed knowledge of the appropriate design features, installation and commissioning.

3) Knowledge of how the system or equipment is, or will be, used, operated, maintained and supported.

4) Knowledge of the condition of the system or equipment at overhaul, including understanding of the actual failure modes, and their effects.

5) Specialist knowledge of constraining influences, e.g. ordnance, Health and Safety, environmental legislation, regulatory bodies, etc.

b. The composition of the team will vary dependent on the specific requirements of each study being undertaken. In particular, detailed knowledge of construction and fatigue theory will be required for the application of the structural method.

7.3 Marine and Combat System Analyses

a. The following engineering expertise should be brought together for analyses using the marine and combat system algorithm:

1) RCM Facilitator. A trained and experienced RCM facilitator to manage the analysis and ensure the correct application of the RCM logic.

2) Operator/Maintainer. An experienced operator/maintainer, familiar with the system or equipment being analysed. The ideal would be a serving or ex RN/RFA operator/maintainer to provide familiarity with general RN/RFA engineering practices.

3) Design Advisor. A design specialist for the particular system or equipment under analysis empowered to represent the design authority. For in-service systems/equipments staff from the SSA/PE equipment specialist section may provide this expertise, or, alternatively, a field engineer from the equipment manufacturer. A representative from the SSA/PE platform section should also be present to provide overall platform expertise.

4) The Maintenance Authority. Personnel with refitting or overhaul expertise to advise on the condition of equipment being returned from service or, for newly designed equipment, the condition of similar equipment returned from service.

b. Team membership should be consistent throughout the analysis, but it is not necessary for all members to attend every meeting. The facilitator should produce a programme of meetings, calling members to attend as appropriate for the topics scheduled for review.

c. It should be noted that:

1) Inputs from the equipment specialist and the operator/maintainer are invariably vital to the success of the RCM process.

2) A clear plan of meeting dates and attendees must be drawn up and followed.

3) Meetings often generate intense, constructive debate and should be programmed to last for no more than 3 hours at a time to avoid fatigue. There should be no more than 2 sessions per day, and usually no more than 3 days per week.

7.4 Structural Analyses

a. The analysis of structural functions (section 11) makes full use of design information, operational experience and available data to identify those items that must be subjected to routine inspection and determine the appropriate survey periodicity. Structural analyses, therefore, start with a theoretical calculation phase that requires considerable specialist knowledge.

b. Participants in structural RCM analyses should include:

1) <u>RCM Facilitator.</u> A trained RCM facilitator, familiar with the structural RCM procedure, to ensure the correct application of the structural RCM logic.

2) <u>Naval Architect</u>. A naval architect for the identification of Structurally Significant Items, assessment of fatigue factors, and definition of threats.

3) <u>Ship Surveyor.</u> A ship surveyor to provide knowledge of inspection techniques and defect types.

4) <u>Platform Manager.</u> For in-service platforms, a representative from the platform section should also be present to provide overall platform expertise.

5) <u>Preservation Specialist.</u> A preservation specialist to provide information on coatings and their performance.

6) <u>Corrosion Specialist</u>. In certain circumstances, it may be prudent for a specialist with intimate knowledge of corrosion in all its manifestations to advise on particular modes of environmental deterioration.

7.5 Technical Secretary.

a. The use of someone skilled in the use of RCM (RN) software to populate the database concurrently with the analysis has advantages in time but is optional. Any proposal to use a Technical Secretary to support an RCM analysis must be sanctioned by the appropriate Project Leader.

7.6 General

a. All full time team members must have been trained in the RCM process prior to starting the analysis. However, specialist expertise brought in for short periods should not be precluded through lack of familiarity with RCM. When necessary, guidance on the RCM process will be given by the RCM facilitator.

7.7 Defence Standard 00-60 Correlation

a. LSA Task Applicability

1) This section has no correlation with any LSA process. It provides guidance on the most suitable composition of an RCM team.

2) The contents of this section are therefore advisory in nature.

8. STRUCTURING THE FAILURE MODES, EFFECTS AND CRITICALITY ANALYSIS

8.1 Key Players

a. The key players in this part of the RCM process are indicated in Table 8.1. All key players should ensure that they understand this section.

Platform/System Status	
New Construction/Procurement	**In-Service**
Warship IPT Leader	Warship IPT Leader
Study Facilitator	Study Facilitator

Table 8.1. Key Players

8.2 Introduction

a. Once the preparatory work detailed in sections 1 to 7 has been completed, the RCM process proper can commence. This process is conducted in 2 stages:

1) The initial stage is to conduct a FMECA to identify all plausible failure modes for the asset being analysed. The FMECA procedure, which documents all plausible failures, determines the effects of each failure on equipment, system and platform operation, and hence assesses its criticality. (The FMECA process is discussed in more detail in section 9.) It should be noted that a FMECA might have previously been generated for the asset during the procurement process. If such a record exists, it should be re-addressed to ensure that it contains sufficient detail for RCM purposes. This section describes the steps to be taken if a suitable FMECA has not previously been generated.

2) The second stage is the application of the Marine and Combat System RCM Decision Algorithm to each plausible failure mode identified during the FMECA to determine a maintenance strategy for that failure mode. (Use of the Decision Diagrams is further discussed in section 10.)

8.3 Functional Asset Groups (FAGs)

a. The functional partitioning process and derivation of functional blocks has been described in section 6. However, a FMECA cannot generally be carried out immediately at functional block level and it is usually necessary to break down the functional block into Functional Asset Groups (FAGs) for more detailed analysis. The top level FAG, (FAG 0), will contain information pertinent to the overall study, whilst subsidiary FAGs will be derived if and when the need arises during an analysis. It should be noted that they are not pre-determined before the analysis starts.

b. The process for deriving FAGs and then conducting FMECAs is illustrated in Figure 8.1.

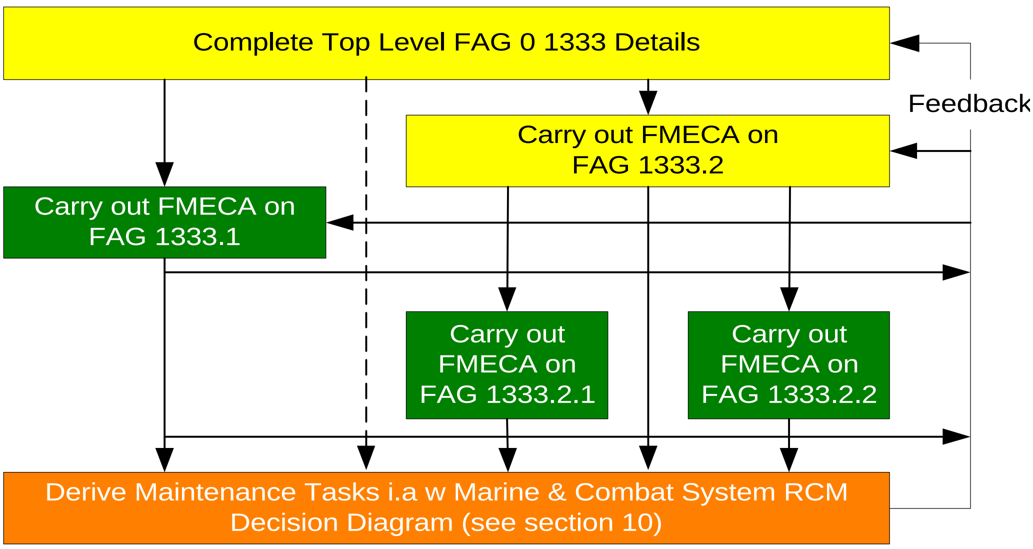


Figure 8.1 - Stages of Deriving FAGs and Undertaking FMECA and RCM Analysis

c. Each FAG should be given a unique LSA Control Number (LCN) which will be an extension to the LCN assigned to its parent functional block. This will enable the results to be located in the RCM (RN) Toolkit.

8.3.1 FAG 0

a. The role of FAG 0 is to act as a repository for top level information on the whole study, and draw together the data held in the ensuing lower level FAGs. It is effectively an index for the whole functional group and should contain the following information:

1) A statement of where and when the study was conducted, with start and finish dates.

2) The names of the Facilitator and all the Group Members, together with expertise they brought to the meetings, e.g. CCWEA MJ Bloggs, WE555, Radar Type 998 Processing.

3) The operating context statement, together with any changes in the light of the study.

4) A list of the global assumptions for the analysis, with any recorded changes as a result of the study.

5) The name of the Auditor and the Analysis Audit Comments.

b. Similarly to FAG 0, lower level FAGs are to contain information relevant to that particular stage of the study, and will include, typically:

1) A list of the detailed assumptions for the FAG, with any recorded changes as a result of the study.

2) A record of the subsidiary FAGs that comprise the analysis together with their LSA/IDEF identification numbers.

3) The names of any specialist group members who may have been co-opted for a specific part of the analysis only, together with expertise they brought to the meetings.

Note that lower level FAGs may also act as indices for FAG at levels beneath them, in the same way that FAG0 acts as an index for the whole analysis.

8.4 Levels of Indenture

a. The level of indenture is of vital importance as it significantly affects the amount of time and effort required to complete a satisfactory analysis. However it requires careful consideration because an analysis carried out at too high a level can become too superficial, while one undertaken at too low a level can become too cumbersome.

8.4.1 Starting at Too Low a Level

a. The tendency is to start at too low an indenture level when applying RCM for the first time. For example: A sheared fuel pump drive shaft is a failure mode that could affect a propulsion gas turbine. The fuel pump is part of the on engine fuel system, so it would seem sensible to address this failure mode by raising an RCM Information Worksheet for the fuel system This might look as shown in Figure 8.2.

	FUNCTION		FUNCTIONAL FAILURE		FAILURE MODE
1	To supply fuel from the service tank to all 8 gas generator burners at a rate of up to 60 litres per minute each.	A	Unable to supply any fuel at all from the service tank to all 8 gas generator burners.	1	No fuel in the service tank.
				3	Fuel filter blocked by particulate build up.
				7	*Fuel pump drive shaft sheared due to fatigue.*
				12	Fuel line severed due to accidental damage.

Figure 8.2 - Fuel System Information Worksheet at Low Level

b. This example indicates that if the analysis were to be carried out at this low level of indenture, the sheared fuel pump drive might be the seventh failure mode to be identified out of a total of perhaps fifteen or twenty. When the decision worksheet has been completed for this sub system, the analysis would proceed to the next system, and so on until the maintenance requirements of the entire gas turbine have been reviewed.

c. The gas turbine as a whole can be divided into many sub systems, only one of which will be the fuel system. If a separate analysis were to be carried out for each sub system, a series of problems can arise:

1) The further down the hierarchy the study progresses, the more difficult it becomes to define the required performance standards and hence the functional failures and failure modes.

2) At a low indenture level, it becomes equally difficult to visualise and hence analyse the overall failure consequences.

3) It becomes increasingly difficult to decide which components belong to which system. (e.g.: does the gas turbine throttle belong to the fuel system or the engine control system?).

4) Some failure modes can cause many sub systems to fail simultaneously (such as loss of electrical power supply). If each sub-system were to be analysed separately, failure modes of this type would be repeated several times.

5) Control and protective devices could become very difficult to deal with, especially when a sensor in one sub system drives an actuator in another through a processor in a third. If attention is not paid to this issue , the same function would end up being analysed three times in slightly different ways, with the same failure finding task being prescribed more than once.

6) A new worksheet would have to be raised for each new sub system. This could lead to the generation of vast quantities of possibly irrelevant data. The associated manual or electronic documentation systems would have to be very carefully structured for the information to remain manageable. In short, the whole exercise would become more extensive and intimidating than necessary.

d. The tendency to undertake RCM analyses at a low indenture level is caused by a mistaken belief that a failure mode which affects a component can only be identified at the level of that component. Failure modes can in fact be identified from any level, as shown below.

8.4.2 Starting at the Top

a. The analysis could be started at the top of the equipment hierarchy. For example; the primary function and desired performance standards of the gas turbine propulsion system is “To generate up to 10MW of power at the propeller shaft at a maximum shaft speed of 250 RPM.” The first functional failure associated with this function would be “Unable to generate any power at all at the propeller shaft”. Four of the failure modes which could cause this functional failure are those already identified in Figure 8.2, except that they would now appear on the information Worksheet as shown in Figure 8.3.

b. The main advantages of starting the analysis at the top level will be:

1) Functions and performance expectations will be much easier to define.

2) Failure consequences will be much easier to assess.

3) It will be easier to identify and analyse control and protective circuits as a whole.

4) The repetition of functions and failure modes will be less likely.

5) There will be no need to raise new worksheets for each sub system so analyses will consume less paper/computer storage.

c. The main disadvantage of performing the analysis at the top level could be hundreds of failure modes resulting in the propulsion system not delivering any power to the shaft. It is therefore more likely to overlook several failure modes altogether in the plethora of data being generated. For instance the sheared fuel pump drive shaft might have been the seventh failure mode out of possibly 15 to be identified in the analysis carried out at the “fuel system” level, whereas, at the propulsion system level, it might have been 73rd out of perhaps 200 failure modes.

	FUNCTION		FUNCTIONAL FAILURE		FAILURE MODE
1	To generate up to a maximum of 10MW at the propeller shaft at a maximum shaft speed of 250 rpm.	A	Unable to generate any power at all at the propeller shaft.	1	No fuel in the service tank.
				42	Fuel filter blocked by particulate build up.
				73	*Fuel pump drive shaft sheared due to fatigue.*
				114	Fuel line severed due to accidental damage.

Figure 8.3 - Fuel System Information Worksheet at High Level

8.5 Choice of Indenture Level

a. The problems associated with both high-level and low-level analyses suggest that it may be sensible to carry out the analysis at an intermediate level. In fact most systems/equipment can be sub-divided into a great many levels and the RCM process applied at any one. For example, Figure 8.5 shows how a gas turbine propulsion system could be divided into at least four levels. It traces the hierarchy from the level of the propulsion system as a whole down to the gas generator fuel pump. It goes on to show how the primary function of the asset might be defined at each level on an RCM Information Worksheet, and how the sheared fuel pump drive shaft can be identified at each level.

b. The choice of Indenture Level should be based on the fact that every system/equipment has been acquired to fulfil specific functions, which will be specified at the top level. Since the objective of maintenance is to ensure that the system/equipment continues to fulfil these functions to the desired performance standard, they must be specified for every system/equipment. For example the reason a platform has a propulsion system is to propel the platform through the water, not to pump fuel to a gas generator. Although the latter function contributes to the former, the overall performance of the propulsion system - and hence the maintenance - should always be judged at the top level.

c. For the reasons mentioned, the FMECA process should always start by specifying the functions and performance standards of the system and equipment at the highest practical level. This approach will also serve to define the context in which all the lower level systems are operating. The next step should be to identify functional failures, followed by failure modes at an appropriate level of detail, discussed further in section 9. This level will vary for different systems and sub systems, although, in general, complex systems, which are likely to suffer from a large number of failure modes, will tend to be analysed at lower levels of detail. For instance, it may be possible to analyse the control system at level 2 as shown in Figure 8.5 but it may be necessary to analyse the gas generator at level 3. The entries for these two sub systems might appear on the top level Information Worksheet as shown in Figure 8.4 ".... analysed separately ...." means that separate worksheets will be raised for the gas generator and control system.

	FUNCTION		FUNCTIONAL FAILURE		FAILURE MODE	LOCAL	NEXT HIGHER	END
1	To generate up to a maximum of 10MW at the propeller shaft at a maximum shaft speed of 250 rpm.	A	Unable to generate any power at all at theprepeller shaft.	1	Gas generator fails.	Gas generator analysed separately in FAG n.		
				2	SSS cluch fails.	*Etc .....*		
						Etc		
		B	*Etc ...*	1	*Etc .....*	*Etc ....*		
				2	*Etc .....*			
2	To be capable of controlling shaft speed between a minimum of 50 rpm and a maximum of 250 rpm.	A	Unable to control shaft speed between 50 and 250 rpm.	1	Control system fails.	Control system analysed separately in FAG ~n.		
		B	*Etc ...*	1	*Etc....*	*Etc ....*		

Figure 8.4 - Propulsion System Information Worksheet

PROPULSION SYSTEM

	FUNCTION		FUNCTIONAL FAILURE		FAILURE MODE
1	To generate up to a maximum of 10MW at the propeller shaft at a maximum shaft speed of 250 rpm.	A	Unable to generate any power at all at the propeller shaft.	1	No fuel in the service tank.
				42	Fuel filter blocked by particulate build up.
				73	*Fuel pump drive shaft sheared due to fatigue.*
				114	Fuel line severed due to accidental damage.

PRIMARY GEARING

MAIN GEARING

SHAFTING

etc

GAS TURBINE

	FUNCTION		FUNCTIONAL FAILURE		FAILURE MODE
1	To generate up to a maximum of 12MW at the power turbine output shaft.	A	Unable to generate any power at all at the power turbine output shaft.	9	No fuel in the service tank.
				16	Fuel filter blocked by particulate build up.
				33	*Fuel pump drive shaft sheared due to fatigue.*
				73	Fuel line severed due to accidental damage.

CONTROL SYSTEM

HP COMPRESSOR

LUB OIL SYSTEM

etc

FUEL SYSTEM

	FUNCTION		FUNCTIONAL FAILURE		FAILURE MODE
1	To supply fuel from the service tank to all 8 gas generator burners at a rate of up to 60 litres per minute.	A	Unable to supply any fuel at all to all 8 gas generator burners.	1	No fuel in the service tank.
				3	Fuel filter blocked by particulate build up.
				7	*Fuel pump drive shaft sheared due to fatigue.*
				12	Fuel line severed due to accidental damage.

LP FUEL VALVE

HP SHUT OFF COCK

FUEL FILTER

etc

FUEL PUMP

	FUNCTION		FUNCTIONAL FAILURE		FAILURE MODE
1	To supply fuel at a pressure of between 10 bar and 150 bar to the gas generator at a rate of up to 60 litres per minute.	A	Unable to supply any fuel at all to the gas generator.	1	No fuel in the service tank.
				2	*Fuel pump drive shaft sheared due to fatigue.*
				3	Fuel line severed due to accidental damage.

Note that at this level of detail, the failure mode "Fuel filter blocked by particulate build up" would now appear in the analysis for the "Fuel Filter"

etc

Figure 8.5 - Functions and Failures at Different Levels

d. As it is easier to identify failure consequences at higher levels, a good general rule is that RCM analysis should be carried out one level higher than at first might seem sensible. Complex sub systems should only be taken to a lower level of indenture when it is absolutely unavoidable. An alternative is to consider the existing level of repair, i.e. at the LRU level. However, care is necessary when taking this approach to ensure that this level is correct.

e. A further aid to the structuring of FMECAs is that experience has shown that, in order to make them more manageable, functions should be limited to about 20-30 failure modes each. If this figure is likely to be exceeded consideration should be given to analysing discrete assemblies separately.

8.6 Defence Standard 00-60 Correlation

a. LSA Task Applicability

1) This section has no correlation with any LSA process. It provides guidance on the application of the Marine and Combat System Engineering Algorithm.

2) The contents of this section are therefore advisory in nature.

9. PROCEDURE FOR CARRYING OUT A FMECA

9.1 Key Players

a. The key players in the RCM FMECA process are indicated in Table 9.1. All key players should ensure that they understand this section.

Vessel/System Status	
New Construction/Procurement	**In-Service**
MoD ILS Manager	HILS(N) RCM Group
Study Facilitator	Study Facilitator
Study Technical/Design Advisor	Study Technical/Design Advisor
Study IPT Representative	Study IPT Representative
	Study D Ops E Representative
	Study Operator/Maintainer
Study Technical Secretary	Study Technical Secretary

Table 9.1 - Key Players

9.2 Introduction

a. At the end of a FMECA, the first 4 questions of the RCM analysis process raised in section 1.4 will have been addressed. This section covers that part of the analysis.

b. The RCM FMECA is to be functionally based, i.e. based upon the outputs required from an asset and not its physical characteristics. The physical characteristics of the assets are only taken into account when deriving its failure mechanisms in the form of failure mode statements. Note, however, that for some items of complex structural assemblies it may be more appropriate to undertake a physical FMECA in accordance with the procedures detailed in section 11. In case of doubt, guidance should be sought from the appropriate authority.

c. RCM is a “zero based” process in that the analysis is undertaken as if nothing is being done to prevent or mitigate failures that could occur (i.e. no maintenance related routines are in place and no spares are available to enable recovery from failure). Similarly, the analysis must be approached with no pre-conceptions of the required maintenance. Within RN analyses there can be confusion as to whether “Rounds” or “Standard Operating Checks” should feature in zero based considerations. It is important that studies are carried out to a common datum, and guidance on these issues is provided in more detail at section 10, paragraph 10.8.1.1.

d. The RCM (RN) Toolkit (detailed at Annex C) shall be used to record all analysis decisions and their justification, either “on-line” by a Technical Secretary during analysis sessions, or immediately after a meeting for validation at the next meeting. This will provide the permanent and auditable record to support the maintenance recommendations proposed by the analysis group members.

e. The FMECA should be completed in three distinct phases, as described in paragraphs 9.3, 9.4 and 9.5.

9.3 PHASE 1: Identification of the Functions

a. The first phase is to carry out a Failure Modes and Effects Analysis (FMEA) to list all the functions associated with the FAG. Each function should be amplified by performance parameters that describe the minimum acceptable performance requirement rather than the equipment design capability, recognising that the “want” should not exceed the “can”. An equipment or a system may have one or more functions, each of which may be described as a capability requirement. Functions are categorised as detailed in the following sub-sections.

9.3.1 Primary Functions

a. These functions are the reasons why the item exists, are fundamental in the determination of maintenance requirements and should be listed first on the Information Worksheet.

9.3.2 Secondary Functions

a. Most items will have secondary functions that will generally be less obvious than the primary, even though their failure can sometimes have more serious consequences. A useful *aide memoire* in defining secondary functions is the mnemonic "ESCAPES":

Environmental Integrity

Safety, Structural Integrity

Control, Containment, Comfort

Appearance

Protection

Economy, Efficiency

Superfluous Functions

b. Arguably the most important of the secondary functions listed above relates to protection and protective devices. Protective devices work in one of five ways:

1) To draw the operators' attention to abnormal conditions.

2) To shut down the equipment in the event of a failure.

3) To eliminate or relieve abnormal conditions that follow a failure and which might otherwise cause more serious damage (e.g. firefighting water drenching system).

4) To take over from a function which has failed (e.g. standby equipment).

5) To prevent dangerous situations from arising in the first place.

c. Protective devices are designed to ensure that the consequences of a failure are much less serious than they would be if there was no protection. As a result, the presence of a protective device usually means that the maintenance requirements of the protected function are much less stringent than they would otherwise be. For this reason protective devices as categorised above must receive special attention. When listing the functions of any item, the functions of <u>all</u> the associated protective devices must therefore be listed.

d. In an RCM context, failure is defined as the inability of an item to meet a desired standard of performance. Consequently, the standards used to define failure form the basis for the remainder of the maintenance decision making process. Therefore, performance parameters need to be clearly defined and, where possible, quantified to assist the group in deciding what constitutes the failed state.

9.3.3 Obscure Functions

a. Within the Secondary Functions described above there may be obscure functions that are applicable to the analysis of certain assets. In some instances these functions may already be covered generically in a separate analysis function identified by the functional partitioning (IDEF) model, and in many studies the obscure functions may not be appropriate. Table F.2 in Annex F provides a list of some such functions for the use of facilitators when undertaking a study.

9.3.4 Method of Working

a. It is suggested that all the functions for a FAG be identified before moving on to the next stage. This will allow the analysis group to focus on this crucial stage of the process. At the end of this process the group will have a good understanding of the totality of the FAG in question.

9.4 PHASE 2: Functional Failures, Failure Modes, and Failure Effects

a. The analysis team should next concentrate on each function in turn identifying:

1) Functional failures.

2) Failure modes.

3) Failure effects in as much detail as possible.

4) Additional functions that may come to light during this process should be added to the original list.

9.4.1 Functional Failures

a. As discussed above, failure in an RCM context is defined as the inability of an item to meet a required standard of performance within its operating context. This covers the complete loss of function as well as situations where performance is outside of acceptable limits. Depending on how precisely the function has been defined, (i.e. number and quality of performance standards), it is very likely that a single function will have more than one associated functional failure.

b. Partial loss of function generally leads to more than one functional failure per function, and is the result of different failure modes with their different consequences/effects. This is why <u>all</u> functional failures affecting each function should be recorded.

9.4.2 Failure Modes

a. Failure modes are the link between the functional and physical worlds and as such will represent the failure characteristics of the asset(s) providing the function(s). A comprehensive list of the failure modes causing each functional failure is to be generated. These include:

1) Failures which have occurred before on similar equipment.

2) Any other failure modes that have not yet occurred but are considered to be probable, including those being suppressed by a current preventive maintenance regime.

b. Failure modes that are possible, but considered unlikely, should be included to show that they have been considered, but should only be subjected to further analysis if their consequences are considered to be significant.

c. The physical causes of functional failures should be recorded in sufficient detail to enable an appropriate failure management strategy to be identified. For the warship case, this may be at a level of detail represented by the Lowest Replaceable Unit (LRU), although this may not be true for all cases. Physical causes may include normal wear and tear (corrosion, abrasion, erosion, fatigue, etc.), contaminant ingress, and inadequate lubrication. Causes due to human error should only be included if firm evidence exists to support such failures, or if operator error can induce significant consequences. Care should be exercised to ensure that subsequent maintenance recommendations address the cause of failure rather than its symptoms.

d. When recording failure modes, it is not only important that the cause (as opposed to the symptom) is described but also that its description is sufficiently detailed to enable failure effects and consequences to be assessed. For example, “bearing seized due to wear” or “bearing seized due to lack of lubrication” are far more useful than “bearing failure”, as an appropriate maintenance task is then more readily identifiable. Statements that include the words “for any reason” are likely to lead to a Default Task with a possibility of a redesign requirement.

9.4.3 Failure Effects

a. The failure effects that accompany each failure mode should be recorded so that an accurate assessment of the consequences of failure may be made. As RCM is a zero based methodology, and since one of the main objectives of an RCM study is to establish whether preventive maintenance is necessary, no assumptions should be made about existing maintenance. The following rules apply:

1) The effects of failure should be described as if nothing was being done to prevent it occurring.

2) No mitigation actions are being taken to reduce the consequences of the failure (but see 9.4.3.2 a. 1)).

3) The failure effect at all levels is to be written as if no spares are carried to restore the function, other than general consumables such as lubricants and commonly used fuses.

9.4.3.1. Failure Effect Categories.

a. Failure Effects fall into 3 categories covering equipment, system and platform:

1) <u>Local Effects</u>. These are the effects local to the failure mode on the asset being analysed, and will include:

- Evidence, if any, that the failure has occurred, or is in the process of occurring, including reduced levels of performance, and covering failure detection methods (such as alarms, built-in-test indications).

- Whether a standby system can provide the function of the failed asset.

2) Next Higher Effects. These are the effects on the system of which the asset forms a part, and will include:

- Potential physical damage to the asset or the system.
- Potential secondary effects to provide an account of any secondary damage to either other equipments in the system or unrelated equipments/systems in the vicinity.

3) End Effects. These are the effects on the platform of which the system forms a part, and will include:

- Ways in which the failure mode may threaten safety or the environment.
- Ways in which the failure mode may affect the operational effectiveness of the vessel.
- The MART of the primary and secondary damage caused by the failure, assuming no mitigation actions are undertaken.

4) The End Effects are to be used to record the following information:

- Initial identification of the spare gear required to undertake the corrective maintenance plus any secondary damage.
- The MART of the corrective maintenance including any secondary damage.

9.4.3.2. End Effect Considerations.

a. The following information is to be included after the end effects statements for subsequent transfer to other RCM (RN) data fields, where appropriate:

1) Mitigation of the Consequences of Failure. This should consider the following:

- Apart from bringing a dedicated standby asset on-line (see 9.4.3.1), how the operator might take mitigating action, other than maintenance (such as re-configuring the system), to reduce the total effects of the failure, once it has occurred.
- An estimate of the time to complete such action.

2) Defective Item Repair Action. This covers:

- The repair action for the original defect and any resulting secondary damage.
- The person undertaking the repair, i.e. the specified member of Ships Staff, Base Staff or Contractor.
- Whether a dry dock or other specialist support facilities may be required. (This information can be used later in the rationalisation process – see section 13).
- Time taken to achieve the repair, including secondary damage, (MART) on receipt of stores, which should be identified by NSN where possible (but see 9.4.3.2 a) 3.).

3) Spare Part Identification. A list of the spares required to carry out the repair action to the asset, identified by NSN where possible, as a result of the failure mode occurring. This information is necessary to enable the analysis team to assess the cost benefits of undertaking a predictive, preventive or detective maintenance task, as well as No Scheduled Maintenance remedial tasks. It may also required for the follow-up RCS (RN) analysis. Note that when secondary damage is extensive, it would be nugatory to attempt to identify the parts required to rectify such damage. If possible, however, a broad estimate of the cost of rectification should be provided.

4) Secondary Damage. Estimates of secondary damage, with associated system downtime and time to repair, in order to justify the assessment of the overall effect of the failure mode on the system and the platform. This will depend on engineering judgement within the team, and will be subjective. Estimates of days/weeks/months are acceptable. Information may also be obtained from sources listed in section 4.

9.4.4 Level of Repair Analysis (LORA).

a. The level at which a failure mode is set and the corresponding task selection process will determine the level at which any maintenance is to be undertaken. This is the first step in a formal LORA undertaken under ILS procedures. Consideration should always therefore be given to existing policies on maintenance and repair levels, but is not to be driven by them. In certain instances the level of repair may be considered to be too high, which is wasteful in assets and not cost effective. In others it may be considered too low, since lengthy repair/replacement procedures, with associated system downtime, could outweigh the costs of using a higher-level replacement unit. The group is to make recommendation for changes to existing levels of repair, where considered appropriate and is to be aware that more than one repair action is allowable, e.g. Operational vs. Economic considerations.

9.5 PHASE 3: Criticality Assessment

a. The purpose of the criticality analysis (CA) is to rank each potential failure mode identified in the FMEA, according to the combined influence of severity classification and its probability of occurrence (POC) based upon the best available data. CA provides a means of comparing each failure mode to all other failure modes with respect to severity. The process will highlight to the appropriate Project Managers/ Project Safety Committees the risk associated with each failure mode.

b. The study team must identify whether accident severity categories, probability ranges and risk classification tables exist for the platform/system under analysis. This information should be sought in the first instance from the relevant platform or system Safety Management System. The tables provided in this section are to be used when a discrete criticality table has not been generated for the asset or system.

c. Ideally, a quantitative analysis should be conducted using failure rate data to determine the POC. However, in many cases this information will not be available, and the analysis group will have to adopt an alternative approach. The three approaches are as follows:

1) Quantitative. Where reliability data is available, the study team can apply the data to the associated criticality matrix. The derived risk classification must then be added to the FMEA. The study team must record the data source(s) and any assumptions in the operating context.

2) Qualitative. In the absence of quantitative data (which is likely to be the case), the team can apply engineering judgement based upon the team's experience to determine the probability of occurrence.

3) Where data is not available and the team is unable to agree any subjective values, do not record a CA code, and continue with the RCM analysis using the generic failure mode category “Plausible”. **THIS IS AN UNSATISFACTORY APPROACH AND SHOULD ONLY BE USED IN EXTREMIS**.

9.5.1 Criticality Matrix.

a. The following procedure is to be followed:

1) Severity Classification. The consequences of each failure mode should be identified and a Severity Category allocated in accordance with Table . The table can consider the consequence of failure from a single or combination of reliability, safety or maintainability aspects. For failure modes whose effects are restricted, e.g. failure of a protective device, the criticality analysis is to take account of the multiple failure and assume that the protected function experiences failure with the protective device in the failed state.

Category	Definition	Consequence (Worst Case)
I	Catastrophic	Death of ships' staff (or own forces in the proximity) The loss of the platform. System or co-lateral damage with repair costs in excess of £500k.
II	Critical	Severe injury requiring hospitalisation leading to long term absence of ships' staff (or own forces in the proximity). Complete loss of, or inability to start the mission. System or co-lateral damage with repair costs in excess of £200k.
III	Marginal	Injury requiring an absence of up to 3 days of ships' staff (or own forces in the proximity). Reduction of system availability below that required by the system specification that would cause degradation of the mission. System or co-lateral damage with repair costs in excess of £10k.
IV	Negligible	An injury, if at all, requiring no more than first aid treatment. Minimal disruption to the system or mission. System or co-lateral damage with repair costs of less than £10k.

Table 9.2 – Example of Severity Classification

2) Probability of Occurrence. The probability of occurrence of each failure mode should then be derived for failure modes identified within the FMEA. POC levels are defined in Table :

Level	Definition	Description
A	Frequent	A probability of occurrence of greater than 1 per 1,000 hours of operation. This is approximately 1 occurrence of the single failure mode under consideration every 40 days or less when taken across the asset population.
B	Probable	A probability of occurrence of less than 1 per 1,000 hours but greater than 1 per 10,000 hours of operation. This is approximately 1 occurrence of the single failure mode under consideration of between 40 days to 1 year when taken across the asset population.
C	Occasional	A probability of occurrence of less than 1 per 10,000 hours but greater than 1 per 100,000 hours of operation. This is approximately 1 occurrence of the single failure mode under consideration of between 1 to 10 years when taken across the asset population.
D	Remote	A probability of occurrence of less than 1 per 100,000 hours but greater than 1 per 1,000,000 hours of operation. This is approximately 1 occurrence of the single failure mode under consideration of between 10 to 100 years when taken across the asset population.
E	Improbable	A probability of occurrence of less than 1 per 1,000,000 hours of operation. This is approximately 1 occurrence of the single failure mode under consideration of greater than 100 years when taken across the asset population.

Table 9.3 – Example of Probability of Occurrence

3) Criticality Matrix. The POC level is overlaid on to the severity category to give the CA code as indicated in Table . This provides a means of identifying and comparing the relative severity of individual failure modes. The position of the failure mode on the matrix indicates its overall criticality.

b. RCM is a resource intensive process if applied to every function and failure mode irrespective of the consequences of failure. Failures towards the bottom right hand corner of the matrix are by implication less important or less likely to occur. Failures assessed to be in the "May be Tolerable" area of the matrix are, therefore, candidates for the analysis category of "Not Considered Further" (NC) if the cost of analysis is likely to be greater than allowing the failure to occur. However, great care must be exercised before this decision is taken, especially when dealing with safety related failures and in particular with protective device types. Liaison between the team and the relevant safety management team will be required to ensure that safety integrity of the system is preserved. The justification for "Not Considered Further" decisions must therefore be recorded in the RCM (RN) Toolkit.

POC / Consequence	FREQUENT (A) > 1 per 1,000 hrs	PROBABLE (B) < 1 per 1,000 hrs > 1 per 10,000 hrs	OCCASIONAL (C) < 1 per 10,000 hrs > 1 per 100,000 hrs	REMOTE (D) < 1 per 100,000 hrs > 1 per 1,000,000 hrs	IMPROBABLE (E) < 1 per 1,000,000 hrs
CATASTROPHIC (I) Death. Loss of Platform. System or co-lateral damage >£500,000.	1 High	2 High	4 High	5 High	8 Medium
CRITICAL (II) Severe injury requiring hospitalisation leading to long term absence. Complete loss of, or inability to start, mission. System or co-lateral damage >£200,000	3 High	6 High	7 Medium	9 Medium	14 Low
MARGINAL (III) Injury resulting in the loss of >3 working days. System availability less than specified. System or co-lateral damage >£10,000.	10 Medium	11 Medium	12 Medium	15 Low	18 May be Tolerable
NEGLIGIBLE (IV) Requires no more than first aid treatment. Minimal risk to system or mission. System or co-lateral damage <£10,000	13 Medium	16 Low	17 Low	19 May be Tolerable	20 May be Tolerable

Table 9.4 –Example Criticality Matrix

9.5.1.1. Use of Generic Criticality Assessment Categories

a. The use of generic criticality assessments as described below is only to be used when consequence and POC information is either not available or cannot be determined subjectively by study team members

b. The analysis can be continued for failure modes based on the criteria of plausibility. Plausible failure modes are those considered to be likely for the asset under consideration. In such cases the criticality is to be shown as "P" and the analysis continued. . It is not to be used either lightly, or as an expedient to hasten the analysis. See also paragraph 9.5.1.b.

c. Other generic indicators are:

1) "Not Plausible". Extremely unlikely failure modes that need not be considered further due to their improbability of occurrence. In such cases the failure mode criticality is to be shown as "NP" and the analysis discontinued. The failure mode should remain in the analysis for audit purposes.

2) "Not Considered Further". Failure modes not considered for further analysis as the cost of further analysis outweighs the costs of the consequences of allowing the failure to happen, provided that the criteria of Para 9.5.1.1.b are met.

d. Spares. For any failure mode categorised as "NC" the requirement for spare parts must be identified as the decision to curtail an analysis does not obviate the need to identify spares to support a remedial task. The appropriate fields in the RCM (RN) Toolkit must be completed for failure modes that will require spares, even though they are not worthy of full analysis.

9.5.2 Use of Non-RCM (RN) FMECA Data

a. Prior to conducting an RCM analysis, it is possible that the equipment design authority (or his representatives) will have already completed a FMECA in support of earlier design and reliability activities. Since the effort required to complete the FMECA phase represents a significant proportion of the total effort required to complete an RCM study, it would be beneficial to both timescale and resources if any earlier FMECA studies are used in support of this phase of an RCM study.

b. However, RCM RN uses a very specific application of the FMECA technique in that it is functionally based, except when analysing complex structure. Therefore, unless the possibility of using FMECA outputs in a future RCM study were identified at the time of performing the previous FMECA, it is unlikely that such a study will be suitable as a straight replacement for the RCM RN FMECA. Furthermore, simply using a previous FMECA runs the risk of losing the study group's commitment to the final product. Problems that may be encountered if existing FMECA outputs are used as the basis for an RCM study are provided in the following sub-section which provides a checklist of important points to be considered when assessing the suitability of such data. The same rules that apply to templating are to be used when using FMECA data from other processes.

9.5.3 Considerations when using Non-RCM (RN) FMECA Data.

9.5.3.1. Level of analysis

a. Section 8 described the importance of selecting the most suitable level of indenture for the subsequent RCM analysis and the problems associated with an analysis carried out at too low a level. If a FMECA has been carried out in support of the design process or other reliability based activities it is quite likely that the level of analysis will be lower than that considered most suitable for an RCM analysis. Furthermore, the FMECA will probably be a piece-part level analysis (i.e. component based) rather than the functional approach required by RCM.

b. A component-based approach will contain failure modes at a level below that at which maintenance will ultimately be carried out. For example, an electric motor will have a number of failure modes (e.g. bearing seizure, brush failures, windings open circuit, etc.). However, if the level at which it is intended to carry out maintenance is by replacing the entire motor as either a preventive measure or on failure, regardless of the root cause failure mode, the RCM functional FMECA would be conducted at the motor level (i.e. failure mode: "motor fails"), rather than the component level. Therefore, if a component based FMECA is used in an RCM study, effort may be required to consolidate the low level failure modes at the correct level.

c. It has been suggested above that existing FMECAs are likely to be component rather than function based. RCM is centred on the principle that the maintenance requirements of any item can only be determined if the functions of an asset are clearly understood.

9.5.3.2. Design FMECAs.

a. A FMECA performed in support of design or other reliability based activities is unlikely to provide sufficient detail on asset function (i.e. different types of function not distinguished and the operating context not considered) and will almost certainly provide very little information (if any) on performance parameters. Without detailed information on function and operating context it will not be possible to assess the true consequences of failure and therefore determine the need for preventive maintenance. In addition, without clearly defined performance parameters, it will not be possible to identify when failure has occurred and therefore the point at which maintenance should be carried out.

b. A FMECA prepared in support of design or other reliability based activities is likely to have been carried out by the design authority with little or no input from the intended user or other interested parties. Sometimes the FMECA may have been performed by an organisation completely removed from even the design authority (e.g. by external reliability consultants). It is therefore very unlikely that the existing FMECA will have been conducted on a team basis.

c. Experience suggests that under these circumstances there will be problems associated with gaining consensus of opinion between the members of the RCM team and the technical validity of the FMECA will inevitably be questioned by one or more members of the team. These problems will need to be resolved before the RCM decision algorithm is applied, a process that invariably requires more time and effort than if the RCM FMECA were to be developed from scratch. The existence of more than one FMECA type for each system or equipment under consideration is to be avoided as this will inevitably lead to a loss of configuration control of the logistic support data.

9.5.3.3. Detail of information provided

a. The failure effects described by a classical FMECA prepared in support of design or other reliability activities are unlikely to provide the level of detail required by the RCM team to assess:

1) The need for preventive maintenance (e.g. is there sufficient information to assess whether preventing the failure is likely to be worthwhile)

2) The most appropriate and effective maintenance strategy (e.g. is there sufficient information to propose a condition based maintenance task together with the associated parameters - i.e. incipient condition to be monitored, interval, etc.)

b. Any non-RCM FMECA will need to be reviewed by the RCM team and modified to the NES45 standard before carrying out the RCM analysis phase. Again, experience suggests that this process will not prove to be as efficient as performing the analysis from scratch.

9.5.4 Summary

a. The above paragraphs indicate that, unless a FMECA is carried out with the knowledge that the outputs will be used in a future RCM study, it is likely that considerable effort will be required before such a FMECA is suitable as the basis for a Naval RCM study. It would not be uncommon for this effort to exceed that required to develop an RCM specific FMECA from scratch. Notwithstanding this, the outputs from an earlier FMECA can provide useful and valuable inputs into an RCM study. For example, the failure modes identified by the earlier FMECA will provide a very useful list against which to cross check the failure modes identified by the RCM FMECA for completeness.

9.6 Example RCM (RN)Toolkit FMECA Worksheet

a. An example FMECA worksheet for the Hunt Class MCMV is at Figure 9.1.

9.7 Defence Standard 00-60 Correlation

a. LSA Process Task Applicability:

1) A FMECA would normally be performed under the control of a reliability programme as defined within Defence Standard 00-40 section 307 and to a specialist FMECA standard such as Military Standard 1629A.

2) The conduct of a FMECA has no direct correlation to any process defined within Defence Standard 00-60. Task 301, subtasks 2.4.1 and 2.4.2 do, however, require a FMECA as input data, and to enable the conduct of an RCM analysis, as detailed in sections 301.3.6 and 301.3.9 of Defence Standard 00-60.

3) The conduct of this activity therefore becomes an integral part of the application of LSA.

Figure 9.1 – Example of an RCM (RN) Toolkit FMECA Worksheet

RCM RN ANALYSIS	PLATFORM TYPE Hunt MCMV	LCN A12134	No 0	Compiled by	Date 11-Dec-96	Sheet 1 of 35
	SYSTEM/EQUIPMENT DESCRIPTION Pressure Level Monitoring & HP Air Distribution		FAG/SSI 4		Date	

Ships Support Agency

	FUNCTION		**FUNCTIONAL FAILURE**		**FAILURE MODE**	**FAILURE EFFECTS**	**Cr**
1	To supply air to the BASCCA Charging Panel and the Decompression Chamber at a nominal 207 bar, conditioned to breathing air standards in accordance with Defence Standard 68-75/3 & BR2807(4)(A).	A	Fails to supply any air at all to any Charging Panel.	1	Either charging panel supply valve seizes shutdue to corrosion.	Not considered plausible due to frequency of use.	NP
				2	Charging panel inlet filter blocked due to build up of particulate matter.	Not considered plausible due to the two previous filtrations.	NP
		B	Fails to supply any air at all to the Decompression Chamber.	1	Charging valve seizes shut due to corrosion.	Local Effects Unable to open charging valve when required. Noticed by operator on setting to work of decompression chamber. Next Higher Effects No conditioned air supply available to decompression chamber. End Effects Unable to treat diver for nitrogen narcosis or other compression related malaise. Operator may be able to reconfigure the system to by pass the valve (approx 30 mins), otherwise there is a risk of death or severe disability to the diver.	ID
		C	Fails to provide any air at all to any BASSCA Panel or Decompression Chamber.	1	3-stage filter absorber blocked due to build up of particulate matter.	Local Effects No supply to BA charging panels or compression chamber, indicated on local gauges and noticed by operator. Next Higher Effects No output at either charging panel or decompression chamber supply points. End Effects Unable to recharge BASCCAs for firefighting or treat diver for compression related malaise. Platform at risk from reduced firefighting capability.	IIC

10. RCM ANALYSIS USING MARINE AND COMBAT SYSTEM ENGINEERING ALGORITHM

10.1 Key Players

a. The key players in this part of the RCM process are indicated in Table 10.1. All key players should ensure that they understand this section.

Vessel/System Status	
New Construction/Procurement	**In-Service**
Warship IPT Leader	Warship IPT Leader
Study Facilitator	Study Facilitator
Study Technical/Design Advisor	Study Technical/Design Advisor
	Study IPT Representative
	Study D Ops E Representative
	Study Operator/Maintainer
Study Technical Secretary	Study Technical Secretary

Table 10.1 - Key Players

10.2 Introduction

a. Having completed the FMECA, each plausible failure mode is then analysed in accordance with the Marine and Combat System Engineering RCM Decision Algorithm. It is recommended that the rest of this section be read in conjunction with Figure 10.5.

b. "Normal Operating Conditions". It should be noted that the answers to the questions posed in the RCM Decision Diagram are only relevant for the operating context under consideration. Operating context issues have been discussed in section 5, particularly how differing operating contexts may have to be derived for some weapon systems having several different states of readiness. The meaning of "Normal Operating Conditions" within a specific study needs to be clearly defined and understood by the analysis team as the RCM Decision Algorithm can drive the maintenance decision down any of the legs of the algorithm for the same system/ equipment given differing operating conditions.

10.3 Consequences of Failure - Hidden/Evident

a. The decision algorithm is divided into two sections: Evident and Hidden failure modes:

1) Evident Failures. An evident failure is one that will eventually become evident to the operating crew under normal operating conditions, i.e. the loss of function will be noticed at some future, indefinite time without any further incident or intervention.

2) Hidden Failures. A hidden failure is a failure that will not become evident to the operator/maintainer under "Normal Operating Conditions". Such failures only become apparent after a second, but related, functional failure or event has occurred, and are commonly associated with protective devices that are not fail-safe. Although there will be no immediate consequences of a hidden functional failure, there will be an increased risk of a multiple failure. The ultimate safety, environmental or operational implications must therefore be considered fully and recorded as "worst case" failure effects.

10.4 Consequences of Failure - Safety/Environmental, Operational or Non Operational

a. Once the Hidden/Evident decision has been taken, the algorithm requires decisions on the nature of the consequences of failure. Each failure mode is examined to determine whether it has Safety/Environmental, Operational or Non Operational effects. The decisions associated with these failures will be specific to the function being analysed and therefore must be agreed by the analysis team and recorded in the operating context.

b. An area of difficulty that may be experienced with some weapon systems is whether certain failure modes should be considered under the Safety or Operational legs of the Decision Algorithm. The rationale in the following examples gives guidance:

1) The consequence of a failure mode in a weapon system that results in the failure to seduce or destroy an incoming missile or torpedo should be considered under the Operational leg.

This is because the absolute effectiveness of the weapon system to intercept hostile projectiles cannot be guaranteed for all engagements. If such failures are treated as Safety items, a considerable number of mandatory redesigns would be generated which would prove to be futile, since the modifications themselves may not guarantee 100% weapon effectiveness.

2) The consequences of a failure mode that results in danger or harm to ship's staff or friendly forces in the vicinity from a ship's own weapons should be considered under the Safety leg, since it must be possible to ensure their safety within the bounds of normal operating conditions.

10.5 Maintenance Tasks

a. The analysis will be driven down one of the six legs on the algorithm. The decision making process then concentrates on selecting the most appropriate maintenance strategy for each failure mode. The derived strategy will be one of the following options:

1) Maintenance tasks:

- Option 1. On-Condition.
- Option 2. Scheduled Restoration.
- Option 3. Scheduled Discard.
- Option 4. Combination. (Safety Consequences only).

2) Default actions:

- Option 5. Failure Finding. (Hidden Failures only).
- Option 6. No Scheduled Maintenance. (Operational/Non-operational Consequences only).
- Option 7. Redesign. (Mandatory for Safety Consequences.)

3) Other actions:

- Option 8. Part of Ship Plan Inspection

4) Supplementary actions:

- Age Exploration

b. Task selection should be undertaken as an iterative process that considers tasks from Options 1, 2 and 3, since there may be several alternative maintenance tasks that could be undertaken to deal with the effects of a single failure mode. Normally, this will be the task that is most effective at reducing the probability of failure to a tolerable level. Another consideration is how the task will be undertaken, e.g. the use of an underwater engineering (UWE) technique may be developed as an alternative to a dry docking. Each needs to be evaluated to ensure that the optimum solution is selected. In new procurement projects, redesign is also an option.

10.6 Maintenance Task Categories

10.6.1 Option 1: On Condition Tasks (Decision Diagram Boxes HS1, HO1, HN1, S1, O1 and N1)

a. On-Condition tasks should be considered first for all patterns of failure since many failure modes will be preceded by an identifiable condition which indicates that a functional failure is either in the process of occurring or about to occur. It is also generally more efficient to conduct downtime on a planned basis rather than to respond to a failure once it has occurred. However, the on-condition task can only be selected if it is both applicable and effective, and each criterion must be considered in its own right.

10.6.1.1 On Condition Task Applicability

a. During the degradation process, the interval between the point where potential failure becomes detectable and the point at which it degrades to a functional failure is referred to as the P-F Interval (see Figure 10.1). This interval is measurable in units appropriate to the event that causes deterioration, e.g. cycles, rounds fired, time out of dock, etc.

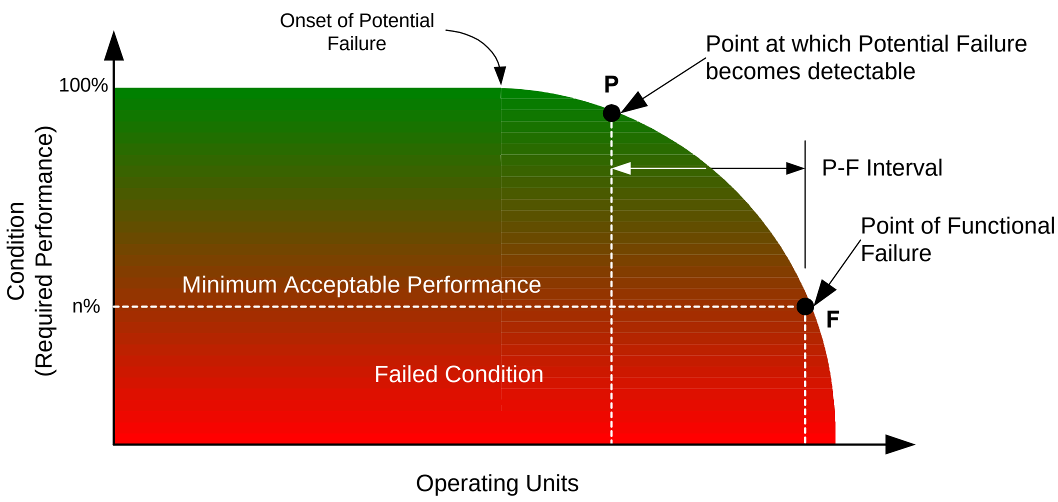


Figure 10.1 - Example of a P-F Interval Curve

b. For an on-condition task to be applicable for a potential failure condition, the following criteria must be satisfied:

1) The deterioration must be detectable. The first consideration is to identify the nature of the deterioration. In other words, does a technique exist that will detect the deterioration of the item caused by the failure mode under consideration? There also needs to be a clear understanding of the nature of the deterioration being detected.

2) The deterioration must be measurable. The progression to failure must be measurable, preferably objectively by a metric, but subjective measurements can be used in appropriate circumstances.

3) The P-F Interval must be long enough to be of use. If an applicable monitoring technique does exist, it will only be of use if it is possible to detect the onset of an impending failure sufficiently far in advance of functional failure to allow something to be done to prevent or mitigate the consequences of the failure.

4) The P-F Interval must be reasonably consistent. Note that "consistent" in this context is not an absolute term, but refers to the practicality of using a wide range of P-F intervals for the failure mode that might be found across a population of similar assets used in the same operating context. Care must be taken to select the optimum value, and for safety related failure modes, the shortest interval identified should be used.

5) Where applicable, the sources of information listed in section 4 should all be considered to determine potential failure points and the P-F interval.

c. The above considerations may lead to the identification of several applicable on condition tasks, all of which should be assessed for effectiveness before the optimum task is selected.

10.6.1.2 On Condition Task Effectiveness

a. In order to be effective, the on condition task:

1) Must reduce the probability of failure to a tolerable level. The maintenance task interval must be set to reduce the risk of an equipment failure to a tolerably low level. Acceptable risk is dependent on the consequence of failure, therefore:

- Safety/Environmental. In order to ensure that the operator/maintainer has a minimum of 2 opportunities to identify the potential failure during the deterioration process, the maintenance task interval must be set at less than half the P-F interval.

- Operational/Non Operational. In order that the potential failure is identified during the deterioration process, the maintenance task interval must be set at less than the P-F interval.

2) Must be Practical to Implement. In all cases it must be practical to carry out the task at the required interval, taking into account accessibility and operational considerations. Although a proposal to introduce new condition monitoring technologies is acceptable, the requirements for capital procurement and running costs such as training will need to be taken

into account and is unlikely to be viable for a single failure mode, unless the consequences warrant it.

3) Must be Cost Effective. The costs of applying the technique over a period of time need to be considered:

- Safety/Environmental. Cost effectiveness is not an issue when considering loss of life (i.e. personnel safety) or environmental implications. However costs should be compared if two or more tasks are applicable and effective and the lowest cost option selected.
- Operational/Non Operational. For Operational or Non Operational equipment failures, the cost of undertaking a task over a period of time (normally the remaining life of the asset) must be less than the total cost of the consequences of failure.

10.6.1.3 Assessment of the Cost of Performing the On Condition Task

a. The costs associated with performing the task should take into account man-hours, spares, tools and facilities. A realistic cost estimate should be allocated to each element and a through life cost assessed.

10.6.1.4 Assessment of the Cost of Allowing the Failure to Occur

a. For failures with consequences other than safety or environmental, the cost of allowing a failure to occur should take account of all activities and materials that will attract cost as a result. This should include an estimate of secondary damage arising from the failure, including direct and indirect repair costs (spares, manpower and facilities). A realistic cost estimate should be allocated to each element and a through life cost assessed. The decision diagram at Fig 10.2 may be used to determine whether or not a task will be cost effective. The most cost effective of the applicable tasks should be selected.

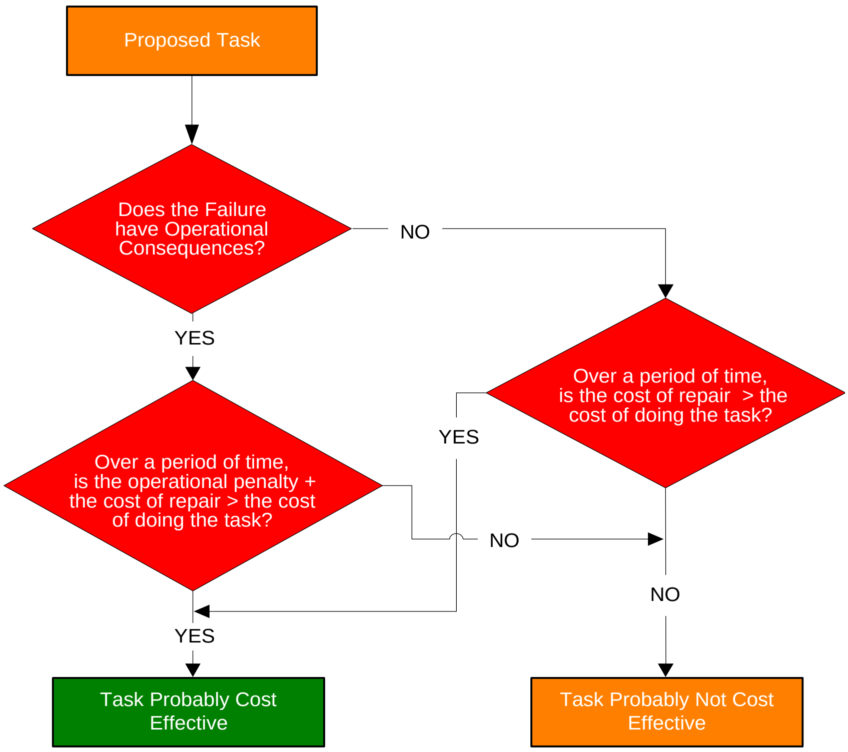


Figure 10.2 - Cost Effectiveness Decision Tree - Operational/Non Operational Failures

10.6.2 Options 2 and 3: Scheduled Restoration and Discard Tasks (Decision Diagram Boxes HS2 & 3, HO2 & 3, HN2 & 3, S2 & 3, O2 & 3 and N2 & 3)

a. If no on-condition task can be identified or justified as being applicable and effective, the analysis should then consider whether a restoration or discard task could be used.

10.6.2.1 Scheduled Restoration and Discard Task Applicability

a. Scheduled restoration and discard tasks are both associated with age-related failures demonstrated in failure patterns A, B & C in Figure 1.1. Therefore there must be a clear "life" for the item, and most items (all items for failures with Safety and/or Environmental consequences), must survive to this life, after which the conditional probability of failure increases significantly. In the case of failure pattern C the life is determined when the increasing probability of failure becomes intolerable.

b. To determine the age at which there is an intolerable increase in the conditional probability of failure, or to carry out an actuarial analysis of the relationship between age and failure, use of the information sources listed in section 4 should be considered.

c. A scheduled restoration task is only applicable if the task will restore the inherent reliability or performance level of the item. Very few items can benefit from this type of task and its use must be considered carefully.

10.6.2.2 Scheduled Restoration and Discard Task Effectiveness

a. In order to be effective, the task:

1) Must reduce the probability of failure to a tolerable level. Both restoration and discard tasks must be carried out at an interval that is less than the age at which the equipment or component shows a rapid increase in its conditional probability of failure (failure patterns A and B) for the failure mode under consideration. For failure pattern C, there is a gradual increase to a point at which the conditional probability of failure of the failure mode becomes intolerable.

2) Safety/Environmental. If the failure has safety or environmental consequences the task must reduce the chance of it occurring to a tolerably low level. Therefore a "safe life" limit must be applied. The safe life normally ensures that all failures are avoided.

3) Operational/Non Operational. For these categories some failures may be tolerable. An agreed level of tolerability, which must be recorded, may be applied

4) Must be Cost Effective. The determination of cost effectiveness is to take the consequences of failure into account:

- Safety/Environmental. Tasks addressing safety or environmental consequences must be effective in preventing failure. Cost effectiveness is not an issue when considering loss of life or environmental implications. However cost should be considered if two or more tasks are applicable and effective and the lowest cost option selected.

- Operational/Non Operational. For Operational or Non Operational failures, the cost of doing a task over a period of time must be less than the total cost of the consequences of failure as previously specified in Figure 10.2.

b. Where applicable, use of the information sources listed in section 4 should be considered to determine the cost effectiveness of a restoration or discard task.

c. Restoration and Discard tasks must both be applicable and effective in reducing the likelihood of failure. The selection between the two will be dependent on a cost comparison to select the most economic solution in accordance with Figure 10.2.

10.6.3 Option 4: Combination Tasks (Decision Diagram Box S4)

a. In some instances it may not be possible to find a single task which on its own is effective in reducing the risk of failure to a tolerably low level. In these cases it may be necessary to employ a combination of tasks such as "on-condition" and "scheduled discard". Each of these tasks must be applicable in their own right and in combination they must be effective. These tasks are applicable to failure with safety consequences, where the probability of failure must be reduced to a tolerable level. In reality, combination tasks are rarely used, and should be considered as a stopgap pending redesign. If the analysis is taken to a lower level it is likely that more than one failure mode will be discovered, obviating the need for a combination task.

10.7 Default Tasks

a. A Default Task must be considered if On-Condition, Scheduled Restoration, Scheduled Discard or Combination Tasks are not considered applicable and effective, (i.e. typically the pattern of failure is random in nature and/or there is a short P-F interval of no practical use - see paras 10.6.1 and 10.6.2. The very fact that an equipment or component has a random failure characteristic means that a scheduled restoration or a scheduled discard task would either be ineffective in preventing failure or costly in resources. This type of task may also increase the likelihood of failure due to the re-introduction of infant mortality.

b. Default Tasks comprise the following 3 options:

1) Failure finding (for hidden failures only).

2) No Scheduled Maintenance (not applicable to safety/environmental failures).

3) Redesign.

10.7.1 Option 5: Failure Finding Tasks (Decision Diagram Boxes HS4, HO4 and HN4)

a. Failure finding tasks are specific to hidden failures and will only be applicable if an explicit task can be identified to detect the functional failure. A failure finding task is a functional check to determine whether an asset would still perform its required function if a demand was to be made upon it. Most of such assets therefore tend to be standby equipments or protective devices.

10.7.1.1 Failure Finding Task Applicability

a. To meet the applicability criteria, a failure finding task must be practical to do at the required interval. In developing such tasks, study groups must ensure that the task:

1) Disrupts otherwise stable systems as little as possible.

2) Does not increase the risk of a multiple failure, i.e. systems are operated without the protection that the hidden function provides or are reconfigured into potentially dangerous conditions.

3) Checks protective systems in their entirety, rather than the individual components that make up the system.

10.7.1.2 Failure Finding Task Effectiveness

a. The failure finding task will be effective if it reduces the probability of a multiple failure to a tolerable level. In practice, it has been found the failure finding process tends to lead to a high level analysis of protective devices, colloquially referred to as “Black Boxing”, the full functionality of which are checked by the failure finding task.

10.7.1.3 Calculation of Failure Finding Intervals

a. The version of the RCM(RN)Toolkit available to MoD and detailed at Annex C has a module that calculates Failure Finding Intervals (FFIs) from information that includes:

1) The availability required of the function delivered by the asset.

2) The probability of a multiple failure.

3) MTBF of the protected device or event. This is also the demand rate on the protective device.

4) MTBF of the protective device.

5) The number of fully redundant protective devices in parallel.

6) The cost of allowing a multiple failure to occur.

7) The cost of conducting a failure finding task.

10.7.1.4 Failure Finding Availability Requirements

a. The asset under consideration should have design (or desired) availability requirements defined in the operating context. (In certain instances the acceptable probability of a multiple failure will have previously been defined in Policy Instructions.) In the absence of this information, and subject to the caveats of mandatory regulations for specific systems and equipments, the following availability requirements may be used to derive FFIs. **THESE FIGURES SHOULD BE USED WITH CAUTION AND STUDY GROUPS ARE TO SATISFY THEMSELVES OF THEIR SUITABLILITY ON INDIVIDUAL CASE BY CASE BASES.**

1) 99.95% for safety and environmental functional failures.

2) 98.0% for Operational functional failures, unless detailed within the system performance specification.

3) 90.0% for Non Operational functional failures.

4) For FFI calculations the data sources listed at section 4 should be accessed to derive the required information.

b. The mathematical formulae to be used to calculate the FFIs are protected by commercial copyright. However, the RN RCMIT will provide a service to calculate FFIs from analysis data using the software module.

10.7.2 Option 6: No Scheduled Maintenance (Decision Diagram Boxes HS5, HO5, HN5, S5, O4 and N4)

a. No Scheduled Maintenance is a maintenance strategy which can only be applied to failures with “Operational” and “Non Operational” consequences.

b. Cost. No Scheduled Maintenance must be considered if the cost of carrying out a preventive task is more than the cost associated with allowing the item to fail (see Figure 10.2).

c. Effectiveness. A maintenance strategy of No Scheduled Maintenance is only effective if the consequences of failure can be tolerated until the item is repaired or replaced. The analysis group should recognise that this may mean appropriate spares have to be available, and spares allocations for failure modes attracting No Scheduled Maintenance should be considered in the analysis. Derivation of the correct spares allocation for a completed RCM schedule is achieved by the use of the RCS (RN) Toolkit, which is discussed further in section 14 and annex D.

10.7.3 Option 7: Redesign (Decision Diagram Boxes HS5, HO5, HN5, S5, O4 and N4)

a. Redesign is a one time change that can encompass the following categories:

1) Physical. This is a modification to an asset or system.

2) Operational. Involves a change in the way in which an asset is used. This will also necessitate a change to the operating context.

3) Procedural. Modifies the way in which an operator or maintainer undertakes a specific task.

4) Spares. Requires modification to the supporting spares to be held.

5) Training. Identifies a shortfall in the capability of an operator or maintainer.

6) It should be noted that any of the above may necessitate a change to support documentation.

b. Requirement. Where no maintenance strategy can be found that is both applicable and effective, a redesign task should be considered to eliminate the failure mode (physical) or mitigate the consequences (operational). For Safety failure modes redesign is mandatory. For Operational and Non Operational failure modes the need for redesign should be carefully considered. It should be noted that, for RCM purposes, a redesign could embrace changes such as warning signs and notices (physical), or modes of usage (operational).

c. Determining the Need for Redesign. The decision diagram algorithm at Figure 10.3 should be used to evaluate the necessity or desirability of initiating design changes. It should be noted that redesign may be expensive if it involves the cost of development and manufacture of new parts or assemblies. In addition, there may also be loss of availability of the platform, system or equipment whilst modifications are being incorporated, and further risks are always introduced when the design of complex equipment is changed. Finally, there is no assurance that the first attempt at modification will eliminate or even alleviate the problem at which the redesign was directed. For the above reasons, it is crucial to distinguish between situations where redesign is necessary and where it is only desirable. The RCM (RN) Toolkit allows for documentation and justification of any redesign requirement.

d. Redesign Responsibilities. The ultimate owner of the maintenance analysis (e.g. WPM, EPM, etc.) is responsible for ensuring that redesign recommendations are addressed. **ANY ANALYSIS ON AN IN-SERVICE EQUIPMENT OR SYSTEM THAT RETURNS A "MANDATORY" REDESIGN RESULT IS TO BE NOTIFIED TO THE PROJECT LEADER AS SOON AS POSSIBLE.**

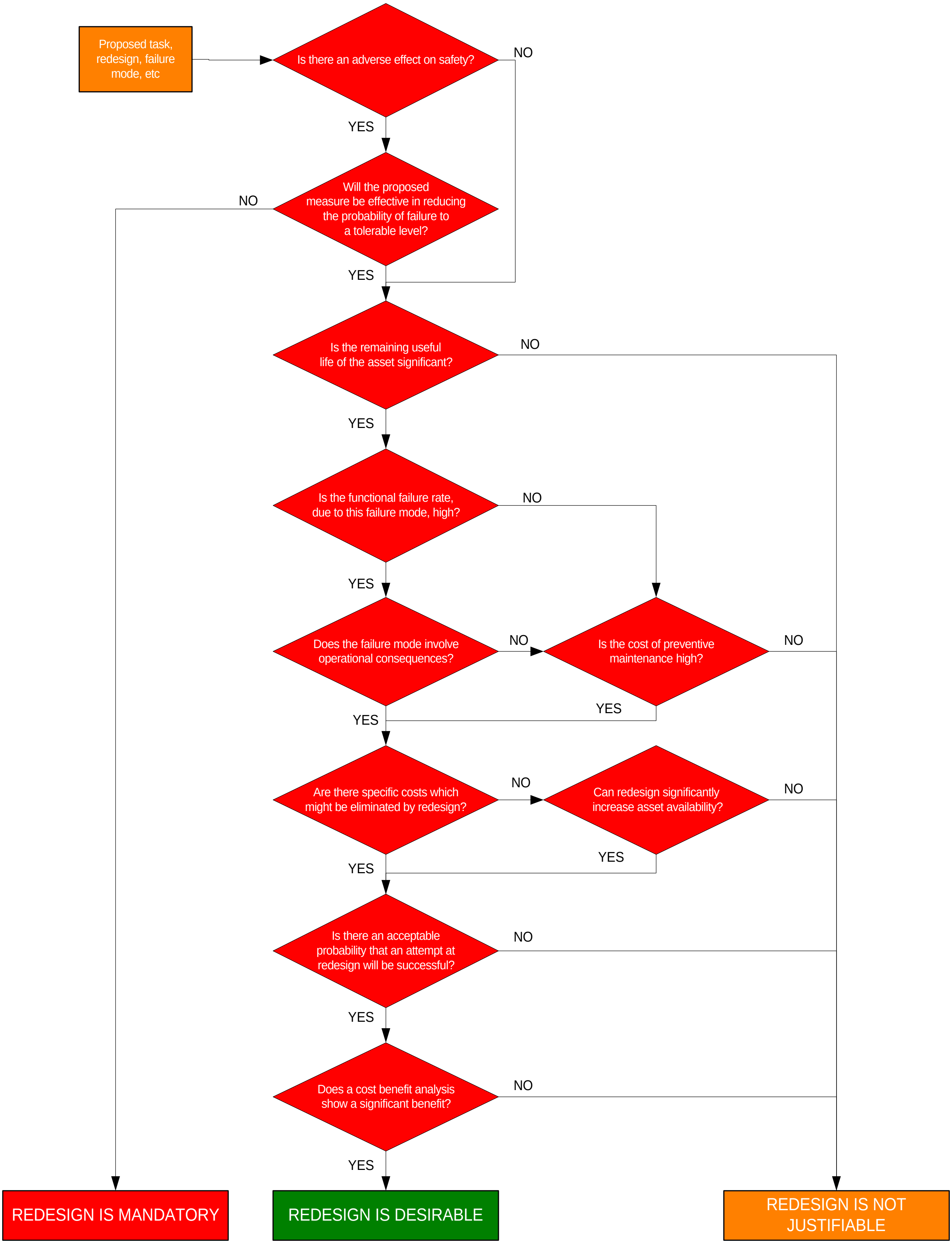


Figure 10.3 - . Redesign Decision Algorithm

10.8 Other Actions

a. Within RCM other tasks are used in support of maintenance or default tasks, namely:

1) Option 8. Part of Ship Plan (PSP) Inspection.

2) Option 9. Age Exploration (AE).

10.8.1 Option 8: Part of Ship Plan (Decision Diagram Boxes HN5 and N4)

a. Part of Ship Plan (PSP). The PSP is a Naval RCM tool that gives the analysis team a means of managing failures for those items that have relatively insignificant failure consequences yet cannot be ignored. Some failure modes that fall within the non-operational legs of the decision diagram and which have minor consequences of no great significance, (e.g. minor leaks, drips) may not justify a separate maintenance JIC. Therefore, if the analysis team is confident that a competent person visiting the compartment would notice the effects of the failure mode, the task should be allocated to the PSP. In an analysis the team needs to record any general policy decisions in the study assumptions, and the lack of a maintenance task for a given Failure Mode noted in the justification field of the RCM (RN) Decision Worksheet. This will give an indication of the potential minor failures that should be looked for, and thus an assessment of the skill level and frequency of rounds required on a class basis. Used properly, the assignment of minor tasks to the PSP can provide a foundation for the establishment or verification of a rounds routine.

10.8.1.1 Rounds and System Operator Checks

a. Rounds. The status of rounds in a “zero-based” maintenance analysis requires careful consideration. As a rule, rounds are not to be considered for the purposes of maintenance analyses carried out using this NES. Any decision to include rounds as part of the "normal circumstances" must be fully justified and ratified by the Study Sponsor before the analysis begins.

b. System Operator Checks (SOCs). Care needs to be exercised not to attempt to justify existing procedures, that are in fact confidence checks, as properly derived and justified maintenance tasks. A typical confidence check is the operation of the helm, or the testing of bridge internal communications, as part of SOCs prior to the ship proceeding to sea. These are not maintenance tasks in themselves, but are undertaken for a variety of reasons, e.g. to confirm the availability of the systems at the time the checks are undertaken or for continuation training. In most circumstances, therefore, SOCs should not be considered as part of “Normal Circumstances” when developing a “zero-based” maintenance strategy, although statutory evolutions will be difficult to reconcile at the commencement of a study. On completion of the analysis, SOCs should be checked against the derived RCM maintenance schedules. At this stage, some SOCs may be found to be bona fide maintenance tasks, but these should be derived by the analysis in their own right.

10.8.2 Option 9: Age Exploration (AE)

a. MOD(N) experience to date with in-service equipment has indicated that the information required to define either the “life” of an asset or the P-F interval for a given failure mode is rarely available from interrogation of current in-service data sources. The study groups estimate of asset life then tends to be based on a combination of:

1) Engineering judgement.

2) Designer’s and/or manufacturer’s recommendation.

3) Tradition.

b. Initial life estimates will therefore be subject to inaccuracies. In due course the RCMS data capture and feedback facilities will enable refinement of these initial estimates, but this will be a relatively slow, iterative process.

c. Lack of data is also particularly applicable to assets in the process of procurement, where any ARM data acquired is likely to have been derived over a relatively short time-scale from a small equipment population. Again, greater accuracy will only be achieved with the increase in sample size over a longer timescale.

d. AE is a systematic procedure for collecting further data to validate RCM maintenance task decisions made without adequate supporting data. As such, an AE recommendation can be made by the study team as an adjunct to any on condition, scheduled rework, scheduled discard, or failure finding task.

e. It is important that the facilitator recognises where life data is unavailable or suspect, and makes the recommendation for an AE in addition to the RCM default decision.

10.8.2.1 AE Priorities

a. Figure 10.4 is to be used to determine the priority that should be applied to the collection of data to validate maintenance tasks and their periodicities.

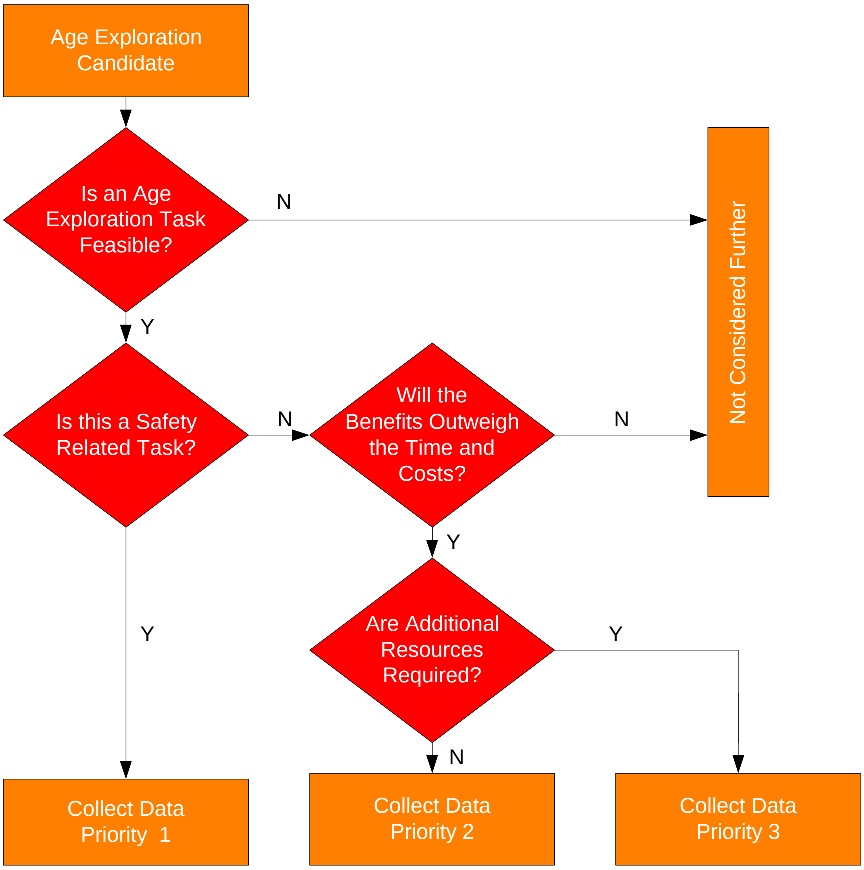


Figure 10.4 - Age Exploration Decision Logic

b. The AE investigation priorities are:

1) Priority 1. Failure modes that involve safety and where the study group has used subjective judgements to determine “life” or P-F interval.

2) Priority 2. Non-safety related failure modes that might realise significant savings, and for which data collection is not resource intensive.

3) Priority 3. Non-safety related failure modes that have the potential to realise savings but for which data collection could be resource intensive.

c. Safety. Careful consideration needs to be given to the implications of conducting an AE for a failure that has safety consequences. Data collection remains a priority one activity to validate safety-related tasks where adequate supporting information to justify initial decisions was not available. However, any proposals to extend task periodicities in the light of such data must recognise that such extensions may lead to an actual failure that has unacceptable failure consequences. It follows that AE undertaken on safety related functional failures must only be carried out under strictly controlled conditions.

d. Non-Safety. AE should be considered for non-safety related items if cost effective to do so. The collection of any data may require effort and commitment from ships staff. Therefore, it is essential that any data capture requirement be fully justified.

e. AE Recommendations. It should be emphasised that any RCM study recommendation for an AE investigation is just that. Unless the AE task is safety related, in which case it becomes mandatory, the MOD Equipment Project Manager (EPM), has the final decision on whether to conduct AE investigations.

10.8.2.2 Use of AE Data

a. The RCM (RN) Toolkit provides the facility to record the justification of any AE requirement.

b. When sufficient data is judged to have been collected, the RCM analysis for the system/equipment under investigation is to be revisited in order to optimise the preventive maintenance task interval or change the task if AE has indicated that the original task selected was unsuitable.

10.9 Consolidating Study Maintenance Tasks (Task Linking)

a. An RCM analysis is based on failure mode analysis. It is therefore possible for several different failure modes within a study to generate similar, or even identical, maintenance tasks on a single item of equipment. These tasks need be consolidated to obviate unnecessary repetition in any ensuing documentation. Once the task duplications have been identified the RCM (RN) Toolkit provides a facility to tag them and hence suppress their inclusion in printed maintenance schedules, without, however, deleting the detail from the record of analysis. For in service platform analyses, the electronic tagging consolidation process is undertaken by study personnel during the analysis. For new procurements the process should be undertaken immediately after the end of an analysis when key study personnel are still available. It should be noted that task linking is necessary for the production of common JICs (section 12).

b. Within the currently defined Defence Standard 00-60 LSAR, multiple occurrences of the same task are precluded by the use of relational database technology. Each task identified within the task inventory is documented once and cross-related to all appropriate failure modes.

10.10 Defence Standard 00-60 Correlation

a. LSA Process/Task Applicability

1) Conduct of an RCM analysis equates to the performance of Defence Standard 00-60 task 301, subtasks 2.4.1 and 2.4.2 in that both corrective and preventive tasks are identified.

2) The definition of RCM as detailed within this section satisfies the task input requirements for detailed RCM procedures and logic as called for in task 301, task input 3.2.

3) The task interval as detailed within task 401 subtask 2.1 is generated from this activity.

Figure 10.5

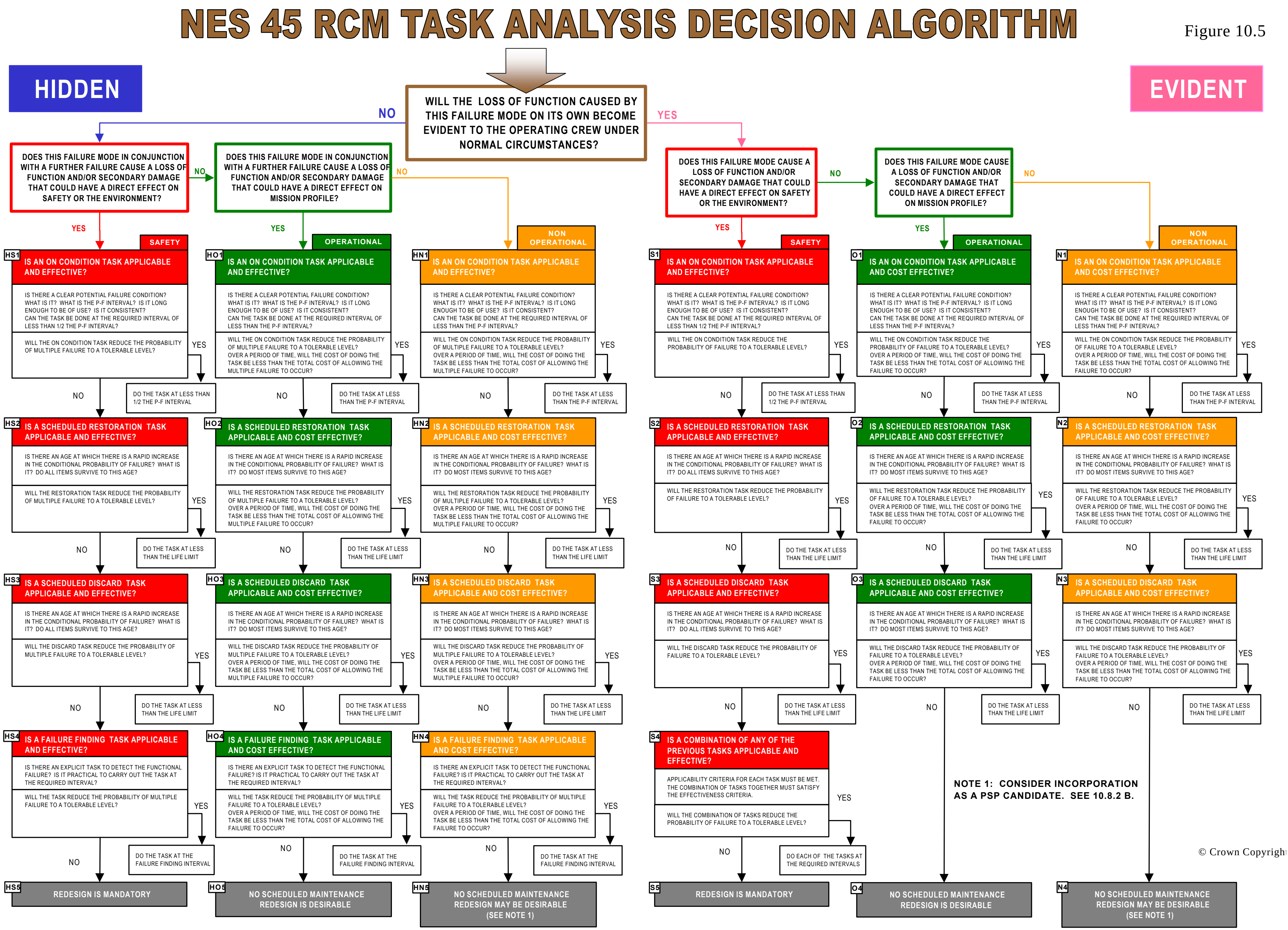


Figure 10.5 - Task Analysis Decision Algorithm

11. RCM ANALYSIS OF STRUCTURES AND STRUCTURAL ITEMS

11.1 Key Players

a. The key players in this part of the RCM process are indicated in **Table** 11.1. All key players should ensure that they understand this section.

Vessel/System Status	
New Construction/Procurement	**In-Service**
Warship IPT Leader	Warship IPT Leader
Study Facilitator	Study Facilitator
Study Technical/Design Advisor	Study Technical/Design Advisor
	Study IPT Representative
	Study Ship Surveyor
Study Technical Secretary	Study Technical Secretary

Table 11.1 - Key Players

11.2 Introduction

a. The FMECA and Task Analysis processes explained in sections 9 and 10 can be used satisfactorily for structural elements when the required functions and performance standards can be determined with accuracy, typically structural components within systems. For elements of complex structure such as the hull girder, however, it can be difficult to determine an absolute range of functions and performance standards and the amount of time required to determine this information from first principles can be prohibitive.

b. Where this is likely to be the case, it is possible to undertake a physical, as distinct from functional, FMECA and apply the RCM decision diagram to the failure modes developed during the process.

c. Whichever style of FMECA is used, however, structural failure modes will fall into 3 categories:

1) Fatigue Damage (FD), caused by the cyclic application of tensile stresses.

2) Environmental Deterioration (ED), where the structural material is attacked and weakened by exposure to its surroundings.

3) Accidental Damage (AD), which is a random event that can accelerate the deterioration processes of FD and ED.

d. Although these instructions have been written with metallic (steel) structures predominating, the process is valid when applied to non-metallic structural materials.

11.3 Complex Structure

a. An item of complex structure can be expected to perform many functions simultaneously. Typically for the hull girder, these functions can be strength, watertightness, electro-magnetic screening and ballistic protection, amongst others. In the normal course of events many, if not all, of these functions are lost when a structural failure occurs.

b. The purpose of an analysis on this type of structure, unlike an analysis conducted on marine engineering or weapon functions, is not to identify explicitly the functions of the structure being considered, but more to understand fully the effects of failure so that the consequences can be accurately predicted. If required, however, the structural functions can be derived from the effects of structural failure, thus completing the loop.

11.3.1 Load bearing Structure

a. Load bearing structures in surface ships and submarines are subject to many types of imposed loads during operation. For the hull, the magnitude and frequency of these loads depend on the nature of the operating environment although, in general, low loads are experienced often whilst peak loads will occur infrequently. There is also the added requirement for survivability from shock loading. Structures must, therefore, be designed in terms of all the load spectra so that they are unlikely to encounter any load that will jeopardise the safety of the ship or its mission.

b. Some forces applied to ship structures are caused by local loading due to the presence of lifting devices, hard points, weapon reaction forces, etc. It is important that such loads are considered at the time of assessing the main structural strength since they can be imposed over and above those global loads that the structure might be expected to resist. Such structure should, however, be analysed as part of the ship function to which it belongs, where the full design parameters and operating context can be better addressed. Any study addressing locally loaded structure should consider fatigue, the method of distributing loads into surrounding structure and environmental deterioration.

11.3.2 Containment

a. Many areas of ship structure have a function, often simultaneous with other functions, of containing fluids such as fuel, water and other liquids together with a requirement to be watertight or gastight.

11.3.3 Other Functions

a. That structure may have other functions to those mentioned above should be recognised during the analysis and identified when considering failure effects.

11.4 Structural Failure Categories

11.4.1 Fatigue Damage (FD)

a. The loads that a structure is subjected to are repeated many times throughout its service life. Although any single load application may be only a fraction of the load carrying capability of any particular structural member, the stress imposed by each load reduces the remaining margin of failure resistance. Eventually, as a result of these cumulative reductions, a small crack will appear in the metal. Initially there will be little change in the strength of the affected member but, as the stresses cause the crack to propagate, the strength is reduced at an ever-increasing rate. The fatigue process, therefore, has two components. Firstly, as time spent exposed to stress increases, the time interval before a crack will appear decreases, i.e. there is a reduction in the time before Crack Initiation (in practical terms this means the time at which the crack becomes detectable). The second component is the reduction in strength associated with Crack Propagation. Both of these components may vary not only with the type of material of which the element is made, but also its shape, size, the manufacturing processes used and the method by which the load is applied.

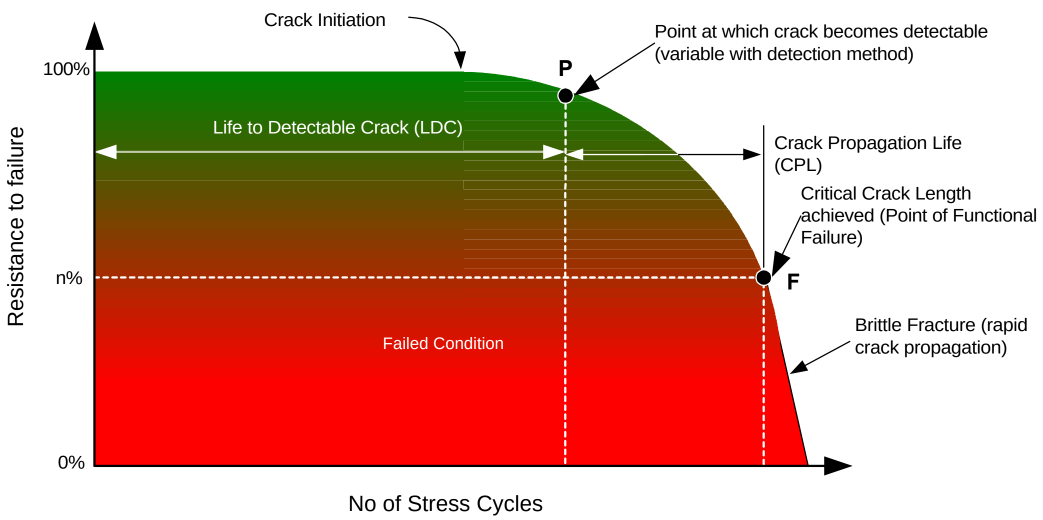


Figure 11.1 - FD Failure Sequence

b. The fatigue process can be portrayed in a similar manner to the P-F curve explained in section 10, albeit with changed nomenclature; this is illustrated in Figure 11.1. For the purposes of structural RCM, therefore, Potential Failure for FD failure modes represents that point on the P-F curve when a crack can be detected. It follows that this point is a variable dependent upon the method used for crack detection. It is possible, for instance, to detect cracks whilst still at the sub-surface stage in thicker metallic sections if radiographic techniques are used. For most ship survey techniques, however, reliance is placed on the ability of the surveyor to detect cracks visually. The length to which a crack will need to propagate before it becomes visually detectable will depend on various factors such as the accessibility of its location and whether corrosion could be expected to initiate quickly. It should also be noted that for some structural applications failure may be occasioned by brittle fracture, i.e. the points at which detectable crack, critical crack length and functional failure are reached may be virtually coincident. The time scale between detectable crack and functional failure may, therefore, be insufficient for any inspection task to be effective.

11.4.2 Environmental Deterioration (ED)

a. ED failure modes can take many forms, but the most frequently encountered type is corrosion, which includes a range of threats from simple oxidation to electrolytic reactions. Corrosion is a complex subject with many facets, all of which can interact and none of which are entirely predictable. Most types of corrosion are observable as surface deterioration, which results in a measurable reduction of cross section of the element. Other forms of deterioration, such as stress corrosion and selective phase corrosion, are much more difficult to detect. Generally the areas that are exposed to contaminants, moisture and heat are the most susceptible to corrosion. Therefore, properly applied and maintained protective systems or coatings are essential to forestall the progression of structural deterioration, except when the material forming the structure has an inherent resistance to deterioration when considered in its operating context. In preparing preventive maintenance regimes, knowledge of the designed corrosion allowances or, more correctly, the minimum values required (i.e. amount of material thickness or chemical change) to preserve an element's mechanical integrity is crucial. These values must be allied with the severity of the hazards that threaten to reduce the expected life, the effectiveness of the protective system(s) and the consequences of failure.

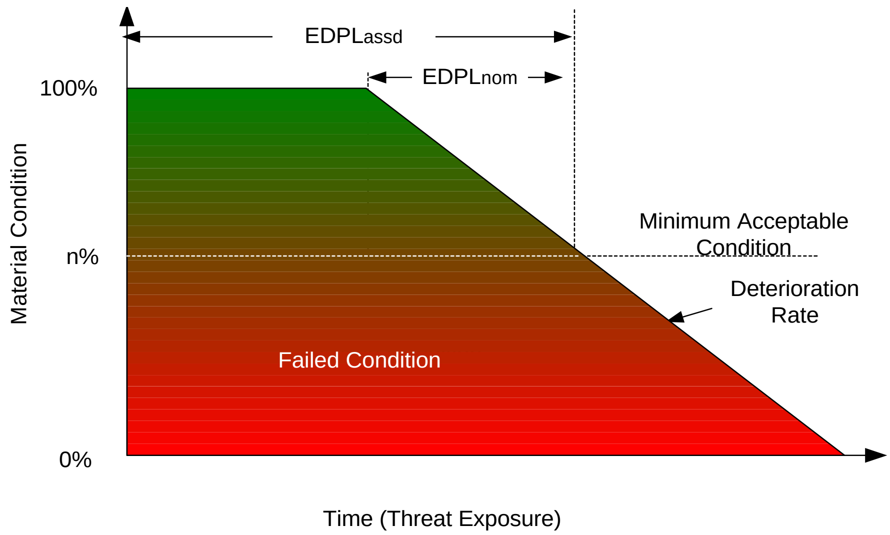


Figure 11.2 - ED Failure Sequence

b. Although not a truism for every case, the deterioration process for unprotected metal can be considered to be linear with time, given unchanging threat. When consideration is taken of the protective system the time interval taken to reduce the metal to its minimum acceptable condition will be increased by the effectiveness of that system, i.e. the time taken for this protection to break down and become ineffective. The time taken to reduce unprotected metal to the minimum necessary to sustain the function is known as the Nominal Environmental Deterioration Propagation Life ($EDPL_{nom}$). The estimated overall time taken when the protective system is taken into account is known as the Assessed Environmental Deterioration Propagation Life ($EDPL_{assd}$).

c. Figure 11.2 illustrates a typical ED failure sequence taking into account an allowance for the protective coating system. For the purposes of structural RCM Potential Failure for ED failure modes represents that point on the P-F interval curve when the anti-corrosion coating becomes ineffective. The performance of such coatings depends on many variables which can be difficult to predict; a notional extension to the P-F interval is appended (discussed in detail below) that takes account of this variability such that, in essence, point "P" is moved away from the origin of the X axis. The location of point "F" on the curve will also require determination and this needs to be based on an understanding of the structural function. Traditionally, a global corrosion allowance of 10% of the designed plate thickness over a defined area of plating was deemed to represent the "failed state". As ship structural designs have progressed towards lighter scantlings, this global allowance may not be suitable in areas experiencing relatively high stress concentrations. Conversely, in areas where the primary structural function is containment, it may be permissible to locate point F further to the right on the curve, representing a larger degree of wastage before the failed state is achieved.

11.4.3 Accidental Damage (AD)

a. The threat from AD is caused by some discrete and random event. The consequences of the event may cause stress risers or cracking reducing the structure's ability to sustain an applied load, or reducing the life of the structure. The event may also reduce the effectiveness of the protective system giving rise to the onset of corrosion earlier than might otherwise be expected. Either type of event will shorten the time taken for the element to deteriorate to its minimum condition since such events can accelerate the FD and ED failure processes.

b. Although these events can occur at any time in the life of the item, major or reported incidents such as grounding or collision are more appropriately dealt with by specific surveys at the time. Structural elements are vulnerable to AD mainly by virtue of their location, allied to a range of threats that might be presented. The role of the survey programme, therefore, is to identify AD attributable to <u>unreported</u> incidents that might require remedial actions to restore the structural design intent.

11.5 Structural Survey and Monitoring

a. The resultant RCM derived inspection strategy is able to determine the extent and frequency of surveys required to detect any deterioration which could impair the load carrying capacity or other functions of the structure. Few failures will be easily identified by the operator when they first occur but the ultimate effect of most structural failures is to impinge directly on safety.

b. The only applicable and effective preventive maintenance task for structural applications is the on condition one, i.e. inspection or survey of the principal items of structure, or the discard task for Safe Life (SL) structure. Because of this fact the focus in developing a RCM based survey programme is not, therefore, a search for applicable and effective tasks, but on an understanding of the failure mechanisms and determining the most appropriate survey interval.

c. All parts of the structure are exposed to the age related processes of fatigue and environmental deterioration, but these processes interact and are not entirely predictable. Thus, even for an asset that embodies well known materials, design practices and production processes, the intervals assigned to an initial programme may be only a fraction of the age at which any deterioration is anticipated. This is especially true for First of Class Ships, which will require intensive inspection as part of an age exploration or fleet leader sampling programme to ensure that the design intent is met.

11.6 The Structural Analysis Process

a. To conduct a structural analysis the stages shown in Figure 11.3 and detailed in the following paragraphs are to be followed.

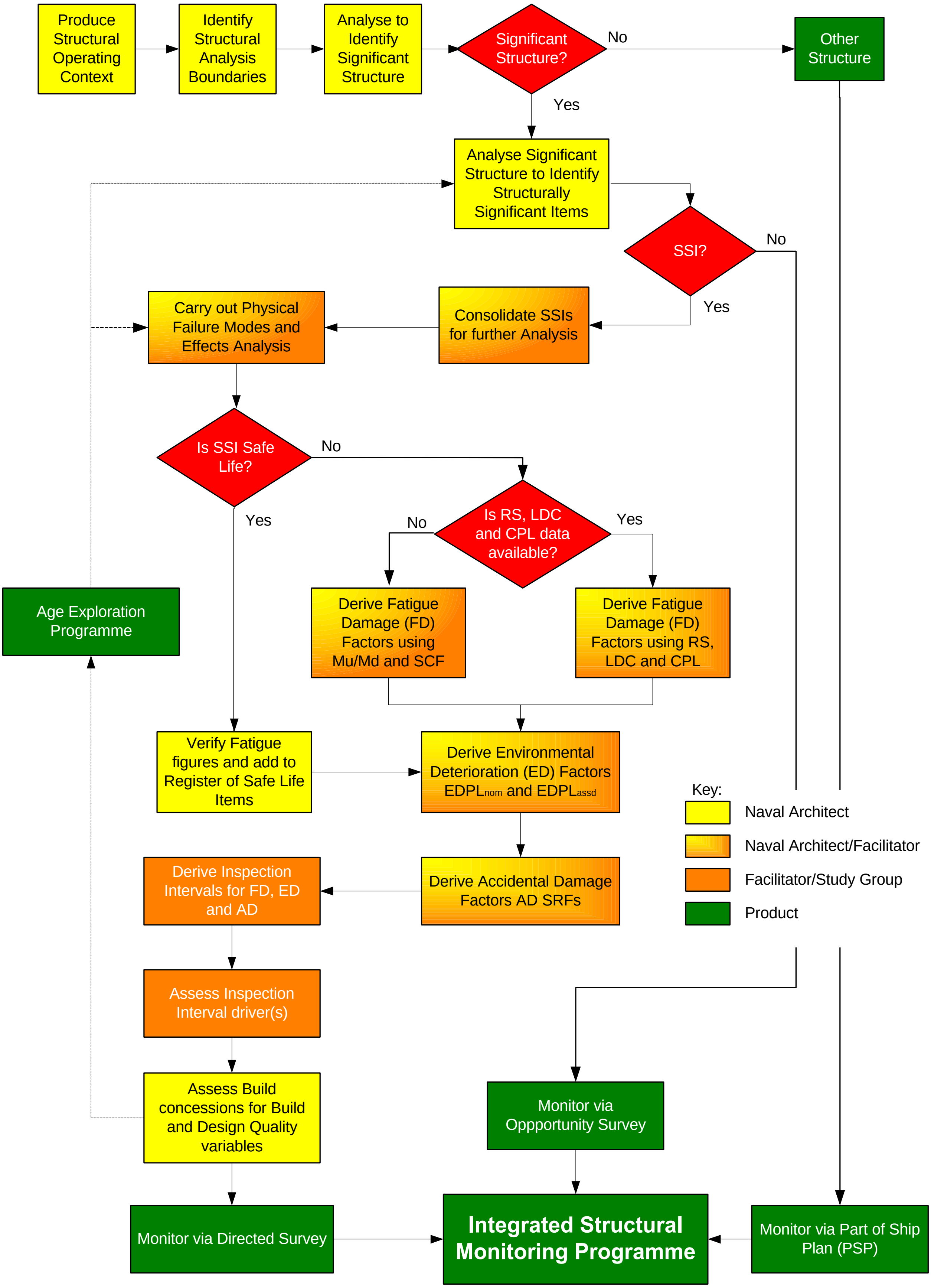


Figure 11.3 - The Structural Analysis Flowchart

11.6.1 Identification of Analysis Boundary

a. Careful consideration needs to be given in relation to the analysis boundary to ensure that all aspects that impinge on structural performance are included. Although when undertaking an analysis of the hull girder the majority of structure to include is fairly obvious, the interaction of other items such as machinery and weapon seatings, together with items giving rise to locally imposed loads such as breakwaters, bollards, fairleads, etc., should not be overlooked. Other items, to name but a few, can include integral daily service fuel tanks, 'A' brackets and local structure in way of rudders and stabilisers.

11.6.2 Operating Context

a. An operating context is to be written for the structure being analysed that should be aligned to the higher level operating contexts for the platform of which the structure forms part. Typically, for the hull girder, the structural design disclosure will be the primary source of information for developing the operating context. In the absence of a design disclosure, the operating context should contain the information shown in Table 11.2.

Primary Statement	Supporting Information Required
Build Standard	Military or Commercial Standard to which the structure is designed.
Design weight distribution	Full weight breakdown (NES 163) with supporting evidence.
Primary design loading	Hogging, sagging (vertical and horizontal) and still water bending moments. Justifications to be based on intended areas of operation.
Naturally induced loads	Sea (transverse, green seas, notional number of wave encounters, etc.), wind, ice, thermally induced.
Artificially induced loads	Berthing, docking and grounding, helicopter and vehicles, shock.
Ultimate strength	Panel collapse curves, sectional properties - sectional area, 2nd moment of area and location of the neutral axis, supporting assumptions - material properties, rolling and corrosion allowances, superstructure effect.
Fatigue	Full fatigue calculations for each selected SSI.
Vibration analysis	Design method used, main machinery excitation frequencies, main modes and frequencies of hull vibration, modes for masts and other structural features.
Environmental protection	Coating schemes, extent of cathodic protection.
Statutory requirements	For systems that include items subject to statutory tests, analysis of supporting structure for concentrated loads, lifting appliances, ramps, RAS points, etc.
Build quality	List of all major and minor build concessions.

Table 11.2 - Operating Context Information Requirements

11.6.3 Categorisation of Structure

a. Like other complex equipment the structure of surface ships and submarines can be sub divided into several different categories, some of which will have a greater significance than others. For the purposes of RCM it is important to identify elements of structure whose failure can impinge on safety or mission capability so that the risk and consequences of failure can be managed through the use of regular inspections. The RCM process commences with an appraisal of the structure to assess the roles and principal functions (i.e. strength and containment) of the various structural elements. This is a crucial process, complicated by many structures being monocoque in construction, and great care is needed to ensure that all structure is considered and subsequently categorised correctly.

11.6.4 Significant Structure and Structurally Significant Items

11.6.4.1 Significant Structure

a. In ship applications significant structure includes that structure which contributes to the strength of the hull girder and resists imposed forces arising from hogging, sagging, racking, etc. as well as loads imposed from docking. In general significant structure will be identified from structure traditionally defined as primary, typically the hull girder, shell plating, decks and watertight/oil tight bulkheads as shown in Figure 11.4. It should also include those items whose failure will result in a reduction in residual strength or loss of load bearing function and those which possess other critical functions such as containment or exclusion of fluids. Significant structure for the HUNT Class MCMV is shown in Figure 11.4.

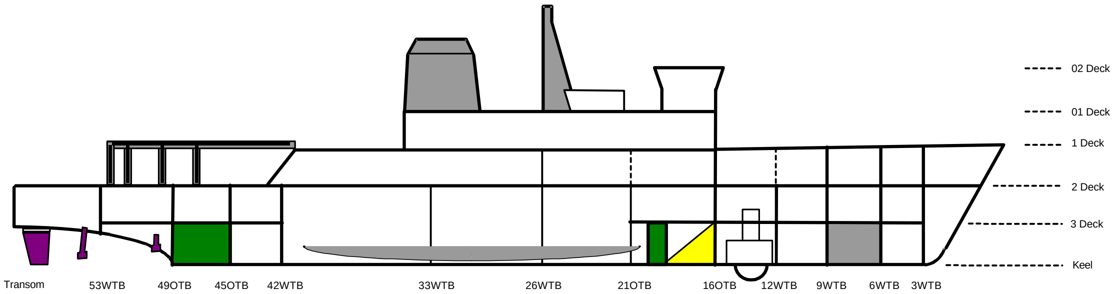


Figure 11.4 - Significant Structure (HUNT Class)

b. In order to identify elements of significant structure engineering judgement may need to be used extensively since, for many items, design intent may not be known fully and considerable work may be required to produce a suitable Design Disclosure Document (DDD). Each case should be considered on its merits with the consequences of failure of each item deciding the extent of data acquisition required, either as a formal task or as part of age exploration.

c. A study of structural drawings and design calculations will help to enable elements of significant structure to be identified, although in cases of doubt the likely consequences of failure should be the deciding factor.

11.6.4.2 Other Structure

a. By default, structure not selected as significant will be termed Other Structure and will only be inspected via the Part of Ship Plan (PSP). By definition an element classed as other structure does not, on its own, have the same level of failure consequences as significant structure. It can however, have an impact on the safety and mission capability of the ship when the failure of several elements combine to cause a more significant failure. In such cases, consideration should be given to consolidating such elements as an item for further analysis. The role of the PSP is discussed in more detail in Para 11.10.2 below.

11.6.5 Identification of Structurally Significant Items (SSIs)

a. For each area of Significant Structure it is necessary to select Structurally Significant Items (SSIs) at appropriate locations. An SSI is a site which may include several structural elements, the entire significant structure itself, or specific regions of the significant structure which are most vulnerable to the failure modes considered and thus likely to exhibit the earliest signs of potential failure.

b. For an efficient survey regime it is important to focus on the selected SSIs as these will be representative of the overall condition of the significant structure. It is therefore wasteful, for instance, when surveying for FD to examine the whole of a longitudinal stiffener when the most likely areas where defects will occur will be in the vicinity of stress raisers caused by structural discontinuities and/or areas subject to the greatest bending moments. It is therefore considered necessary only to examine those areas most likely to exhibit failure characteristics, whether due to FD, ED or AD. Typical SSIs for a watertight bulkhead are illustrated in Figure 11.5.

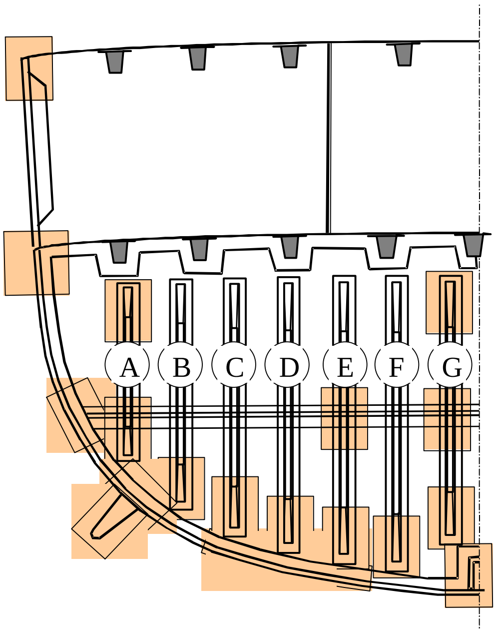


Figure 11.5 - HUNT Class O/WTB 26 SSIs (shown shaded)

c. The selection of SSIs is a critical process and information/data will be required from many different sources. Since the design intent of each item of structure may not be fully known engineering judgement may have to be used extensively to ensure that candidate SSIs are identified. Ultimately the consequences of failure must feature highly in this decision process and in certain instances it may be necessary to introduce an Age Exploration (AE) programme to help with this decision. SSI selection needs to be justified as per Table 11.3 in order to sustain the audit trail.

Significant Structure	Structurally Significant Item	Justification
O/WTB 21	Upper snaped end of stiffener A	Direct support of 2 deck girder
	Lower snaped end of stiffener A	Dissipation of shell impact energy. Overpressurisation of fuel tank.
	Intersection of bhd with 2 deck and shell plating.	Dissipation of shell impact energy. Strength integrity - longitudinal bending.
	Lower snaped end of stiffener B	Overpressurisation of fuel tank.
	Lower snaped end of stiffener C	Overpressurisation of fuel tank.
	Lower snaped end of stiffener D	Overpressurisation of fuel tank.
	Turn of bilge bhd/shell connecting angle adjacent to lower snaped end of stiffener C, D and E	Dissipation of shell impact energy. Strength integrity - longitudinal bending. Overpressurisation of fuel tank.

Table 11.3 - SSI Justification Table

11.6.5.1 Consolidation of SSIs for Analysis

a. The process of SSI selection will undoubtedly identify many items for further analysis, many of which will be from lower levels of detail. Some of these items may be:

1) adjacent to one another and subject to similar if not identical stresses and threats and which ultimately will result in identical failure effects.

2) considered as a single entity when the practicalities of inspection are considered, e.g. the work in wake required would require an inspection of all structure local to a particular area to be undertaken.

b. As a consequence it may be possible to identify individual lower indenture level SSIs capable of being grouped together to form one for the purposes of the RCM analysis, thus reducing the number of SSIs for analysis whilst still retaining their individual identities for later inclusion into the survey plan.

c. This process of consolidation will speed the analysis and may realise advantages for the conduct of structural surveys. It is possible that for some locations several individual SSIs will have been identified, e.g. the conjunction of a vertical weld, athwartships weld, fore and aft weld, deck plating, vertical plating and athwartships plating, all centred around one location. In this instance it may be possible to group these separate entities together to form one consolidated SSI and therefore attract a single survey inspection. The individual elements will be subjected to an initial assessment to identify those element(s) that provide the “worst case” drivers for each of the FD, ED and AD failure categories. The results of this initial assessment are to be recorded within the analysis proper to sustain the audit trail. By adopting the worst case scenario a margin of safety is introduced at an early stage in the RCM process. As an example of the benefits of this consolidation process, the suitable packaging of individual SSIs inside a fuel tank will allow a survey of the tank at a frequency driven by the worst case, thus reducing the number of times the tank is required to be opened.

d. Not all SSIs will either require or be suitable for consolidation in this way and it is important that careful consideration is taken when using this approach and engineering judgement will be required throughout. In cases of doubt the individual SSI must be analysed as a separate item.

e. For the subsequent management of the derived survey plan, SSIs will need to be added to the asset hierarchy within the RCM (RN) software.

11.6.5.2 Structural Failure Modes Effects and Criticality Analysis (FMECA)

a. Because RCM analysis of complex structure focuses on an understanding of the consequences of failure rather than a functional performance statement in order to determine appropriate survey inspections, it is necessary to adopt a revised approach to the compilation of the FMECA. Although the same layout of the FMECA worksheet is used as with the Marine and Combat System Engineering RCM analyses, the data is entered to accord with the Significant Structure/SSI hierarchy as shown in Figure 116. This is, in effect, a physical rather than a functional FMECA.

b. The data fields shown allow a tailored FMECA to be established which identifies clearly the ultimate effects of failure of any particular SSI, whether singular or consolidated, and allows failure mode considerations for FD, ED and AD.

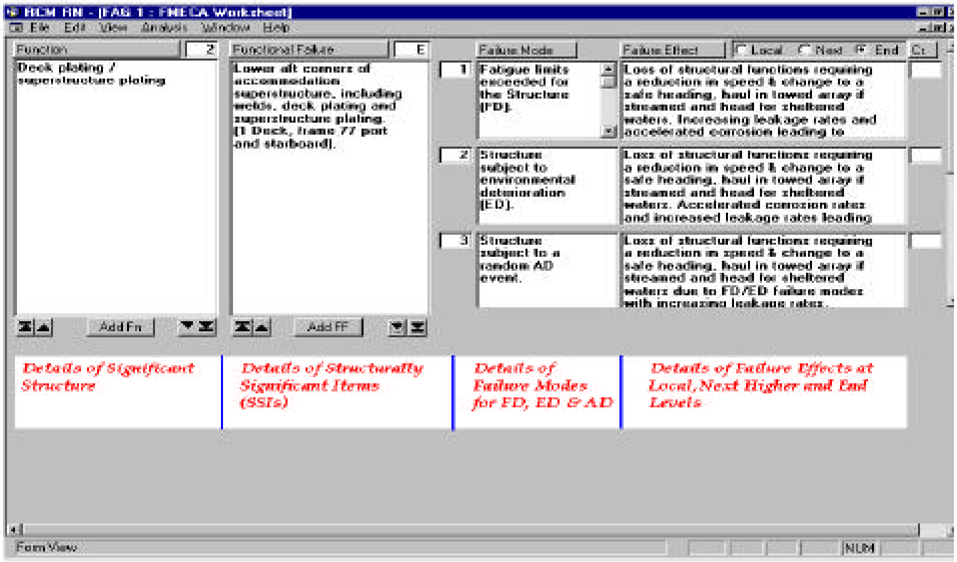


Figure 11.6 - Example Data Entry for Structural FMECA

c. This revised approach has necessitated data being established in fields, which traditionally record functional FMECA data, as shown in Table 11.4. This approach enables all aspects of the failure modes to be considered.

FMECA Field	Revised Data
Function	The detail of the Significant Structure is documented in this column.
Function Failure	A Detailed description of the SSI (singular or consolidated) for analysis is recorded in this column.
Failure Mode	Failure modes for FD, ED and AD are entered in this column.
Failure Effects	Failure effects at local, next higher and end levels are entered into this column.

Table 11.4 - Structural FMECA Data Input

d. By adopting this approach it reduces considerably the work required of a more conventional FMECA which concentrates on identifying all functions, functional failures and their failure modes. This structural FMECA is able to show the importance, in safety and mission terms, of each SSI and provides a sound basis from which to launch the RCM analysis based on the effects from which loss of function can be identified.

11.6.6 Safe Life (SL)/Damage Tolerant (DT) Assessment

a. When undertaking the FD assessment, the further categorisation of an SSI into ‘Safe Life’ or ‘Damage Tolerant’ structure must be made. This decision is based on the following criteria:

11.6.6.1 Safe Life Definition

a. SL structure is designed to survive its intended service life without undergoing fatigue failure and is generally ’one shot’ structure, i.e. failure is characterised by rapid crack propagation to critical length once cracks have been initiated, strength is reduced to zero and the structural function(s) is/are lost completely. Generally this is true for monolithic structure subject to cyclic stresses that include significant amounts of tensile loading. When the cyclic loads are mainly compressive, the likelihood of catastrophic failure is reduced. The profile for fatigue failure with tensile cyclic stresses is illustrated by Figure 11.7.

b. Great care will have to be exercised at the design stage to ensure that structure designated to be SL will achieve the required fatigue life in service. A monitoring process must be established such that, once this fatigue life is consumed, the element is removed from service and replaced. Such structural intervention can be prohibitively expensive and the structural SL might be expected to extend beyond or coincide with the End Item Design Life (EDL).

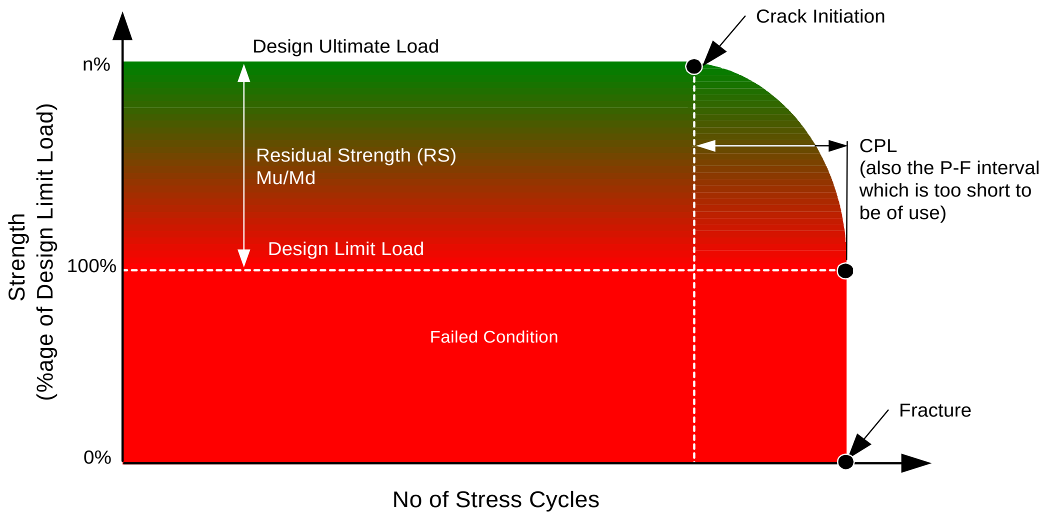


Figure 11.7 - Typical Failure Sequence for SL Structure

c. Although SL structure will not be subject to static or fatigue failure, it will be subject to ED and AD which might lead to premature failure (i.e. stress corrosion). Further analysis of these threats will be required since these aspects cannot be fully controlled or accurately predicted by the designer.

11.6.6.2 Damage Tolerant Definition

a. DT structure is characterised by one of the following characteristics:

1) a relatively slow rate of crack propagation once a crack has been initiated or,

2) the possession of multiple load paths, i.e. following crack growth to critical length and loss of structural function of a component, the ability of the remaining effective structure to permit continued safe operation with the proviso that either of these criteria must occur over a period of time that is long enough for an effective survey to be undertaken.

b. If, following RCM analysis this is not found to be true, the structure should be re-classified as SL and analysed as such.

c. An idealised failure sequence is shown in Figure 11.8. This structural redundancy, termed Residual Strength (RS), is provided by alternative load paths that direct imposed loads to adjacent structure, although the fatigue life of the adjacent structure is obviously shortened proportionally.

d. There is a finite limit to this process which is defined by the Design Limit Load (DLL). Once this threshold has been crossed the rapid onset of structural collapse can be expected if loads are sustained at or near the DLL. For example, if the design of a surface ship consists of a series of longitudinals to counter the effects of hogging and sagging, there may be adequate structural redundancy such that one or, perhaps, more of these members may fail without compromising the overall structural integrity of the ship below the DLL. It follows that the fatigue process in the remaining structure is, however, accelerated. As a general rule, the main structure of surface ships can be considered as DT, since fatigue cracking of structural members can be expected during the ship's life. The DT approach enables the maintenance engineer to produce more cost effective survey plans by ensuring that SSIs, representing vulnerable points of the structure, are inspected as warranted by their importance and predicted performance, instead of attempting to examine all structural elements at fixed intervals.

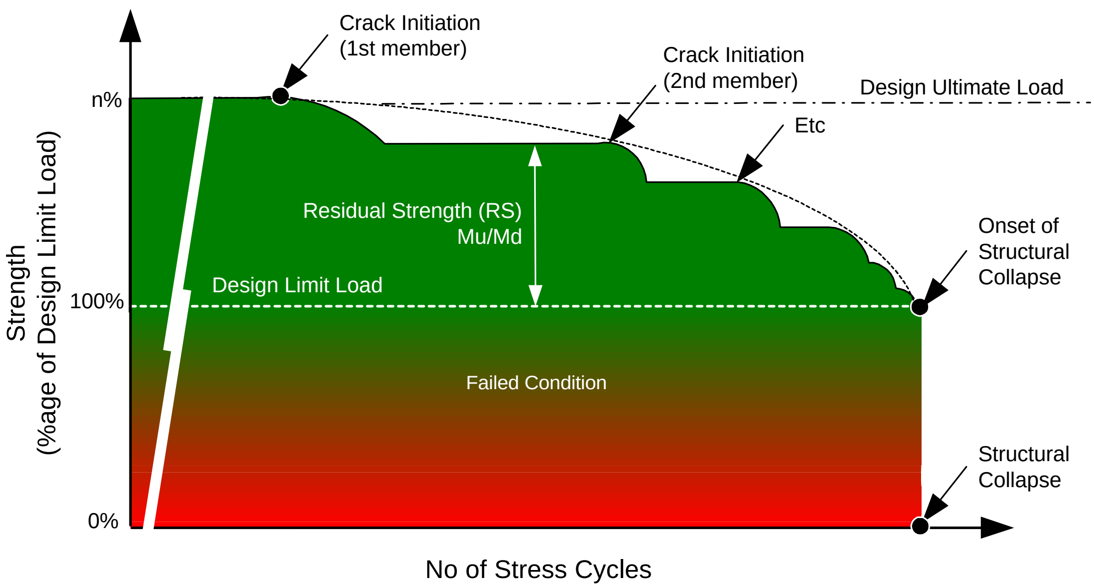


Figure 11.8 - Typical Failure Sequence of DT Structure

11.7 Data Analysis

a. Once the initial stages of RCM analysis have been carried out, identification of significant structure, identification and consolidation of SSIs and the completion of the structural FMECA, it is necessary to use this information to derive suitable inspection intervals for each SSI or consolidated SSI based on gathered data. RCM dictates a thorough analysis of each SSI and considers the effects of failure for FD, ED and AD failure modes. The following paragraphs detail the required data elements together with the analysis process for each SSI.

11.7.1 Fatigue Damage Analysis

a. Because of the differences in the design philosophy of SL and DT structure it is necessary to analyse each using a different set of criteria. The analysis of SL structure concentrates on the verification of the fatigue data to ensure adequate factors of safety have been applied at the design stage. On the other hand, for DT structure the emphasis is placed more on the failure mechanisms and a requirement to identify failures before they progress to a loss of structural function.

NOTE: It is not customary for naval surface vessels to use a measurement system to manage the consumption of fatigue life, since reliance is based on conservative factors of safety and corrosion margins.

11.7.1.1 Safe Life Analysis

a. When the structure being considered has been designed to SL criteria, a check should be made to ensure that appropriate factors of safety have been applied so that the structure will not fail either by static loading or by cracking induced by fatigue. The item should be added to a register of SL items and a structural sampling AE programme initiated to ensure that the initial life of the structure, possibly established through the manufacturer’s test programme, will be achieved in practice. The collection of operational data will help with this process. Once this life is established a suitable discard task can be implemented such that when the fatigue life is consumed the item is removed from service and replaced. In addition, RCM requires that SL structure is also analysed for ED and AD failure modes, this is discussed in Paras 11.7.2 and 11.7.4.

11.7.1.2 Damage Tolerant Analysis

a. Structure designed as DT should be assessed for its susceptibility to FD. Data processing is necessary to either derive values for or validate a range of fatigue criteria. The necessary criteria will include the following:

1) <u>Residual Strength</u> - This is a measure of the structure’s damage tolerance expressed as a percentage of the items DLL. This is akin to the ultimate collapse calculation Mu/Md.

2) <u>Life to Crack Initiation</u> - In practical terms this is the Life to Detectable Crack (LDC) and is to be calculated for each SSI. The LDC is the life up to the point at which it is considered unlikely that an existing crack will be missed by an on-condition inspection. A crack length of 100mm is normally assumed to be the minimum that is visually detectable for surface ships, although this figure should be reviewed for individual analyses. The life at which a crack can be expected to occur is dependent upon several factors, these include design complexity, manufacturing method, material and the stress concentration factors (SCF) based on physical location within the ship. For instance, an item made from a material possessing good ductility can be expected to crack sooner if its location is near the mid ships point on the weatherdeck, where SCFs are high, than it would if its location was nearer the neutral axis, where SCFs are much lower.

3) <u>Life to Critical Crack Length (Crack Propagation Life (CPL))</u> - The time it takes for a crack to grow from a detectable crack length to critical crack length in SSIs. This time may vary depending on the method of detection used. Hence, the CPL for a crack to be detected by NDE will be longer than if the crack is to be detected visually. The critical length is reached when the structural element can no longer sustain the required strength function. In a similar way to that of LDC, a crack which initiates in a SSI of a particular design will propagate more quickly if located in an area with a higher SCF. The CPL may also be accelerated if the fatigue process is complicated by the presence of corrosion.

b. <u>Inspection Interval</u> - Inspection intervals for each SSI are dependent on the method by which CPL is calculated. For vessels built to the Sea Systems Publication No 23 (SSP23) structural standard, this is equivalent to the CPL. This is due to SSP23 using a statistical approach which already possess a degree of conservatism; there is no need to add an additional factor by using a divisor to reduce the interval further as required by the Marine Engineering and Combat Systems methodology. For fatigue calculations based on other criteria, the design authority will specify whether a suitable divisor is to be used. Since fatigue failures are ultimately safety or mission related this value is normally <2 (e.g. CPL/3), although this can be modified if the structure has a sufficient level of redundancy. The start time for the commencement of inspections is derived from a sub division of the LDC. This value is again to be decided by the design authority, but LDC/1.5 may be suitable initially. The resultant is referred to as FD deferred (FD_{def}). This will ensure that the appropriate inspection interval is in place before the time when a crack is expected. AE of the first in class, fatigue life fleet leader and sampling techniques are all valid methods which can be adopted to establish suitable values. This method will ensure that sufficient inspections are carried out during the time when progression to failure might be expected.

11.7.1.2.1 An Alternative Approach for the Analysis of Fatigue Damage

a. Suitable quality information regarding RS, LDC and CPL may be difficult to acquire. In these circumstances a revised method may be adopted with the approval of the design authority. If a revised method is adopted AE will be required to validate many of the findings. The following is intended to demonstrate possible acceptable variances to the preferred method of assessing inspection intervals for DT structure.

b. If it is agreed that there is a sufficient factor of safety for the structure it may be acceptable not to calculate RS. This will be indicated by the value of Mu/Md.

c. It is possible, using fracture mechanics and fatigue assumptions published in SSP23, to calculate LDC and CPL for a range of different weld profiles. Using engineering judgement in consultation with the design authority a particular weld profile may be adopted as representing the structure as a whole.

d. For each type of weld the values are affected by the SCF associated with the physical location, both horizontally and vertically, within the ship. Because this type of information is likely to be available only for specific locations it may be necessary to apply linear interpolation to weld profile values at various key longitudinal ordinates to avoid an overly conservative approach being adopted for inter-ordinate locations. This will allow the removal of large step differences between adjacent ordinate locations. Where a SSI extends across a range of ordinates the values for the worst case should be adopted, thus applying a factor of safety. Further to this, engineering judgement will be required when deciding which SCFs are applicable at specific Deck locations.

e. Because this method is based on the statistical methods used by SSP23, (which carry a safety margin of mean minus two standard deviations), it may, with the approval of the design authority, be possible to deviate from the normal RCM procedure and not have a divisor for the CPL. The derived Inspection Interval should not, however, exceed the CPL other than in exceptional circumstances and only then with the approval of the Design Authority.

11.7.2 Environmental Deterioration Analysis

a. Each SSI is to be assessed for its susceptibility to ED failure modes irrespective of whether it is SL or DT. Generally, ED is material loss (wasting/pitting) due to corrosion but other mechanisms, such as selective phase corrosion, may also occur. ED failure modes are assessed against a wide range of threats and varying exposure levels because of the uncertainty surrounding the expected total exposure to the threat and the likelihood that undetected ED will cause premature failure of the element. For each SSI proceed as follows:

11.7.2.1 Stage One

a. The first stage in the assessment for ED failure modes is the determination of the age at which the material deterioration reaches its minimum acceptable condition. This condition can be defined as the point at which the function can no longer be guaranteed, particularly if shock survival is paramount. The period is based on the rate of deterioration of unprotected material and is called the Nominal Environmental Deterioration Propagation Life ($EDPL_{nom}$) and is given by Equation 1:

$$\frac{(\text{Design Condition} - \text{Min Acceptable Condition})}{(\text{Deterioration Rate} \times \text{Threat Factor})} = EDPL_{nom}$$

Equation 11.1 - Calculation of $EDPL_{nom}$

b. The values required to perform the above calculation can be obtained from the DDD or structural drawings (designed condition) or the Design Authority (minimum acceptable condition). The minimum acceptable condition is often regarded as 90% of the original thickness, but each analysis will require an individual judgement. In certain circumstances it may be necessary to include in the minimum acceptable condition an allowance for rolling thickness tolerances in plate metal. This is because in some instances the rolling thickness tolerances of plate are large enough to reduce the metal to its minimum acceptable condition before the ship enters service. If this is found to be the case the Design Authority should be consulted to determine the way forward. An example of how this could be interpreted is reflected in Equation 2:

$$(\text{Design Condition - Rolling Tolerance}_{\text{min}}) \times 0.9 = \text{Min Acc Condition}$$

Where Rolling Tolerance$_{min}$ is the Minimum Rolling Under Thickness Tolerance

Equation 11.2 - Calculation of Minimum Acceptable Condition

c. Deterioration rates and threat factors are more problematical but basic rates for corrosion in sea water can be taken from Def Stan 01-2, which quotes figures for a variety of alloys; typically unprotected steel has a corrosion rate of 0.15mm/year in salt water. This situation is further complicated, however, by the range of different threats to which individual SSIs may be subjected to, for instance the threat to an SSI at the waterline is substantially greater than the threat to an SSI in an internal dry compartment. This needs to be accounted for when determining the EDPL$_{nom}$.

d. The threat to the structure from ED has many components and they may interact in unforeseen ways. It is important, therefore, that engineering judgements reflect this situation and take cognisance of the threat to the material, dependant not only on operating context but also the item's physical location. It is important to realise at this stage that although the tendency is to reflect the corrosion rate of unprotected steel in sea water this is in fact a baseline measurement which is effectively adjusted depending upon type of threat and physical location. This factor will ultimately determine whether the value of EDPL$_{nom}$ tends towards EDPL$_{min}$ or EDPL$_{max}$, Figure 11.9 shows this in practice.

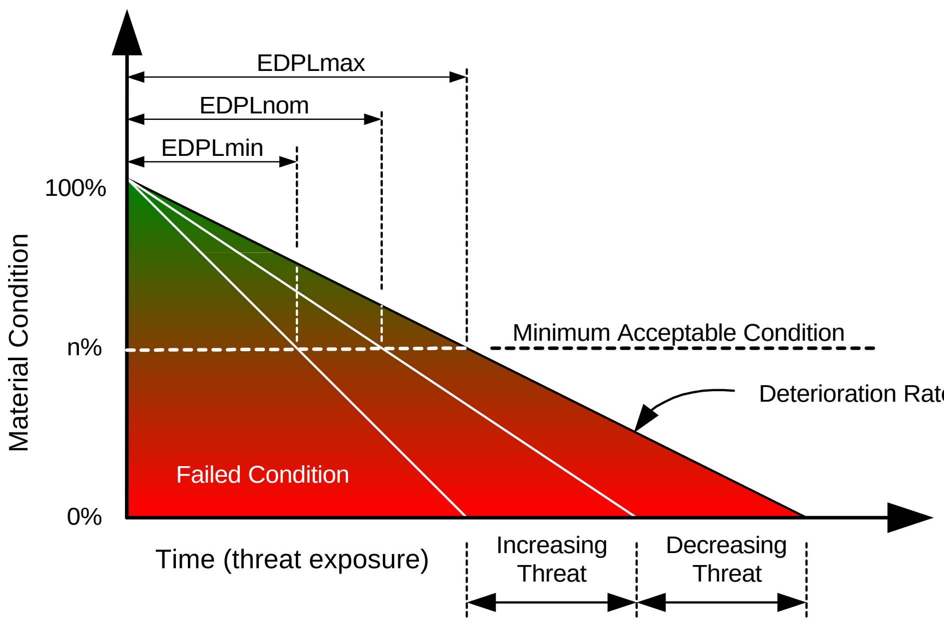


Figure 11.9 - Effect of Changing Threat

e. All likely threats are to be considered and the following notes are intended for guidance only, they are not exhaustive and appropriate judgements are to be made:

1) Sulphate Reducing Bacteria (SRB) are virulent in many environments, but are found mainly in Water Compensated Fuel System (WCFS) dieso and sewage tanks. Their presence leads to pitting corrosion which can become extremely aggressive.

2) Galvanic corrosion results when materials of different electrode potential are connected together in a conducting electrolyte such as sea water. The more noble metal (cathode) does not corrode, whilst the less noble metal (anode) is consumed. The rate of consumption of the anode is affected by many parameters including the potential generated between the dissimilar metals and geometry, etc. The anodes and cathodes may be of different metals as is the case with bi-metal corrosion, or even of similar atoms perhaps only a few inter-atomic distances apart as is the case of corrosion of a single metal.

3) Selective phase corrosion is a form of galvanic corrosion which affects some alloys containing two or more microstructural phases. An internal galvanic cell is created between the different phases resulting in the dissolution of one or more phases. This can degrade the mechanical properties of the material. This type of corrosion is extremely difficult to detect visually, i.e. de-zincification of brass.

4) Some materials suffer from stress corrosion cracking (SCC) when subjected to tensile stresses in a corrosive environment, e.g. stainless steel.

5) Caustic or acidic agents are self explanatory and can occur in battery rooms and dirty oil tanks. Sewage is caustic (pH 8 - 8.5), particularly on vacuum collecting systems.

6) Alternate wetting and drying by sea water, as well as salt laden air, can cause accelerated corrosion. When structure is continuously wetted, the rate of corrosion often depends upon the oxygen content of the sea water. The threat from fresh water, although significantly less than that posed by sea water, should not be overlooked.

7) Excessive heat often exacerbates corrosion and can emanate from many sources, including steam lines, funnel uptakes and weapon exhaust efflux. Drying rooms can also experience excessive corrosion.

8) Condensing humidity, generally found in galleys and bathrooms. The cycles of heating and cooling, wetting and drying, can cause serious corrosion.

9) Cavitation, erosion and abrasion are particular types of wastage and represent mechanical damage rather than corrosion, although a synergy of the two can lead to accelerated wastage. The action of propellers, anchor cables, etc. should be considered.

f. Table 11.5 identifies example threat factors. Values less than 1 indicate a reduced threat whereas values above 1 indicate an increased threat, they are not exhaustive and are intended as guidance only. Each ship application will require its own separate sub division and assessment of the individual threat values relating to its operating context.

AREA	THREAT FACTOR
Underwater	1.1
Waterline	1.5
Above Water Hull	0.8
Dirty Oil	2.0
Masts & Funnel	1.7
Internal (Wet)	1.0
Internal (Dry)	0.25

Table 11.5 - Example Threat Factors

11.7.2.2 Stage Two

a. During Stage One the value of $EDPL_{nom}$ was established, providing a baseline for the likely elapsed time unprotected metal will take to degrade to the minimum acceptable condition. It is recognised, however, that protective systems, which can vary from simple paint schemes to cathodic protection, have a major role to play in influencing $EDPL_{nom}$ effectively extending the period taken for the metal to degrade from its design condition to its minimum acceptable condition. Figure 11.10 shows how the time to failure can be extended by an effective protective system. Whilst it is accepted that the ideal conditions of coating application needed to realise the full longevity of the protective coating are difficult to achieve in the shipbuilding environment, it can be assumed that there is an adequate level of protection for at least a proportion of that time. It is also accepted that the protective coating can become damaged by random accidental events, therefore potentially reducing or eliminating its effectiveness, this section considers only the gradual erosion of the protection due to ED. The connection between protective coating effectiveness and AD is discussed in Para 11.7.4

b. The effectiveness of the protective coating is assessed by determining Design Coating System Life and a confidence factor based on type of coating and its location. This confidence factor, based on experience with the coating system in use, is used to determine what percentage of the Design Coating System Life that can be expected given in any particular operating context. Equation 3 is used to determine the period of time to be used as the P-F interval to develop task periodicities. The resultant of this equation is called the Assessed Environmental Deterioration Propagation Life ($EDPL_{assd}$):

$$EDPL_{nom} + (\text{Design Life of Coating} \times \text{Confidence Factor}) = EDPL_{assd}$$

Equation 11.3 - Calculation of $EDPL_{assd}$

c. Figure 11.11 identifies the various elements required to calculate $EDPL_{assd}$ and demonstrates how its value can be affected by the influence of any one of these

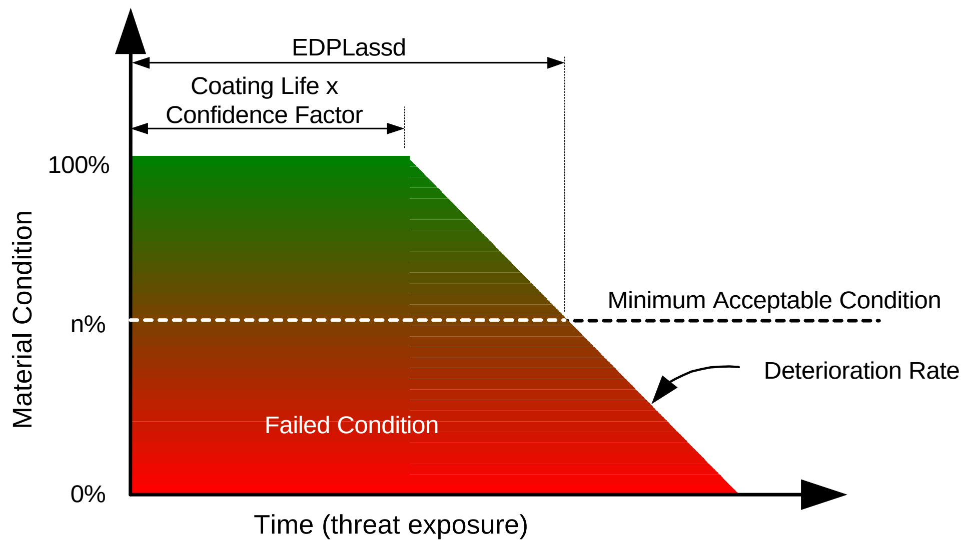


Figure 11.11 - Effect of Threat and Protective Coating

d. Engineering judgement and actual operating data will be required during this stage of analysis. Whilst considering the confidence factor the effectiveness of any cathodic protection must also be considered especially in areas which have no protective coatings or those areas whose protective coatings are considered inadequate. In cases of doubt the confidence factor must be assessed as low and in extreme cases it may be zero, thus effectively making $EDPL_{nom}$ and $EDPL_{assd}$ identical.

ITEM	PAINT LIFE (Yrs)	CONFIDENCE FACTOR
Waterline	10	0.6
Weather Deck	5	0.6
Superstructure	10	0.3
Dirty Oil Tank	0	0.0

Table 11.6 - Example Paint Life and Confidence Factors

e. Table 11.6 identifies example paint system lives and relevant confidence factors, they are not exhaustive and are intended as guidance only. Each ship application will require its own separate sub division and assessment of the individual values relating to that situation.

11.7.2.3 Stage Three

a. Stage three of the analysis for ED requires that a suitable inspection interval is identified. This is achieved by dividing the $EDPL_{assd}$ by a factor ('n') in accordance with standard RCM procedures. This provides a further margin of safety for the SSI under analysis (the primary safety margin is provided by the defined Minimum Acceptable Material Condition).

11.7.3 Threat Reduction Effect Of Cathodic Protection

a. The presence of either passive or active cathodic protection can influence significantly the threat ranged against metallic structures in the presence of a suitable electrolyte, e.g. seawater. For the purposes of the RCM study of structure, such protection is to be assumed as effective in fulfilling its design intent, i.e. reducing or eliminating the ED failure mode of “corrosion due to electrolytic action”. It follows, therefore, that cathodic protection, in all its forms such as zinc protectors and Impressed Current Cathodic Protection (ICCP), need to be the subject of a separate study in which their respective failure modes are addressed.

11.7.4 Accidental Damage Analysis

a. The threat from AD is largely due to random events which have the effect of modifying the timescales of FD and/or ED failure modes. This is primarily because any such occurrence is likely to result in the following situations:

1) Deformation of the structure and introduction of stress raisers or changed load paths, which will accelerate the onset of FD failure modes.

2) Damage to the coating system which allows the introduction of corrosive elements and therefore accelerates ED failure modes.

b. Further, the assessment of AD is in essence a judgement of the risk of damage to the structure and therefore has two components:

1) The probability of the structural element occasioning damage likely to be disadvantageous to its expected failure from FD or ED, and

2) The plausible consequences of such damage.

c. When considering purely FD failure modes, the LDC value assumes a resistance to failure until a crack actually appears. When assessing FD in an AD context, the likely scenario is that LDC is consumed instantaneously and a crack initiated. It follows, therefore, that in these circumstances CPL becomes the more dominant factor in determining the task interval.

d. Figure 11.11 identifies the AD task interval link to FD failure modes.

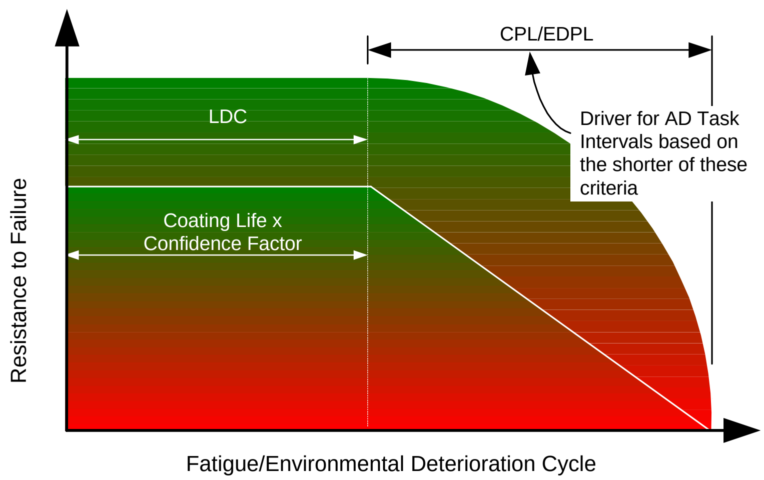


Figure 11.11 - AD Task Interval Link to FD and ED Failure Modes

e. When assessing purely ED failure modes allowance will have been made for the fact that the protective coating is providing some measure of failure resistance which is reflected in the ED task interval. When assessing ED in an AD context it is assumed that the protective coating is damaged and has become ineffective spontaneously. It follows, therefore, that in these circumstances $EDPL_{nom}$ becomes the more dominant factor in driving the task interval because the protection afforded by the protective coating purely for ED failure modes cannot be assumed. This makes allowance for the fact that there will be no protection to the SSI and assumes that ED will commence immediately. The $EDPL_{nom}$ is then sub divided by a factor that is determined by the design authority in accordance with standard RCM procedures. This introduces an element of conservatism, which ensures that for SSIs vulnerable to severe failure consequences it is possible to achieve at least one survey inspection during the progression to failure.

11.7.4.1 Damage Tolerant Structures

a. During the course of applying RCM to DT structures both CPL and $EDPL_{nom}$ will have been previously identified as part of the assessment for FD and ED failure modes. The results of these previous analyses will allow identification of which failure modes, FD or ED, will drive the AD task interval by determining which is smaller, CPL or the $EDPL_{nom}$. By using the data established during the analysis of FD and ED failure modes in this way it is possible to assess AD failure modes and derive task intervals which reflect the nature and characteristics of the failure mode in relation to its most likely outcome.

11.7.4.2 Safe Life Structure

a. Because of the characteristics of FD failure modes in relation to SL structure, CPL will not have been calculated in these cases. Although this initiates a different approach, the identification of Structural Risk Values and subsequent calculation of Structural Risk Factors is still required. When determining the suitability of AD tasks, driven by FD failure modes, engineering judgement in consultation with the design authority is essential. Guideline procedures are detailed in Para 11.7.4.6. For ED failure modes the same principles as those for DT structure can be adopted.

11.7.4.3 Accidental Damage Risk Assessment

a. Each SSI is to be assessed for the severity of failure consequence and POC against a range of threats likely to be encountered by the ship during its operational life. Table 11.7 has been designed to allow a quantitative assessment to be made regarding the risk posed by the consequences and frequency of AD.

b. After determining the POC assessment, the appropriate value for consequence can be derived using the values in Table 11.8. The consequence value is derived subjectively for each application, and can be based on the amount of historical data available for actuarial analysis or from engineering judgement. It follows that for new construction, it may be appropriate to assess data generated from incidents involving similar types of ship with parallel operating contexts.

POC / Consequence	FREQUENT (A) > 1 per 1,000 hrs	PROBABLE (B) < 1 per 1,000 hrs > 1 per 10,000 hrs	OCCASIONAL (C) < 1 per 10,000 hrs > 1 per 100,000 hrs	REMOTE (D) < 1 per 100,000 hrs > 1 per 1,000,000 hrs	IMPROBABLE (E) < 1 per 1,000,000 hrs
SEVERE (I) Creation of a stress riser or crack in the vicinity of the impact and/or damage to the protective coating exposing bare metal to salt water, salt-laden air or other corrosive agent.	1 High	2 High	4 High	5 High	8 Medium
NOTABLE (II) Damage to the protective coating and exposure of bare metal to fresh water or other relatively benign substance.	3 High	6 High	7 Medium	9 Medium	14 Low
MARGINAL (III) Damage to the protective coating and exposure of bare metal to internal conditioned ambient atmosphere or other benign substance.	10 Medium	11 Medium	12 Medium	15 Low	18 May be Tolerable
NEGLIGIBLE (IV) Only cosmetic damage accumulated over the frequency period.	13 Medium	16 Low	17 Low	19 May be Tolerable	20 May be Tolerable

Table 11.7 - AD Risk Matrix

	Typical Threats			
Consequence Potential	**Severe**	**Notable**	**Marginal**	**Negligible**
Typical Threat (related to likely energy input)	Tug handling Weapon reaction Wave action Ice/Flotsam Slamming	Machinery /Ammunition shipping route Heat extremes RAS dump site Docking Abrasion	High duty personnel transit /access route Stores handling /loading route Indirect maintenance activity	Low duty personnel transit /access route

Table 11.8 - Typical AD Threats

11.7.4.4 The AD Task Interval

a. The actual interval for AD rated tasks is derived by determining an AD Structural Risk Factor (SRF) for each SSI using the values derived in Equation 11.4:

$$\log\left(\frac{\text{Risk Value}_{max}}{\text{Assessed Risk Value}}\right) = \text{AD Structural Risk Factor}$$

Where Risk Value$_{max}$ is the maximum value for AD risk (20) indicated in 11.7.

Equation 11.4 - Calculation of Structural Risk Factor

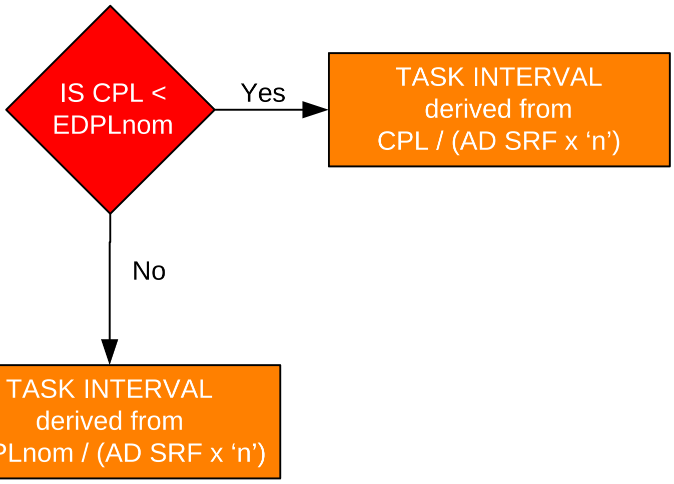
b. The “log” influence reverts the calculation to provide an exponential curve, compliant with the POC criteria. This ensures that only those elements with a high/medium risk assessment are capable of providing drivers for inspection intervals.

11.7.4.5 Damage Tolerant Structure - Accidental Damage Task Interval

a. For each SSI the calculated AD SRF is applied using the algorithm detailed in Figure 11.12. Depending on which is the shorter, CPL or the EDPL$_{nom}$ a suitable AD task interval can be determined using the same criteria as that for purely FD and ED failure modes, i.e. further division by a value ('n') of either less than the “P-F” interval or less than half the P-F interval depending upon the consequences of failure, to build in the factor of safety required by RCM principles.

b. The benefits of this method of analysis is that the SSIs exposed to the greatest threat and have the highest probability of an AD event are inspected to ensure that, should the LDC be consumed by a random AD event, further FD will be discovered before any crack has developed to critical length. Similarly, for ED failure modes it ensures that corrosion initiated by the AD event will attract an inspection before the deterioration reaches the minimum condition. For each type of failure possibility, therefore, AD tasks help to give confidence in the LDC and protective coating elements of FD and ED failure modes respectively.

c. For SSIs which are identified as having low consequence and probability values to AD events, then FD or ED failure modes will be dominant and hence drive the task interval.



Where ‘n’ is the divisor specified by the design authority

Figure 11.12 - AD Driver and Task Interval

11.7.4.6 Safe Life Structure Accidental Damage Task Interval

a. Because CPL will not have been calculated for SL structure a more subjective assessment of FD failure modes induced by an AD event, based on the SRF, is required. In the same way that DT structure exposed to a high consequence rating and a high probability of an AD event occurring is given prominence, so must similarly rated SL structure. In consultation with the design authority a method based on SRFs for the SSI, linked to a percentage of the SSI design SL, may be adopted. By linking the risk factor of the individual SSI to its SL an AD task can be considered which will give due prominence to high risk items. Percentage values identified in Table 11.9 are intended as guidelines only, each analysis and SSI will require individual assessment.

SRF	Task Interval as a percentage of the SSI Design Life
< 0.07	50%
0.07 - < 0.2	25%
0.2 - 0.5	10%
> 0.5	5%

Table 11.9 - SL Structure FD Task Interval Guidelines

b. SL structure is automatically assessed for ED failure modes because of the uncertainty surrounding the likely exposure during its service life and to help give confidence that the design life of the SSI will be achieved in practice. When considering ED failure modes in an AD context, therefore, the same criteria as for DT structure can be adopted, i.e. base the task interval on ($EDPL_{nom}$ / (AD SRF $\times$ 'n')).

c. Once both FD and ED assessments have been made a suitable AD task will be driven by the one with the shorter interval. This will be recommended for comparison against purely FD and ED failure modes in order to determine the most effective inspection interval driver.

d. It should be noted that the method of establishing a survey regime for unreported AD has been derived from the application of the results of analysis of damage information collated from a necessarily limited sub-set of existing survey data. The equations have been designed to meet the following criteria:

1) For structural elements showing a high risk of AD (i.e. high Consequence x high POC), the resulting inspection interval must not to be higher than the lower of CPL or $EDPL_{nom}$. This ensures that these SSIs will become the survey inspection interval driver.

2) For structural elements with a negligible risk of AD, it must not produce the survey inspection interval driver.

11.7.5 Deriving the Natural Inspection Interval

a. Once the data gathering and analysis has been carried out it is necessary to interpret the findings and derive a suitable inspection interval and regime. The interval will be based on the driver element, i.e. the failure mode prescribing the most frequent inspection. This situation can become more complex than simply identifying the failure mode with the shortest task interval and a full appraisal of all parameters will be required.

b. FD - For DT structure only, this failure mode identifies two inspection intervals, one based on FD_{def} and the other based on CPL. FD_{def} is the time to first FD inspection or the amount of time by which the first inspection is deferred followed by a repeat inspection interval based on CPL (but see 0 e). SL structure is not assessed for FD in the same manner but a 'Safe Life' will have been determined which will insure against failure from fatigue during the life of the SSI.

c. ED - The inspection interval determined by the analysis of ED failure modes will be considered alongside inspection intervals derived by FD and AD failure modes. If this interval is identified as the shortest then this will be adopted for the inspection.

d. AD - Similarly AD failure mode analysis will identify an inspection interval which if found to be the shortest will be adopted as the inspection interval.

e. A possible scenario faced by the analyst is given in Example 11.1.

An SSI has a LDC of 21 years with a CPL of 2 years thereafter. FD_{def} is, therefore, set at 14 years, based on LDC/1.5, with a repeat interval of 2 years, based on CPL. For the same element an ED inspection atl 3 years has been deemed appropriate, with an AD inspection interval of 6 years. In this scenario, a survey interval of 3 years would be recommended for the first 14 years to cater for the ED threat after which the FD inspection interval (2 yearly) would become the driver for the inspection. The surveying instructions for the actual inspection will direct the surveyor to report the condition due to the active driver at that time, with concurrent opportunity inspections to look for other modes of structural failure. Figure 11-13 shows this in practice.

Example 11.1 - Identifying Driving Inspection Intervals

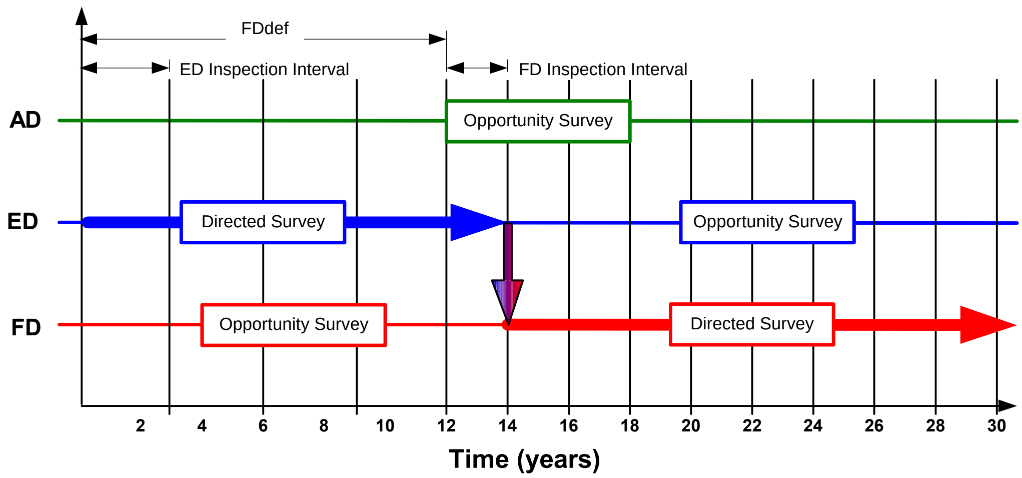


Figure 11.13 - Example Change of Task Interval

f. Figure 1.14 shows the inspection driver and interval selection process.

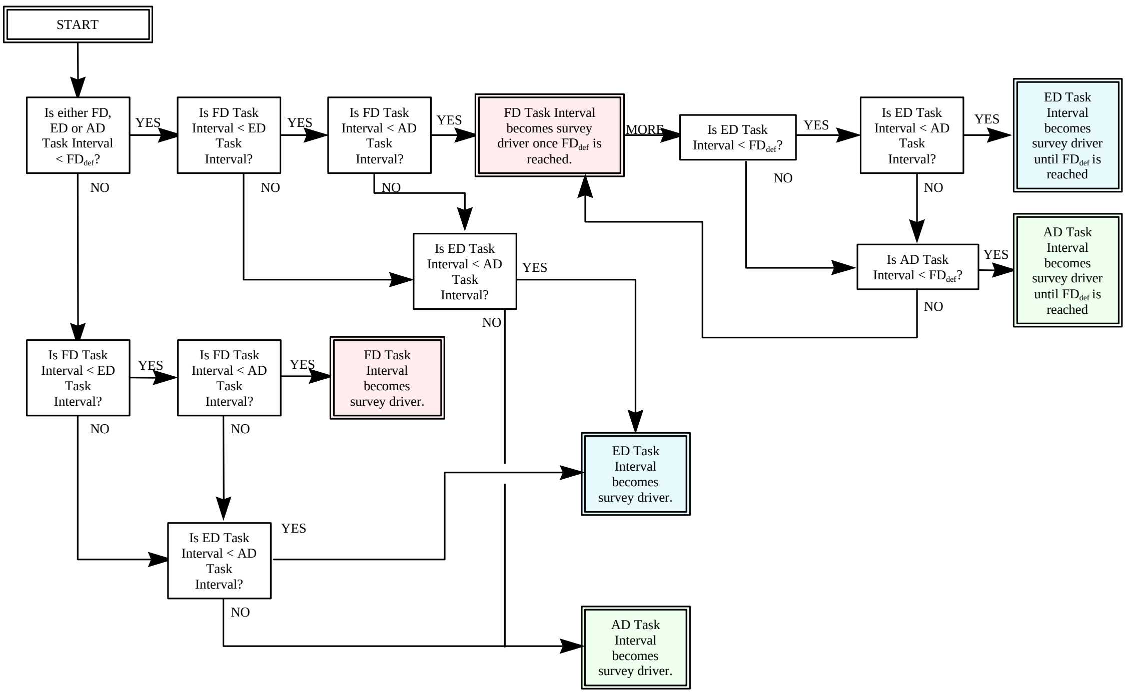


Figure 11.14 - Inspection Driver and Interval Selection

g. Once all of the inspection intervals have been derived it will be found that there will be a plethora of frequencies and in order to form a cohesive and manageable survey schedule some form of packaging or consolidation process will be required. This process is discussed in section 13.

11.7.6 Influence of Build and Design Quality (B&DQ)

a. As a category, B&DQ can influence FD, ED and AD and, although the primary effects will concern FD, each category should be assessed against known build concessions, as shown in Table 11.10. Build quality is generally easier to assess than design quality and can manifest itself in many ways from incorrect materials to misalignment of weldments. This assessment is mainly subjective, based on build quality assurance and other appropriate sources. For the effects on ED, engineering judgements should be made and AE programmes instituted if considered necessary.

B&DQ Assessment	FD Influence	Comment
Major concession or defect outstanding, to be rectified	Unacceptable	AE is to be undertaken until rectification action is completed. For system components, withdraw from service
Major concession or defect accepted	Conditional acceptance	AE is to be undertaken until satisfactory level of confidence is established. For system components, consider withdrawal from service
Minor concession or defect accepted	Unconditional acceptance	Consider AE programme in marginal cases
Within design/build tolerances	Nil	None

Table 11.10 - Build and Design Quality Influence

11.8 Age Exploration Considerations

a. During the application of RCM to structures many decisions made will be based on engineering judgement. This will, in certain circumstances, allow the scheduling of conservative maintenance requirements based on insufficient information. AE provides the analyst with a systematic procedure for collecting the information necessary to reduce or eliminate this gap in knowledge. It is, therefore, a method of data accumulation and analysis, the results of which are input back into the RCM process to validate the need for, or the frequency of, a particular maintenance inspection.

b. AE is explained in section 10 (para 10.8.3) of this standard which shows that the basic principles apply equally when applied to structures as they do when applied to systems. The main considerations are as follows:

1) Initiated for all safety cases where sufficient supporting data was not available.

2) Considered for all non-safety cases where it is cost effective to do so.

3) Collect data for only as long as it takes to gather the information required to give confidence in the outcome of the analysis.

4) Where possible AE must make use of existing support equipment and manpower, without the need for re training or the purchase of special support equipment.

5) Use appropriate samples, sampling techniques and analysis methods to determine required information.

6) Reapply the RCM process using the new information to validate original task and task interval.

c. With structures, AE may form the basis of a formal structural sampling programme which could last for the lifetime of the platform and provide valuable data for future projects or if the life of the vessel is extended beyond the initial projection.

11.9 Applying the RCM Algorithm

a. So far the process has concerned itself with the identification of suitable survey inspection intervals as determined by the ‘driver element’ during analysis. Once this has been achieved it is necessary to select an appropriate survey inspection task which will fulfil the requirements of the structural survey plan. This is accomplished with the limited use of the Marine and Combat Systems Engineering - RCM Decision Algorithm. However, the only appropriate applicable and effective preventive maintenance tasks are the on condition ones, i.e. survey of the selected SSIs, or the hard time discard task for SL structure. The effective use of the algorithm is also constrained by the fact that failures of SSIs are categorised as evident failures possessing safety or mission failure consequences and therefore must be assessed accordingly.

11.9.1 On Condition Tasks

a. On condition tasks are primarily aimed at detection of reduced failure resistance prior to actual functional failure occurring. When applied to structures they are the survey inspections able to detect a crack before it propagates to critical proportions or identify corrosion before the minimum acceptable condition is achieved and hence prevent the consequences of the loss of structural functions. For the on condition task to be applicable and effective the characteristics of the failure process of structure must accord with the criteria required of on condition tasks detailed in section 10. Therefore, for DT structure, the on condition task is capable of detecting FD, ED and AD failure modes because the mechanisms of failure demonstrate measurable deterioration during the failure process. Effectiveness is assured by the ability of the inspection task to reduce the probability of failure to a tolerable level. It follows that ship surveyors need to be adequately trained to ensure that they are capable of detecting and identifying the various failure modes and drawing appropriate conclusions. For SL structure the on condition task is only applicable for ED and AD failure modes since the characteristics of FD to SL structure do not lend themselves to inspection during the failure period, i.e. rapid crack propagation/brittle fracture.

11.9.2 Discard Tasks

a. Discard tasks are applicable only to fatigue damage of SL structure. Because the point of crack initiation and propagation to critical length can happen almost coincidentally, discard tasks are scheduled at a point before the time where there is a high probability that a crack will occur. The role of the discard task is, therefore, not to detect the rate of crack propagation but more to remove the item from service before the age at which a crack is expected to initiate. These tasks must be supported by management system to record the consumption of fatigue life and appropriate on condition tasks for ED and AD failure modes.

11.10 Compiling the Structural Inspection Plan

a. The RCM analysis of structures identifies vast amounts of data and analyses this to determine an effective preventive maintenance programme appropriate to ensure the functional capability of the structure under analysis. This maintenance programme will consist of directed maintenance tasks for the inspection of SSIs at periods determined by the RCM analysis and more general monitoring activities based around the PSP.

11.10.1 Directed Inspections

a. The term ‘directed inspections’ identifies tasks determined by the detailed RCM analysis of SSIs. This process reflects maintenance activities built around items of significant structure that will exhibit early signs of failure and forms the cornerstone of the structural survey plan. The output from the RCM process is essentially a list of recommended maintenance activities and their associated intervals. These intervals are called the ‘natural’ intervals and may, if left unrefined, present the maintenance management authority with an impractical survey programme. Rationalisation of Maintenance Packages is discussed in section 13 of this Standard.

11.10.2 Part of Ship Plan (PSP)

a. The PSP provides for a monitoring process of the whole ship and is able to include significant and non-significant (Other Structure) structure in one general inspection process. This process is not intended to replace the more detailed and specific RCM derived structural surveys of SSIs but is designed to complement and support these. There exists, however, quite distinct reasons for the PSP, as follows:

1) <u>Monitoring</u>. PSP provides a complete monitoring activity which can assess the condition of all items categorised as Other Structure and consequently not analysed as part of the formal RCM process.

2) <u>Identification of Significant Failure Modes</u>. PSP provides the opportunity to identify any unforeseen modes of failure of significant items. For example this may be as a result of operating the platform outside its intended operating environment.

3) <u>Accidental and Environmental Degradation</u>. PSP will provide a monitoring tool able to identify accidental damage and environmental deterioration of all items, including SSIs.

4) <u>Multiple Insignificant Failure Modes</u>. This allows the monitoring of Other Structure to ensure that several seemingly insignificant failures do not, in fact constitute a major hazard.

11.11 Analysis of non Metallic Structural Materials

a. Although non-metallic structural materials such as Glass or Fibre Reinforced Plastics (GRP/FRP) can be analysed using the same RCM process as that used for metallic structures, the nature of the materials may require special consideration. Note that experience to date has been with the GRP structure of the HUNT Class MCMV; ships or structural components fabricated from other non-metallic materials should be considered on their individual merits taking full cognisance of their structural performance and modes of failure. The discussion that follows, therefore, necessarily deals solely with the current experience on GRP as used in the HUNT Class.

b. The three-failure mode approach (FD, ED and AD) is valid for these materials but with differing failure characteristics:

1) <u>FD</u>. For GRP, this is generally agreed by material technologists to be immune from FD during the expected life of the ship. Indeed, the P-F interval associated with GRP is confidently predicted to be in 100’s of years. For the most part, therefore, the RCM analysis will return a non-plausible failure mode and need not be considered further. This decision, if taken, should be fully justified for each SSI.

2) <u>ED</u> - ED failure modes manifest themselves as either osmosis or wicking in those areas of structure either immersed or subjected to frequent wetting:

- Osmosis - Osmosis only occurs in GRP lay-ups completed with a gel coat. Since this method of construction is not used for constructing naval vessels, it can only occur when employed as part of a repair. As such, these individual repair areas should attract a survey dependant upon the results of the analysis. It is known that osmosis will cause structural degradation over a period of time, but the time span to failure is dependent upon many factors and can be difficult to estimate accurately. The consequences of the failure should be assessed and an initial inspection interval established and a formal age exploration programme instituted which can verify or modify the initial interval.

- Wicking - Wicking is precipitated by the exposure of reinforcing fibres to water or moisture. A capillary process is initiated which involves the moisture forcing the individual strands of fibre out of their embedded resin. If the manufacturing process is completed satisfactorily, this process can only be initiated via an AD event. Once initiated, the progression to structural failure is, as with osmosis, very difficult to estimate.

3) <u>AD</u> - By its random nature, AD needs to be handled carefully if the correct conclusions are to be drawn from the analysis. The AD event, in a similar way to metallic structures, can initiate other failure modes that can then lead to structural failure. In the main, the FD failure modes need not be considered, since even in the damaged condition considered by AD analysis, FD is unlikely to be valid. ED failure modes can, however, be initiated which can then cause increasing structural degradation. In these conditions, therefore, the AD inspection interval should be based on the shortest period derived from the ED failure modes considered.

c. All valid GRP defects types are detailed in NES 752 Annex C and their categorisation as structural failure modes are given in Table 11.11.

GRP Defect Type	NES 752 Annex C Reference	Structural Failure Mode	Comments
Abrasion	6.2	AD	May give rise to wicking/osmosis dependent upon threat exposure.
Air Voids in Lay up	6.3	None	Build "original sin" and may be subject to concession. AE programme may be valid. Not normally considered in RCM unless contributory to FD/ED.
Blast Damage (Abrasive)	6.4	AD	Repair action taken after damage incurred will rectify defect. Not considered by RCM.
Cracking (surface)	6.5	AD	May give rise to wicking/osmosis dependent upon threat exposure.
Crazing	6.6	AD	May give rise to wicking/osmosis dependent upon threat exposure.
Debonding	6.7	AD	May have significant strength reduction implications.
Delamination	6.8	AD	Usually "original sin" which becomes evident as a result of an impact.
Fracture	6.9	AD	May give rise to wicking/osmosis dependent upon threat exposure and nature of fracture. May have significant strength reduction implications
Inclusions	6.10	None	Build "original sin" and may be subject to concession. AE programme may be valid.. Not normally considered in RCM unless contributory to FD/ED.
Leaking Fasteners / Fittings	6.11	AD	Usually caused by overtightening of fasteners that may give rise to wicking/osmosis dependent upon threat exposure.
Osmosis	6.12	ED	Only occurs on vessels where a gel coat has been used to effect a repair. P-F interval variable but mainly measured in years.
Pitting	6.13	ED	Only occurs on vessels where a gel coat has been used to effect a repair. May give rise to wicking/osmosis dependent upon threat exposure and depth of pits. P-F interval variable but mainly measured in years.
Root Whitening	6.14	FD/AD	Can be caused by overstressing of a joint or flexure under cyclic loading. The mechanics of failure are the subject of a research programme at the time of writing (Sep 97).
Wicking	6.15	ED	Invariably initiated by an AD event. P-F interval variable but mainly measured in years.
Wrinkling	6.16	None	Build "original sin" and may be subject to concession. AE programme may be valid.. Not normally considered in RCM unless contributory to FD/ED.

Table 11.11 - GRP Defect Types

11.12 Defence Standard 00-60 Correlation

a. LSA Process/Task Applicability:

1) Conduct of an RCM analysis equates to the performance of Defence Standard 00-60 task 301, subtask 2.4.1 and 2.4.2 in that both corrective and preventive tasks are identified.

2) The definition of RCM as detailed within this section satisfies the task input requirements for detailed RCM procedures and logic as called for in task 301, task input 3.2.

3) The task interval as detailed within task 401 subtask 2.1 is generated from this activity.

12. MAINTENANCE SCHEDULES AND JOB INFORMATION CARDS

12.1 Key Players

a. The key players in this part of the RCM process are indicated in **Table** 12.1. All key players should ensure that they understand this section.

Vessel/System Status	
New Construction/Procurement	**In-Service**
Shipbuilder Equipment Manager	Warship IPT Leader
Warship IPT Leader	D Ops E
Study Facilitator	Study Facilitator
Study Technical/Design Advisor	Study Technical/Design Advisor
Study Operator/Maintainer	Study D Ops E Representative
	Study Operator/Maintainer
	HILS(N) RCM Group
Study Technical Secretary	Study Technical Secretary

Table 12.1 - Key Players

12.2 General

a. It is essential that a full set of comprehensive maintenance documentation be generated to cater for the platform and its associated systems throughout the phases of their life (both operational and during any prolonged upkeep periods). The documentation must also reflect the possible variation in the maintenance requirement under differing operating contexts, and during the various mission phases where appropriate. (These issues were discussed in more detail in section 5 and should have been covered by following the subsequent FMECA analyses as described in sections 8, 9 and 10.)

b. The RCM process documented by the RCM (RN) Toolkit will create the following maintenance documentation:

1) Originating and Remedial Job Information Cards (JICs). Note that it is possible to have more than one remedial JIC for a given failure mode.

2) Maintenance Schedules.

3) Maintenance Packages are generated from Task and Task Interval information by the co-ordinating Maintenance Management System, e.g. RCMS.

12.3 Job Information Cards

a. JICs are detailed instructions for carrying out maintenance tasks. The JIC is formatted by the RCM (RN) Toolkit, but the technical content is the responsibility of the appropriate EPM/WPM. BR 1313 Chapter 2 gives general guidance on the level of technical content for a JIC.

b. Bullet Point JICs will be written by the analysis team during the study for the guidance of the JIC author.

c. The finalised JIC will be written by a selected MOD contractor, or by the equipment Design Authority. Information contained in these JICs can be used as input to the MOD Reliability Centred Stockholding (RCS RN) Toolkit to determine the associated stores holdings.

12.4 Production of Bullet Point Job Information Cards

a. RCM analysis project managers and facilitators should use RCM analysis group expertise to the full when producing Bullet Point JICs. To avoid nugatory work, RCM study teams should consolidate maintenance tasks by undertaking task linking within the RCM RN software to remove repeated or common tasks prior to writing JICs. Particular attention should be paid to the following points:

1) The facilitator should delegate the task of verifying or drafting Bullet Point JICs to the appropriate group member/ contractor/ technical author. The full RCM analysis group should always verify Bullet Point JICs.

2) If a JIC already exists which is applicable to a derived task then the technical accuracy of the JIC should be validated by the RCM study team before its entry into the RCM (RN) Toolkit.

3) When defining the JIC for an on condition task then both the on-condition check and the remedial task are to be addressed. The labels “Originating” and “Remedial” should be used in the task details field of the RCM (RN) Toolkit to identify the two parts of the on condition task. Remedial JICs also need to be written for failure modes that have attracted No Scheduled Maintenance. The requirements for the differing JIC types are illustrated by Table 12.2.

	JIC Type	
Task Type	**Originating**	**Remedial**
On Condition	✓	✓
Scheduled Restoration	✓	
Scheduled Discard	✓	
Combination	✓* ✓*	✓* ✓*
Failure Finding	✓	✓
No Scheduled Maintenance		✓

* Dependent upon the selected combination of task types

Table 12.2 - JIC Requirement Matrix

4) The following fields in the RCM (RN) Toolkit JIC input form must be completed to allow subsequent maintenance costing and rationalisation of spare parts to be undertaken.

- Periodicity.
- Skill level, e.g. Mechanic, Technician, etc..
- Man hours.
- Spare gear, identified by NSN where possible.
- Tools required.
- General Naval Stores (consumables), identified by NSN where possible.
- Handbooks.
- Related maintenance, i.e. other tasks that might usefully be undertaken at the same time.

12.5 Finalised Job Information Cards

a. Qualified technical authors shall prepare the finalised JICs for on-condition and remedial activities, based upon the bullet point JICs and risk assessment carried out in accordance with Defence Standard 00-56. The format for finalised JICs is described at Annex H, and examples of finalised JICs for on condition and remedial activities taken from the Hunt Class MCMV, are provided at Figure 12.1 and Figure 12.2 respectively.

12.6 Personnel Safety Assessments

a. The overall MoD ship safety objective is that the levels of risk of accidental death or injury to the crew or other parties and damage to property or the environment due to MoD shipping activities are as low as reasonably practicable. Safety assessments shall be conducted to eliminate or otherwise control as many hazards as possible and prioritise hazards for corrective action.

b. Study group members are unlikely to be experienced in risk assessment procedures and will not, be expected to undertake personnel safety assessments. However, the rationale that is taken by the group to reach appropriate equipment maintenance decisions should be recorded within the analysis comments. The Technical Author preparing the finalised JIC will use the analysis comments and information obtained from the relevant safety case/safety management system to conduct the personnel safety assessment for consolidated tasks.

12.6.1. As Low As Reasonably Practical (ALARP)

a. The ALARP principle means that it is not sufficient merely to meet the safety target for the system if an additional safety margin can be provided at reasonable cost. The application of the ALARP principle in practice requires different definitions of what is "intolerable" to be set for different groups of people at risk. Within the RCM process, the ALARP principle must be applied to the derived maintenance tasks to reduce the risk to personnel carrying out the activity.

b. Formal safety assessments and appropriate control measures to reduce the risk to ALARP levels are to be applied as a mandatory requirement for new ship or equipment designs, or when significant changes are made to existing designs or operating profiles.

12.6.2. Personnel Risk Safety Assessments

a. The technical author must refer to the existing safety case/safety management system to identify any relevant safety analysis tables before carrying out any risk assessment. The tables shown in this section are for illustrative purposes only and have been taken from Defence Standard 00-56. For further details on carrying out risk assessment the reader should refer to the defence standard indicated.

1) Accident Severity Categories The consequences to personnel undertaking each maintenance task should be identified and an Accident Severity allocated in accordance with Table 12.3.

Definition	Description
Catastrophic	Multiple deaths.
Critical	A single death; and/or multiple severe injuries or severe occupational illnesses.
Marginal	A single severe injury or occupational illness; and/or multiple minor injuries or minor occupational injuries.
Negligible	At most a single minor injury or minor occupational illness.

Table 12.3 – Accident Severity Categories

2) Probability Ranges. The probability of occurrence should then be derived for each failure mode using the definitions in Table 12.4.

Accident Frequency	Occurrence *during operational life considering all instances of the asset*
Frequent	Likely to be continually experienced
Probable	Likely to occur often
Occasional	Likely to occur several times
Remote	Likely to occur some time
Improbable	Unlikely, but may exceptionally occur
Incredible	Extremely unlikely that the event will occur at all, given the assumptions recorded about the domain and the system

Table 12.4 - Probability Ranges

3) Risk Classification Scheme. Table 12.5 shows the risk class of each accident severity and probability combination. For the purposes of the accident risk classification scheme, accidents are considered as single events.

	Catastrophic	Critical	Marginal	Negligible
Frequent	A	A	A	B
Probable	A	A	B	C
Occasional	A	B	C	C
Remote	B	C	C	D
Improbable	C	C	D	D
Incredible	C	D	D	D

Table 12.5 – Risk Classification Scheme

4) Risk Classification. Table 12.6 shows the four risk class definitions.

Risk Class	Interpretation
Class A	Intolerable
Class B	Undesirable, and shall only be accepted when risk reduction is impracticable
Class C	Tolerable with the endorsement of the Project Safety Review Committee
Class D	Tolerable with the endorsement of the normal project reviews

Table 12.6 – Risk Classification

5) Hazards that are identified as Risk Class A are unacceptable. Class A risks shall be reduced by the use of safety features and the system should be subject to further analysis to confirm the reduction in risk classification. Further reduction measures that are considered but rejected should also be clearly recorded to demonstrate that the ALARP principle has been adequately applied. Safety features are described in Section 12.6.3.

6) The results of the risk assessment must be reviewed by the associated PM to enable changes to be recorded within the hazard log and/or the safety case.

12.6.3. Risk/Hazard Control Strategy

a. The personnel safety assessment process does not end with the estimation of the risk category above. In many instances, the personal safety assessment for the proposed maintenance task may involve the recommendation for a redesign task to either reduce the accident severity or extend the accident frequency. Any redesign requirement must accord with the risk estimation from which it was identified.

b. Proposed redesign actions affecting personnel will tend to fall into 3 categories, which need to be clearly defined by the team:

1) Prevention.

2) Control.

3) Remedial.

4) Table 12.7 illustrates some typical redesign mitigation actions covering all 3 categories, and should be referred to during the analysis.

Category	Sub Category	Illustrative Measure
Prevention	Eliminate the hazard	Provide alternative equipment Avoid/retime the task
	Reduce the risk by substitution	Provide substitute for hazardous substances Change to solvent base (water etc)
	Reduce the Demand Requirement	Reduce power (volts, amps, HP) Reduce pressure
	Isolate the danger	Segregate completely Provide barriers / guarding Consider remote operation Enclose equipment (full or partial)
Control	Safety redesign	Provide additional safety measures – silencers/ducts/local exhaust ventilation Personal - work permit/inform/instruct/train/supervise/monitor Medical - examination/inoculation Exposure - job rotation/reduce number of people exposed
	Provide additional personnel Protective equipment	Last resort/temporary measure/emergency use
	Personnel discipline	Approved procedures/rules Self discipline
Remedial	Post event containment	Consider evacuation procedures Plan medical response/first aid equipment.

Table 12.7 - Redesign Action Categories affecting Personnel

12.7 Maintenance Schedules

a. Maintenance schedules are a collection of maintenance tasks to be carried out a specified interval by personnel with specified skills. Maintenance schedules should be rationalised in accordance with section 13 prior to inclusion in the overall maintenance strategy.

b. Sample Maintenance Schedules, taken from the Hunt Class MCMV, are provided at the Figures at the end of the section.

12.8 Defence Standard 00-60 Correlation

a. General.

1) Within Defence Standard 00-60 there are two discrete areas that are concerned with technical documentation: the LSAR as defined by Part 0, and the Common Source Database (CSDB) as defined by Part 10. The level and scope of data recorded within a JIC can be adequately stored within the LSAR using the task, subtask, personnel, provisioned item and support equipment data elements and their associated relationships. Migration from the LSAR into the CSDB can be achieved on a largely automated basis.

2) The use of the CSDB requires that a formal Standard Numbering System (SNS) be used, with a predefined SNS for Ship System being provided within Part 10 of Defence Standard 00-60. This SNS is however not currently used within the RCM IT for RCM studies, nor is an equivalent SNS in use.

3) The future utilisation of the CSDB is currently being investigated by HILS(N). Revisions to the method of documenting maintenance tasks will be undertaken at a later date if necessary.

b. LSA Process/Task Applicability.

1) The preparation of maintenance documentation, including JICs, equates to the Defence Standard 00-60 task 401 subtask 2.1 (task analysis).

Job Information Card
RCM Reference: **To Provide Steering Capability**

Project Code		**LCN** MCMA113	
Platform Type	Hunt M		**CMV** FAG/SSI
NBCD Code		**PMS**	**MOP No.** 0

Ref. Task Description

1B4 Carry out rudder rate trials and record results.

To be done by Technician **Time to do task**

RELEVANT HANDBOOK **ISSUE**

BR6509 (108), BR 2000(20)

RELATED MAINTENANCE

NAVAL STORES

Spare Gear List on remedial task JIC

SAFETY PRECAUTIONS

1. BEFORE THE STEERING GEAR IS OPERATED IN ANY MODE, ENSURE THAT ALL MOVING PARTS ARE FREE FROM OBSTRUCTION.

TASK DETAILS AND JOB DESCRIPTION

2. With the ship proceeding at maximum speed, carry out the following checks on overall performance standards.

3. Measure and record the mean rate of rudder operation rudder rate from 35^{o} Port to 30^{o} Stbd, and then from 35^{o} Stbd to 30^{o} Port. Both tests should be carried out with the steering gear pumps under the following operational modes:

   a. With both steering pumps in operation.

   b. With port steering pump only in operation.

   c. With Stbd steering pump only in operation.

4. The performance standards that must be achieved are:

   a. With 2 pumps running, the time taken to travel through each range should not exceed 11 seconds.

   b. With 1 pump running, the time taken to travel through each range should not exceed 22 seconds.

Figure 12.1 - Example of an On Condition JIC

Remedial Job Information Card
RCM Reference: **To Provide Steering Capability**

Project Code		**LCN**	MCMA113		
Platform Type	Hunt MCMV	**FAG/SSI**			
NBCD Code		**PMS**		**MOP No.**	0

Ref. Remedial Task Description

1B4 RENEW main steering pump.

To be done by Technician **Time to do task** 8 hours

RELEVANT HANDBOOK **ISSUE**

BR6509 (108), BR6570 (028), BR 2000(53), BR 2000(20)

RELATED MAINTENANCE

NAVAL STORES

SPARE GEAR

Part Number	**Part Description**	**Quantity**
D900-995581130	Pump, Radial Piston	1

SAFETY PRECAUTIONS

1. ENSURE STEERING GEAR PUMP MOTORS ARE STOPPED AND ELECTRICAL SUPPLIES ARE ISOLATED.
2. SECURE WARNING NOTICES ON THE STEERING PUMP MOTOR STARTER BOXES.
3. ENSURE THAT ALL PORTABLE/FIXED LIFTING EQUIPMENT IS IN DATE FOR TEST BEFORE USING AS PART OF REMOVAL PROCEDURE.
4. OBSERVE THE STANDARDS AND PRACTICES LAID DOWN IN BR 2000(53)(2) WHEN CARRYING OUT WORK ON HYDRAULIC SYSTEM.
5. BEFORE THE STEERING GEAR IS OPERATED IN ANY MODE, ENSURE THAT ALL MOVING PARTS ARE FREE FROM OBSTRUCTION.

Figure 12.2 - Example of a Remedial JIC (Introduction)

Remedial Job Information Card RCM Reference: **To Provide Steering Capability**	SSA Ships Support Agency

Project Code		**LCN**	MCMA113		
Platform Type	Hunt MCMV	**FAG/SSI**			
NBCD Code		**PMS**		**MOP No.**	0

TASK DETAILS AND JOB DESCRIPTION

Maximum cleanliness is essential to prevent the ingress of dirt into the system. When breaking hydraulic pipe connections, it is essential that all exposed orifices are sealed.
Main pumps will be supplied filled with preserving oil. They are to be kept filled until required for installation, when they should be flushed and pressure tested. The preserving oil is to be drained and replaced with OM33 oil using an approved flushing/filtering test rig until the level of contamination in a 100ml sample taken from the effluent is not greater than RN Class 15000.
On receipt of a pump from stores, ensure that it is correctly "handed", i.e. the pump control spindles are fitted to the correct side before being installed. If necessary, it may be required to change over the control and dummy spindles ensuring that the spindles with the guide blocks and floating ring move freely before putting the pump into service.

1. Ensure electrical supplies to the relevant steering motor are isolated at source and warning notices secured in position (TAG OUT).
2. Release and remove drive belt guard. Remove drive belt.
3. Release and remove pump shaft support bracket.
4. Disconnect hydraulic pipework from the pump and seal off all orifices.
5. Release and remove the pump holding down bolts.
6. Disconnect pump spring control gear at control spindles, slide pump away from the control gear and using suitable lifting gear, lift pump clear.
7. Lift new pump into position and locate and secure control spindle in spring control gear.
8. Install holding down bolts but do not tighten.
9. Remove seals from exposed orifices and reconnect hydraulic pipework.
10. Reinstall the pump shaft support bracket.
11. Check alignment of pump and motor pulleys using straight edge across the pulley faces. Adjust pump position to ensure correct alignment and then tighten holding down bolts.

Figure 12.3 - Example of a Remedial JIC (Task Details)

Review of Maintenance

RCM RN Maintenance Schedule Page 1

SSA Ships Support Agency

RCM Reference: Pressure Level Monitoring & HP Air Distribution

Project Code **LCN** A12134

Platform Type Hunt MCMV **FAG/SSI** 4

Ref.	Task	To be done by	Interval
1C1	Renew 3 stage BA filter elements.	Technician	6 months
1D1	Renew 3-stage BA filter carbon medium	Technician	6 months
5D8	Test Deltic air start, engine room LP or generator room LP high pressure alarm circuit using portable test equipment.	Technician	6 months
5D9	Test Deltic air start, engine room LP or generator room LP high pressure alarm circuit using portable test equipment.	Technician	6 months
5D10	Test Deltic air start, engine room LP or generator room LP high pressure alarm circuit using portable test equipment.	Technician	6 months
5D14	Check Deltic air start low-pressure sensor circuit operation at ship shutdown.	Technician	6 months
5D15	Check Deltic air start low-pressure sensor circuit operation at ship shutdown.	Technician	6 months
5D16	Check Deltic air start low-pressure sensor circuit operation at ship shutdown.	Technician	6 months
5D19	Check SSD clutch pressure low alarm sensor circuit operation at ship shutdown.	Technician	6 months
5D20	Check SSD clutch pressure low alarm sensor circuit operation at ship shutdown.	Technician	6 months
5D21	Check SSD clutch pressure low alarm sensor circuit operation at ship shutdown.	Technician	6 months
5D24	Check Generator air start pressure low alarm sensor circuit operation at ship shutdown.	Technician	6 months
5D25	Check Generator air start pressure low alarm sensor circuit operation at ship shutdown.	Technician	6 months

Figure 12.4 - Example of a Maintenance Schedule

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TASK RATIONALISATION AND DERIVATION OF THE PLATFORM UPKEEP CYCLE

13.1 Key Players

a. The key players in this part of the RCM process are indicated in Table 13.1. All key players should ensure that they understand this section.

Vessel/System Status	
New Construction/Procurement	**In-Service**
Shipbuilder Platform Manager	HILS(N) RCM Group
Shipbuilder Equipment Managers	Warship IPT Leader
	D Ops E
Equipment Commodity Manager	Equipment Commodity Manager
Warship IPT Leader	
Warship Type Commander	Warship Type Commander
Squadron Staff	Squadron Staff
CFM Warship Manager	CFM Warship Manager

Table 13.1 - Key Players

13.2 Introduction

a. The task rationalisation phase of the generation of a platform upkeep cycle removes task duplications within individual studies and optimises their maintenance schedules to form the platform level maintenance schedule. This process is conducted by following strict rules to ensure that any changes made to individual analysis results do not compromise platform or personnel safety or affect the environment, or cause an unacceptable degradation to platform operational requirements. When this process is applied to new procurements it is normal practice that the information generated is fed back into the Use Study, and this may result in changes to the design.

13.3 Rationalisation of Maintenance Schedules

13.3.1 Maintenance Task Category Grouping

a. The following work is to be undertaken prior to commencing the rationalisation process:

b. Each task is to be categorised at its appropriate level of maintenance. Unless otherwise directed within the contractual documentation of a specific project, or other documentation such as the ILS Plan or Use Study, the following maintenance categories are to be used:

1) Cat A: Can be undertaken at sea (Ship's Staff).

2) Cat B: Tasks requiring the use of external resources other than dry docks or shiplift, e.g. craneage (Ship's Staff, Base Staff, Contractor).

3) Cat C: Tasks requiring a dry dock or shiplift facility. These tasks are candidates for UWE procedures.

c. It should be noted that these categories are solely for use in RCM analyses and do not correlate with "Lines" or "Levels" of maintenance as defined in Defence Standards 00-49 or 00-60.

d. Tasks for each category of maintenance are to be grouped together to ensure that the rationalisation process captures relevant tasks. Two of the aims of an RCM upkeep strategy are to optimise asset availability and minimise maintenance costs. In the case of the RN, the asset is a warship platform. RCM objectives are therefore to reduce platform downtime alongside for maintenance purposes, and to reduce the requirement for external support, subject to operational and personnel loading considerations. Each task should therefore be categorised by technical suitability, with Cat C tasks kept to a minimum.

13.3.2 Rationalisation Rules

a. The rationalisation process can only be applied to Cat B and C tasks.

b. The natural task intervals produced by an RCM analysis are based upon P-F intervals, equipment life and calculations of failure finding intervals, or, in certain instances, combinations of the above. Task periodicities within and between RCM studies do not, therefore, automatically align with the accepted calendar-based intervals of current maintenance planning systems. The number and spread of these natural RCM task intervals would be extremely cumbersome, if not impossible, to manage individually on a platform basis and would lead to unacceptable losses in platform and system availability if they were not to be rationalised. It is necessary, therefore, to align these task intervals into a common schedule that provides a safe maintenance strategy, with minimum platform or equipment downtime. Moving natural task intervals to the left increases cost, moving them to the right increases risk. This is illustrated in Figure 13.1

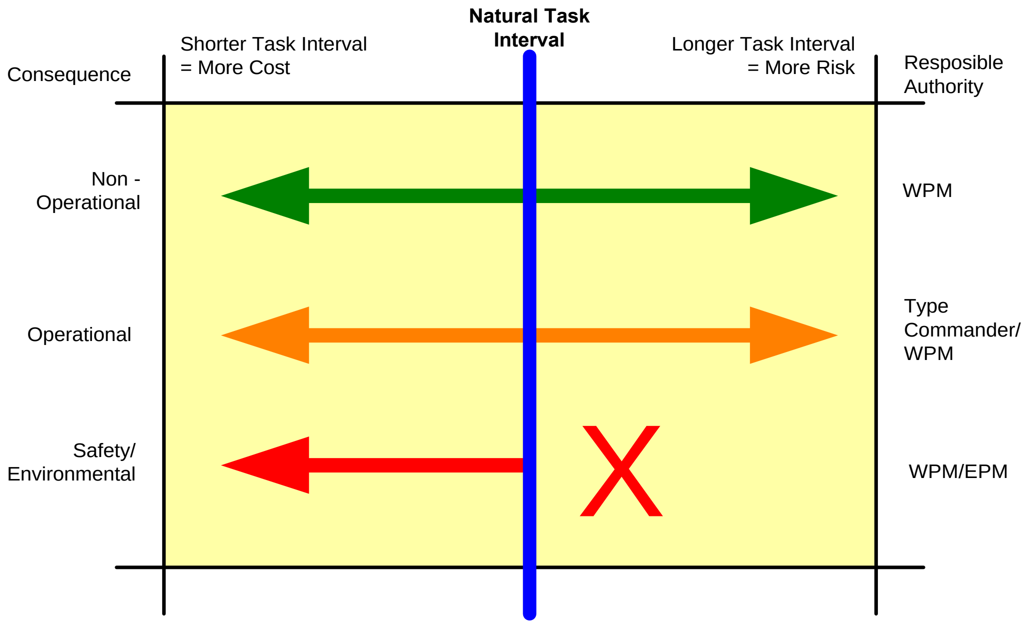

Figure 13.1 - Risk vs Cost Considerations

c. Rationalisation is achieved by converting individually derived task periodicities to a common time base, and then aligning their periodicities to achieve the optimum upkeep cycle. For new procurements, platform availability requirement is defined in platform Staff Targets or Staff Requirements are to be considered initially, but are likely to be subject to change proposals arising from the RCM process. Also, when selecting a suitable common measurement base care must be taken to ensure that the Defence Standard 00-60 LSAR data element definitions are adhered to. In particular the common measurement base must correspond to one of the defined task interval codes of Defence Standard 00-60.

d. Rules. Task periodicity adjustments can only be undertaken within the limitations of the following rules (see Figure 13.1):

1) Scheduled tasks with safety/environmental related consequences requiring base support are to be considered first.

2) The natural task intervals as derived by the study groups are to be used in the first instance.

3) Tasks with safety/environmental consequences can only be moved to the left, (i.e. reduced periodicity), to ensure that safety is not compromised. Changes are to be sanctioned by the WPM and/or EPM.

4) Tasks with Operational consequences can be moved to the left with WPM approval, but only to the right (i.e. extended periodicity) by agreement with the Type Commander. The involvement of the Type Commander is essential to ensure that any changes are compatible with the engineering risk associated with the conduct of operational tasking.

5) Tasks with Non-Operational consequences can be moved to the left or right subject to WPM approval.

6) Task analyses for any task interval that is changed may need to be revisited to revalidate the risk.

13.4 Individual Study Rationalisation

a. The rationalisation process is therefore initially constructed around those Category B and C scheduled maintenance tasks intended to avoid failures with safety or environmental consequences. Operational and Non-operational items are overlaid to check for significant mismatches, as shown in Figure 13.2. However, it may not be possible to rationalise some tasks within any of the Safety or Environmental, Operational and Non-operational categories. In these instances, tasks will either have to be addressed at their natural RCM interval or may need to be revisited to investigate opportunities for changes to their periodicities.

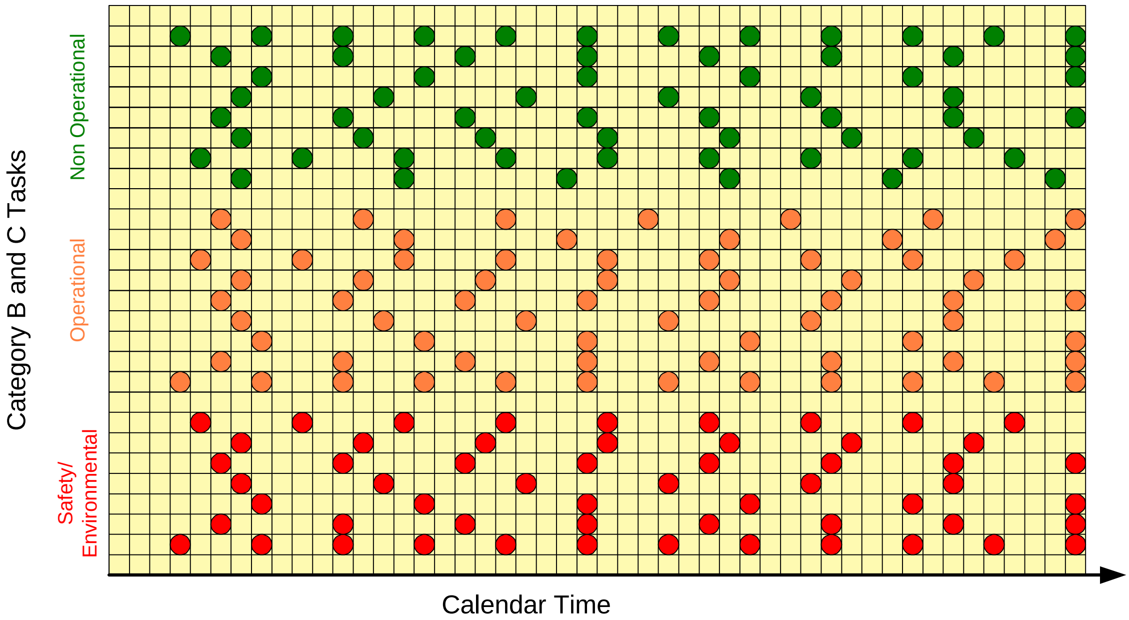

Figure 13.2 - Tasks at Natural Intervals

13.5 Upkeep Cycle Derivation

a. Once the maintenance schedules of the individual analysis studies have been rationalised, the overall upkeep strategy for the vessel can be derived. This process utilises all the information generated for the platform during RCM analyses and requires access to the entire RCM (RN) Toolkit, which is currently only available to the RN RCMIT.

b. The platform process is a development of that applied to individual analyses. Using the Rationalisation module within the RCM RN software, Category B and C tasks are first overlaid on a common timeframe to look for the cycle drivers that require the vessel to be alongside. The process also looks for tasks dependent on facilities external to the vessel, such as dry docks and specialist manpower requirements. This could be for tasks involving base support assistance, for ships staff tasks, or for a combination of the two. However, a dry dock or shiplift should be regarded as a support facility, rather than a cycle driver in its own right. The docking should be integrated with other more frequent cyclic alongside activities. Building on the task rationalisation process already completed, a ship upkeep strategy is developed that will ensure that the required facilities and level of effort necessary to maintain the material of the vessel are made available throughout the life of the vessel. The task frequencies post-rationalisation are shown in Figure 13.3.

c. The above process will define the frequency of the RCM upkeep periods, but not their duration. Only schedulable tasks can be planned, and they will generate the initial duration. However, RCM derives relatively few schedulable tasks. Upkeep period duration could well be driven by the size of the remedial tasks that the on condition tasks have generated, or by the defect rectification work arising for equipments operating under a No Scheduled Maintenance policy. Feedback obtained by operating the class under the RCM upkeep cycle will allow the initial upkeep period duration estimates to be refined.

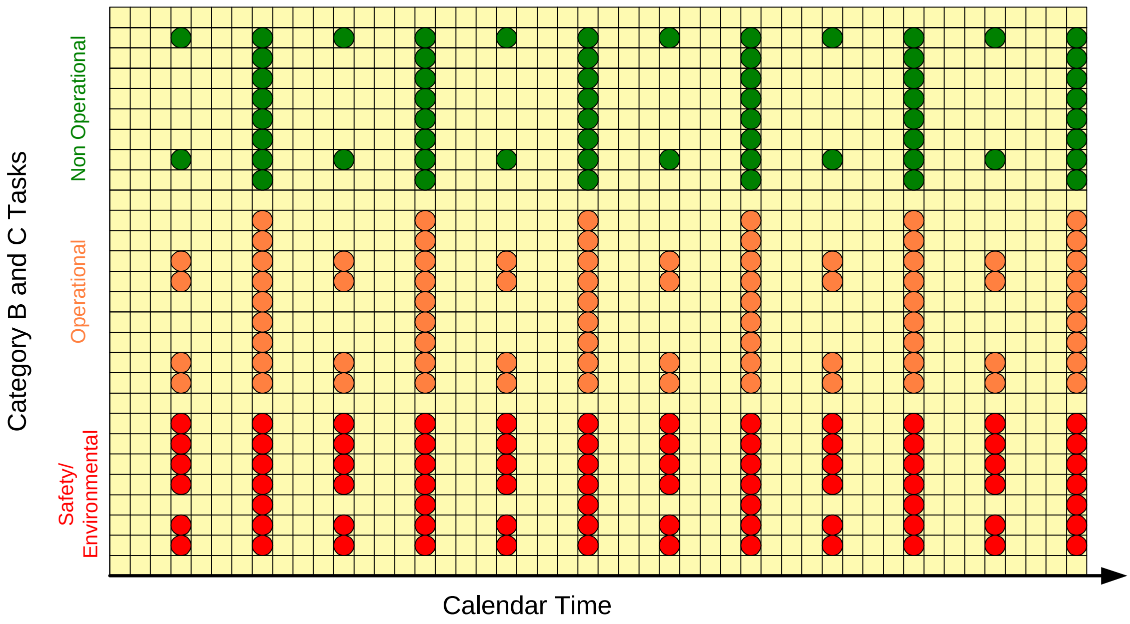

Figure 13.3 - The Upkeep Cycle (Grouped Rationalised Tasks)

d. The above process describes the formulation of the ideal RCM upkeep cycle for the platform. However, external factors outside RCM, such as manpower resource limitations, availability of shoreside assets, changing operational requirements, etc., will all influence, if not dominate, the upkeep cycle for a vessel at certain times. RCM is a dynamic process and can manage such changes safely by giving ship managers visibility of the risks inherent in deferring maintenance.

13.6 Recording Rationalised Task Intervals

a. Following completion of the rationalisation process for an individual study the rationalised task intervals for equipments should be recorded in the RCM (RN) Toolkit alongside the derived natural interval. By recording both intervals a full audit trail is maintained. Finally, on completion of the platform rationalisation process, any further changes are to be recorded.

13.7 Spares

a. For the proposed maintenance schedules to be viable, it is essential that the spares that are to support the identified maintenance tasks are available in the appropriate timescales. On completion of the RCM analysis, the spares requirement will have been determined using the RCS (RN) Toolkit (see section 14 and Annex D) to determine the optimum spares holdings necessary to deliver the required Operational Availability. It may well be that such a pass through the stores analysis process will make recommendations for changes to the upkeep strategy for systems that could affect the total platform. Several passes may be necessary before the optimum, viable platform upkeep strategy is derived.

13.8 Management of Royal Navy Vessels within an RCM Derived Upkeep Cycle

a. Once the initial upkeep cycle has been derived, it will need to be reviewed periodically to take account of the maintenance data feedback acquired when running the class on RCM, and also the requirement for capability updates.

1) RCMS. Feedback on the performance of the derived RCM maintenance schedules will be acquired from the data collected by the Reliability Centred Maintenance System, (RCMS). The success of derived P-F intervals in deriving on-condition monitoring periodicities, and the results of age explorations in increasing equipment life, should be constantly reviewed. Analyses should be revisited as necessary, and maintenance schedules changed accordingly.

2) Alterations and Additions (As & As). All major new systems and equipments will in future be subject to RCM analysis on introduction into service. The requirement for these platform capability updates could well drive the platform upkeep cycle if they cannot be contained within the designated RCM platform upkeep periods. If possible, the actual A&A installation should be undertaken on a progressive basis. However, it may at some stage be necessary to break the upkeep cycle to complete a specific installation. In such a case the

worth of the platform capability update will need to be weighed carefully against the loss in platform availability.

3) Capability Enhancements and Mission Sensitive Equipment Fits. For equipment fitted on vessels as a temporary measure, the specific maintenance can only be integrated into the host vessel's maintenance management system if it has its own RCM derived maintenance schedules. If this is not the case, the equipments generic maintenance will need to be handled on an ad-hoc basis.

4) Equipment Modifications (Mods). By definition, Mods are undertaken on equipments only, and should not involve ship wiring or ship systems. Smaller Mods are in general undertaken by ship's staff, and larger Mods by contractors. If possible, large Mod packages should be undertaken on a progressive basis, so as not to interrupt the RCM platform upkeep cycle. Any Mod package that significantly alters the operating context or performance parameters of a system should result in revisitation of the FMECA and RCM analysis.

13.9 Defence Standard 00-60 Correlation

a. LSA Process/Task Applicability

1) The development of the task inventory equates to the performance of Defence Standard 00-60 task 301, subtask 2.4. Although this subtask does not explicitly cover the rationalisation of common tasks, details of each identified task stored within a Defence Standard 00-60 LSAR must be allocated a unique task code. This ensures that identical tasks will be allocated the same basic task code with only a task sequence code used to differentiate between them. The task-coding regime would otherwise cause a duplicated task code.

2) Activities conducted under Task 303 (Evaluation of Alternatives and Trade-off Analysis) will require at least an outline maintenance schedule to be developed. This may be documented as a task inventory (Table CK) within a Defence Standard 00-60 LSAR, or as a stand-alone report.

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14. SPARES HOLDING DETERMINATION

14.1 Key Players

a. The key players in the spares determination part of the RCM process are indicated in Table 14.1. All key players should ensure that they understand this section.

Vessel/System Status	
New Construction/Procurement	**In-Service**
Shipbuilder ILS Manager	HILS(N) RCM Group
Warship IPT Leader	Warship IPT Leader
D Ops E	D Ops E
Equipment Commodity Manager	Equipment Commodity Manager
MoD ILS Manager	
Study Facilitator	Study Facilitator
Study Technical/Design Advisor(s)	Study Technical/Design Advisor(s)
	Study Operator Maintainer
Study Technical Secretary	Study Technical Secretary

Table 14.1 - Key Players

14.2 General

a. For the proposed maintenance schedules identified within section 13 to be viable, it is essential that the on board spares required to support the identified maintenance tasks can be made available in the appropriate timescales. On completion of the RCM analysis, therefore, the maintenance spares requirement is to be subject to a Reliability-Centred Stockholding (RCS) analysis to determine the optimum on board spares holdings necessary to deliver the required Operational Availability. The Stockout Consequence Decision Algorithm is illustrated at Figure 14.1. This can be achieved using the RCS RN Toolkit detailed at Annex D.

b. It is to be expected that a first pass through the stores analysis process may suggest modification to the proposed Upkeep Strategy before an optimum solution can be achieved. The process of developing a complete upkeep strategy including maintenance policy and spares support is therefore an iterative one.

c. During procurement, the process will be performed by equipment or platform manufacturers, supported by MOD(N) personnel, as part of the design process where predicted or estimated utilisation data can be used. Later analyses, particularly for in-service equipments, will be performed by MOD(N) to ensure a fleet wide optimisation in support of the fleet wide upkeep strategy, and using measured data where available.

14.3 The RCS Process

a. The RCS RN Toolkit has been specifically developed to complement the NES 45 RCM process. It supports the following activities:

1) Reading in of the parts needed to undertake tasks to correct each failure mode identified within the RCM study, including parts required for remedial work to correct 'on condition', failure finding, and 'no scheduled maintenance' failures. This can be automatically achieved where the RCM study has been documented using the RCM RN toolkit.

2) Documenting the effects of a stockout of a particular part during a mission.

3) Determination of whether usage of each part can be anticipated. In a forthcoming revision the toolkit will also allow a list of such parts to be compiled. At present this should be done manually.

4) Where usage cannot be anticipated, determination of the number of spares which must be held to achieve the desired operational availability, and their position in the logistics chain.

5) Revision of the RCM outcome to determine alternative maintenance or design strategies where it is not possible to achieve the desired operational availability simply by adjusting the stockholding of spares.

b. The process of determining the necessary spares holding is fully described within the RCS RN Toolkit User Guide that accompanies the software package. Although the use of the RCS RN Toolkit is not prescribed, organisations conducting spares ranging and scaling activities by other means must ensure the processes utilised yield results suitable to replace those provided by the RCS RN Toolkit.

c. A full set of example RCS RN Toolkit output reports are contained within the User Guide.

14.4 Defence Standard 00-60 Correlation

a. General.

1) Within Defence Standard 00-60 there are two discrete areas that are concerned with spares support:

2) The LSAR as defined by Part 0.

3) The Initial Provisioning List (IPL) as defined by Part 20 and AECMA S2000M.

4) The level and scope of data recorded within the RCS RN Toolkit can be adequately stored within the LSAR using the Packaging and Provisioning Requirements series of tables (H series) and their associated relationships. Migration from the LSAR into the IPL can be achieved on a largely automated basis with most of the required spares related information being extracted as predefined LSAR Reports.

5) The future utilisation of the IPL is currently being investigated by HILS(N). Revisions to the method of providing spares support will be undertaken at a later date if necessary.

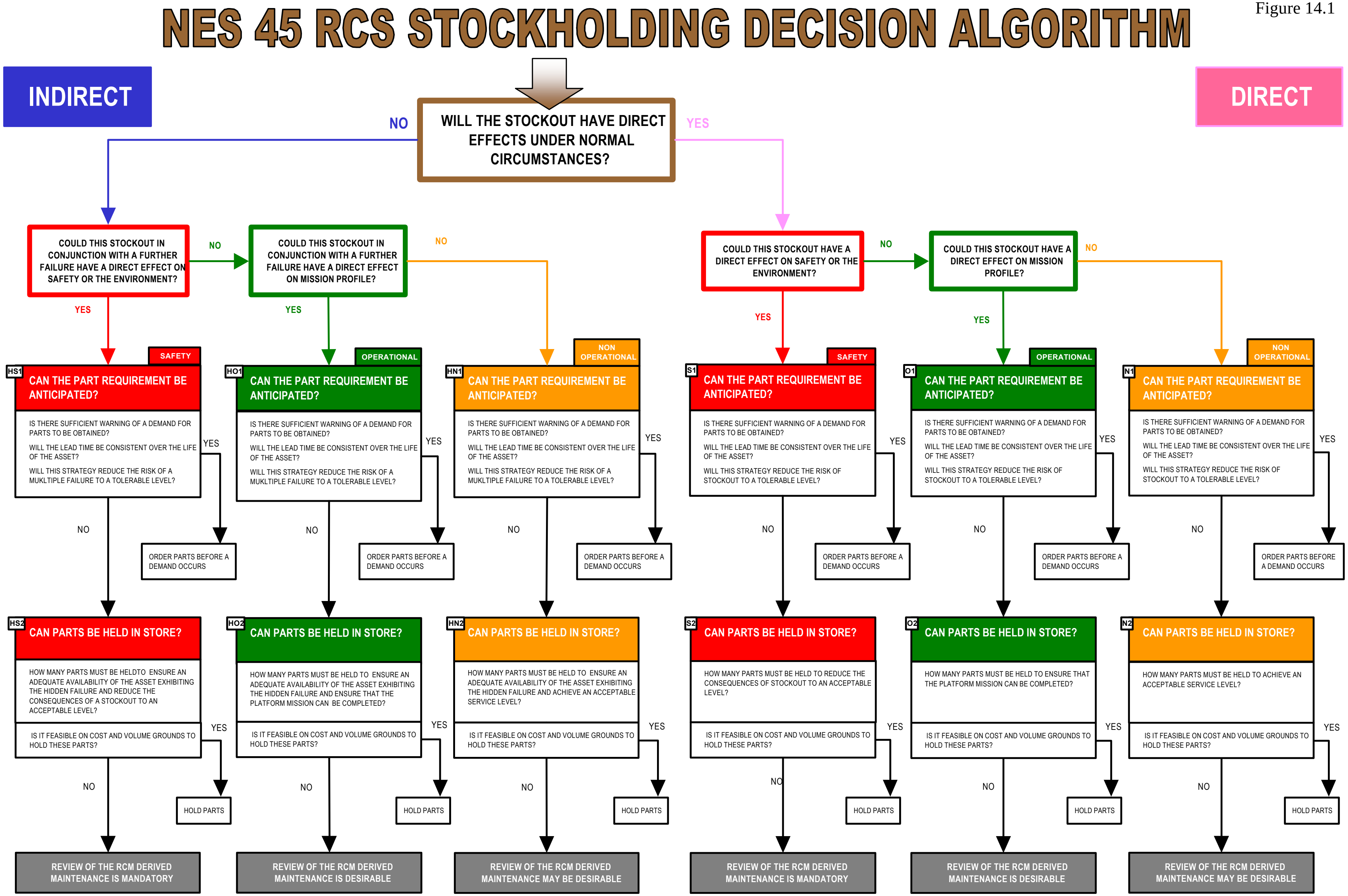
b. LSA Task Applicability.

1) The conduct of spares holding determination equates to the Defence Standard 00-60 task 401 subtask 2.8 (provisioning technical documentation).

14.3

Figure 14.1 - Reliability Centred Stockholding Decision Diagram

Figure 14.1



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15. AUDITING

15.1 Key Players

a. The key players in the RCM auditing process are indicated in Table 15.1. All key players should ensure that they understand this section.

Key Roles	
New Construction/Procurement	**In-Service**
Shipbuilder ILS Manager	HILS(N) RCM Group
Warship IPT Leader	Warship IPT Leader
D Ops E	D Ops E
Independent RCM Auditor	Independent RCM Auditor
Facilitator	Facilitator

Table 15.1. Key Players

15.2 Purpose

a. The purpose of the RCM analysis audit process is to ensure that:

1) An analysis is compliant in all respects with NES 45.

2) The analysis provides an audit trail to future users, providing a record of the logic, assumptions, study analysis boundaries, and recommendations that were used by the original study group.

3) The technical decisions made by the study team are appropriate.

b. The audit process is illustrated at Figure 15.1.

15.3 Audit Definitions

a. The following definitions are used in the audit process:

1) Deficiency. A documentation entry in the audited RCM study that has been omitted, or does not comply with the requirements of NES 45.

2) Corrective Action Recommendation (CAR). An action placed on an auditee during the course of an audit to rectify a Deficiency.

15.4 Audit Guidelines

a. Initial, Continuation and Completion Audits, together with formal reviews and acceptance procedures for the study JICs, are to be conducted in accordance with the programme detailed in Table 15.2.

Audit Level Entry Point	Type of Audit	Conducted By
Analysis Details Analysis Operating Context Analysis Comment Block Diagram	Initial Audit	Study Auditor
FMECA Decision Worksheet 1 Decision Worksheet 2	Continuation Audit	Study Auditor
Bullet Point Originating JICs Bullet Point Remedial JICs	Completion Audit	Study Auditor EPM Section
Draft Originating JICs	RCM Review	SSA Project Leader PE Equipment Section
Draft Remedial JICs	Technical Review	EPM Section PE Equipment Section
Completed Study Final Originating JICs Final Remedial JICs	Formal Acceptance	EPM Section Head PE Section Head

Table 15.2. RCM Analysis Audit Programme

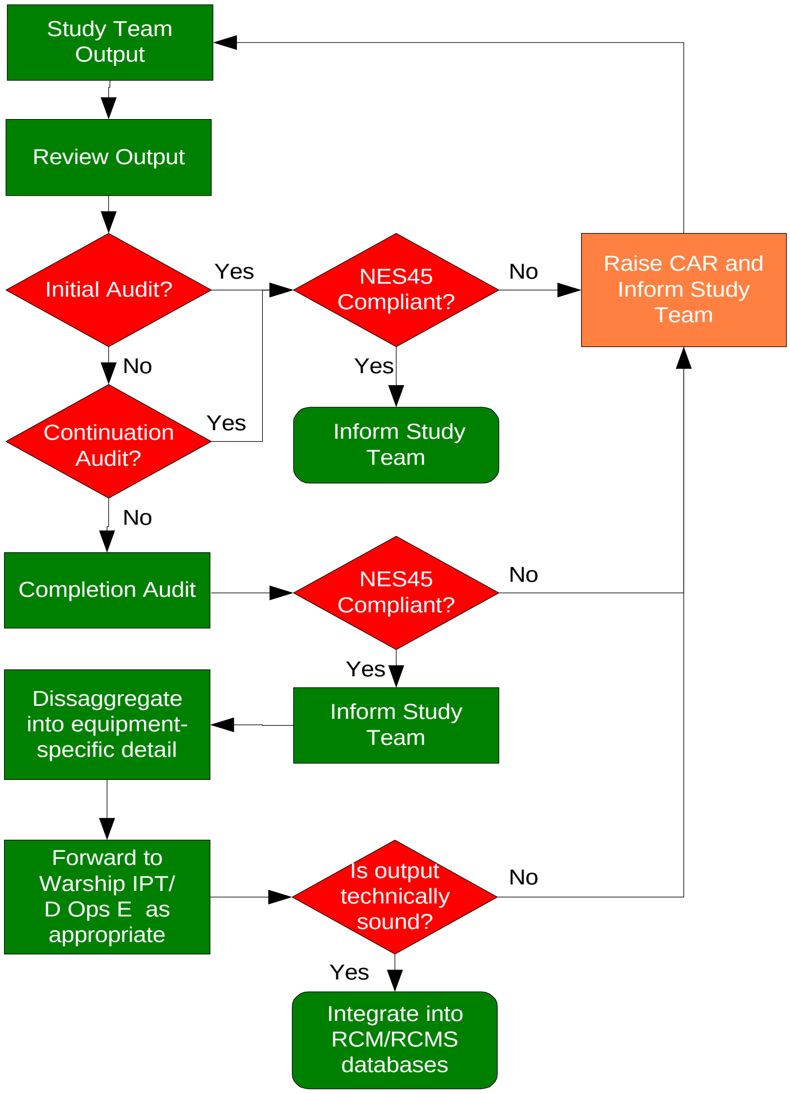

Figure 15.1 - The RCM/Technical Audit Process

b. The progress of the each Audit shall be recorded within the study using the RCM (RN) analysis audit page. The record is to include the audit date, audit stage, and name and qualification of the auditor. A report is also to be included detailing any observed deficiencies.

c. Audits are to be conducted by personnel, trained to or experienced at Facilitator or Practitioner level, who have had no direct involvement in the study presented for audit. Wherever possible, the same auditor shall be employed at each audit stage.

d. An audit report shall be created at the start of each audit. The report is to continue through each entry point until all CARs have been completed. No deletions of CARs shall be made. If the recommendations are not implemented a statement of the reason shall be added to the CAR. CAR completions are to be tracked by the project leader and hastened as necessary.

e. The completed audit report is to be presented to the SSA or PE project leader at a meeting attended by the RCM study contractor (if applicable), study facilitator and group members, as decided by the project leader.

f. The completed report, together with any correspondence pertinent to the audit raised by the Project Leader, shall be filed in the applicable SSA or PE project/PE contractor pack.

g. Audit reports shall be held on file until acceptance of the complete platform RCM maintenance package into service. Major amendments made after acceptance of a specific analysis need not be subject to audit, but a record of change is to be added to the analysis audit page within the software.

h. An illustrative audit check sheet is shown at Table 15.3 at the end of the section.

15.5 Defence Standard 00-60 Correlation

a. LSA Task Applicability

1) This section has no correlation with any LSA process. It provides guidance on procedures for auditing RCM analyses.

2) The contents of this section are therefore advisory in nature.

DOCUMENT	NES 45 Section	CHECK REQUIRED	CORRECTIVE ACTION RECOMMENDATION	COMPLETED
		Top Level		
Analysis Details	7	Completed correct to stage		
Analysis Operating Context	5	Is there an operating context in the database for the FAG being undertaken? Does it adequately determine what the operators want it to achieve? (Note that a brief description of the Equipment is acceptable, but the prime consideration here is to gain the understanding of how it will be used by the Operators) Is there a Block diagram or amended system schematic diagram attached (non software)?		
Analysis Operating Comments	3,4,6	Are the system boundaries and limitations stated? Is there a list of parts / components? Is there a diary of the study to date? Are any stated assumptions and assertions that may influence the outcome of study documented?		
		FMECA		
Functions	8,9	Is there a comprehensive list of functions, including obscure functions - For example: To allow safe access for operation and maintenance To Protect from corrosion (environmental deterioration To enable Lifting Facilities To allow Removal of equipment If not, are the locations within other FAGs or Studies documented? Are the parameters for each function identified or recognised?		
Functional Failures		Are the functional failure statements the antithesis of the function statements and, if multiple performance standards are used, does each have its own functional failure statement?		
Failure Modes		Are the failure modes comprehensive? Is the failure mode correctly constructed, i.e. it is not an effect? Is each failure mode a singular entity? Are they identified at a sufficient level of detail to enable a viable maintenance strategy to be discerned?		
Local Effects		Does the Local Effect fully describe what happens at the equipment, specifically: Evidence that the Failure has Occurred or is Occurring, including detection methods (reduced performance, alarms - remote, local, visual and audible - etc)? Is there standby plant available to provide the lost function?		

15.5

DOCUMENT	NES 45 Section	CHECK REQUIRED	CORRECTIVE ACTION RECOMMENDATION	COMPLETED
Next Higher Effects		Does the Next Effect fully describe what happens at system level, physical damage to the asset/system/other unrelated systems?		
End Effects	8, 9	Does the End Effect fully describe the effects to the platform/personnel safety, environmental threats and operational capability, including own forces in the vicinity? Have actions to be taken by the Operator to isolate or mitigate the failure been identified, including the time required to complete such actions? Has the MART for the primary and secondary damage caused by the failure been identified, (from receipt onboard of all stores required)? Have Part Nos. for the ultimate remedial tasks, including secondary damage, been identified (if possible and as appropriate)? Note that a manufacturers Part no is acceptable where no NSN can be identified.		
Failure Mode Details		Where applicable have the following been entered: PF Interval, Life, MART, and where appropriate, MTBFs for the protected function, protective device and for the multiple failure?		
		Task Analysis		
Decision Worksheet 1	10	Is the analysis audit function clear of inconsistencies?		
Decision Worksheet 2		Are the justifications consistent with the RCM Decision Algorithm? Is the CSSS Box annotated where required? Is the PSP Box annotated where necessary? Has the P-F interval, Age or Life been identified and justified in support of any proposed maintenance task? Has the task been allocated to a Task Category?		
		Other Documentation		
Originating JICs	12	Have the Part Nos been annotated? Has the Skill Level been entered? Has the Task Category been identified? Has the Rationalised Interval been left blank?		
Remedial JICs		Have the Part Nos been annotated? Has the Skill Level been entered? Has the Task Category been identified? Has the Rationalised Interval been left blank?		

Table 15.3. Audit Check Sheet

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ANNEX A

RELATED DOCUMENTS

A.1. The following documents and publications are referred to in this NES:

Reference	Title	Section No
AECMA 1000D	International Specification for Technical Publications with a Common Source Database	2, 14, Annex B
AECMA 2000M	International Specification for Materiel Management	2, 14, Annex B
BR 1313	Maintenance Management in Ships	4, 11, 12
Def Stan 00-41	Reliability and Maintainability - MOD Guide to Practices and Procedures	4
Def Stan 00-60	Integrated Logistic Support	1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15
Def Stan 01-2	Guide to engineering alloys used in Naval service: data sheets	11
Green & Bourne	Reliability Technology, John Wiley & Sons, ISBN 0 471 324809	4
HMSO	Health & Safety at Work Act	Foreword
HMSO	Control of Substances Hazardous to Health Regulations (COSHH)	Foreword
MIL HBK 217F	Reliability Prediction of Electronic Equipment	4
MIL HBK 338	Electronic Reliability and Design Handbook	4
MIL-STD-1388	Integrated Logistic Support	2
MIL-STD-1629A	Failure Modes, Effects and Criticality Analysis	9
MIL-STD-2173 (AS)	Reliability-Centred Maintenance for Naval Aircraft, Weapons and Support Equipment	Foreword, 2
MSG-3	Airline/Manufacturer Maintenance Program Development Document	2
Moubray, John	Reliability-centred Maintenance (Butterworth-Heinemann ISBN 0 7506 3358 1)	Foreword
NAVAIR 00-25-403	Management Manual – Guidelines for the Naval Aviation Reliability Centred Maintenance Process	Foreword
NES 752	GRP Survey Handbook	11
Nowlan & Heap	Reliability-Centered Maintenance. Department of Defense, Washington, D.C. 1978. Report Number AD-A066579.	Foreword
NPRD	Non-electronic Parts Reliability Data - 1995	4
RR&A document RRA18453	Safety Justification for the Application of NES 45 RCM Methodology on NSRP	Annex E
Smith, D J	Reliability, Maintainability and Risk (Butterworth Heinemann, ISBN 0 750 60854 4)	4
SSP 23	Design of Surface Ship Structures (Vols. 1&2)	11

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ANNEX B

ABBREVIATIONS AND DEFINITIONS

B.1. For the purpose of this NES the following abbreviations apply:

Abbreviation	Meaning
A's & A's	Alterations and Additions
AAW	Anti-Air Warfare
AC	Agreed Characteristic
AD	Accidental Damage
AE	Age Exploration
AECMA	Association Européene des Constructeurs de Matériel Aérospatiale
ALARP	As Low As Reasonably Practicable
ARM	Availability, Reliability and Maintainability
ASUW	Anti-Surface Warfare
ASW	Anti-Submarine Warfare
B&DQ	Build and Design Quality
BR	Book of Reference
CA	Criticality Assessment
CAR	Corrective Action Recommendation
CB	Charge Book
CBM	Condition Based Maintenance
CCR (LP) / (HP)	Combined Communication Room (Low Power) / (High Power)
CCTV	Closed Circuit Television
CCWEA	Charge Chief Weapon Electrical Artificer
CDPI	Chief of Defence Procurement Instruction
CIWS	Close In Weapon System
CM	Condition Monitoring
COSHH	Control of Substances Hazardous to Health
COTS	Commercial Off The Shelf
CPL	Crack Propagation Life
CSDB	Common Source Database
CSSS	Certificate of Safety Structural Strength
D Ops E	Director Operations Equipments
D Ops P	Director Operations Platforms
DDD	Design Disclosure Document
DLL	Design Limit Load
DNA	Director, Naval Architecture
DT	Damage Tolerant
EBS	Equipment Breakdown Structure
ED	Environmental Deterioration
EDL	End Item Design Life
EDPL	Environmental Deterioration Propagation Life
EPM	Equipment Project Manager *(Obsolete)*

Abbreviation	Meaning
EW	Electronic Warfare
FAG	Functional Asset Group
FD	Fatigue Damage
FFI	Failure Finding (Task) Interval
FMD	Failure Modes Distribution
FMEA	Failure Modes and Effects Analysis
FMECA	Failure Modes, Effects and Criticality Analysis
FRP	Fibre Reinforced Plastic
FSU	Forward Support Unit
GRP	Glass Reinforced Plastic
HAT	Harbour Acceptance Trial
HF	High Frequency
HILS(N)	Head of ILS (Navy)
HO	Hull Outfit
HP	High Pressure
HSE	Health and Safety Executive
HTT	Hard Time Task
ICCP	Impressed Current Cathodic Protection
IDEF (IDEF0)	Integrated computer aided manufacturing Definition
IF	Identification Friend or Foe
ILS	Integrated Logistic Support
IMC	Inventory Management Code
IMO	International Maritime Organisation
IPC	Illustrated Parts Catalogue
IPL	Initial Provisioning List
IPT	Integrated Project Team
JIC	Job Information Card
JSP	Joint Service Publication
LCN	LSA Control Number
LDC	Life to Detectable Crack
LF	Low Frequency
LORA	Level Of Repair Analysis
LSA	Logistic Support Analysis
LSAR	Logistic Support Analysis Record
MART	Mean Active Repair Time
MCA	Marine and Coastguard Agency
MCMV	Mine Countermeasures Vessel
MCO	Main Communications Office
MEM	Marine Engineering Mechanic
MF	Medium Frequency
MHE	Mechanical Handling Equipment
MILSM	MOD ILS Manager

Abbreviation	Meaning
MOD	Ministry Of Defence
MOP	Maintenance Operation
MPL	Melamine Plastic Laminate
MTBF	Mean Time Between Failures
MTTR	Mean Time to Repair (see MART)
MW	Megawatts
NAAFI	Navy, Army and Air Force Institute
NBCD	Nuclear, Biological and Chemical Defence
NC	Not Considered (Further)
NDE	Non Destructive Examination
NES	Naval Engineering Standard
NP	Not Plausible
NSC	Naval Support Command
NSM	No Scheduled Maintenance
NSN	NATO Stock Number
NSRP	Nuclear Steam Raising Plant
OASIS	Onboard ADP Support In Ships
OPDEF	Operational Defect
P	Plausible
PC	Personal Computer
PIL	Part Identification List
PMS	Planned Maintenance Schedule
POC	Probability of Occurrence
POL	Petroleum, Oil & Lubricants
PRA	Probabilistic Risk Assessment
PSP	Part of Ship Plan
RADHAZ	Radiological Hazard
RAM	Reliability, Availability & Maintainability
RAS	Replenishment At Sea
RCM	Reliability Centred Maintenance
RCMS	Reliability Centred maintenance Management System
RCS	Reliability Centred Stockholding
RFA	Royal Fleet Auxiliary
RN	Royal Navy
RN RCMIT	Royal Navy Reliability Centred Maintenance Implementation Team
RPM	Revolutions Per Minute
RRSS	Rapid Reaction Spray Systems
RS	Residual Strength
SA	Stores Accountant
SAT	Sea Acceptance Trial
SCC	Ship Control Centre
SCC SCF	Stress Corrosion Cracking Stress Concentration Factor

Abbreviation	Meaning
SNS	Standard Numbering System
SOCs	System Operator Checks
SR	Staff Requirement
SRB	Sulphate Reducing Bacteria
SRD	Safety and Reliability Directorate
SRF	Structural Risk Factor
SSI	Structurally Significant Item
SSP	Sea Systems Publication
SUDS	Submarine Upkeep Data System
SW	Saltwater
TI	Task Interval
UHF	Ultra High Frequency
VHF	Very High Frequency
WCFS	Water Compensated Fuel System
WPD	Warship Policy Document
WPM	Warship Platform Manager *(Obsolete)*

B.2. For the purpose of this NES the following definitions apply:

Term	Definition
Accidental Damage	A discrete, random event that can cause functional failure directly or initiate a deterioration process that can lead eventually to functional failure.
Age Exploration	The process of determining age/reliability relationships through controlled testing and analysis of deterioration or failure.
Analysis Boundary	A limit line that groups the assets to be the subject of a particular RCM study. They are generally assets that combine to produce a defined function or related functions.
Applicable Task	A task is said to be applicable if it is technically viable and practical to apply.
Assessed Risk Value	The value awarded to a SSI following assessment of the probability and consequences of an Accidental Damage event. This figure is non-dimensional.
Asset	A physical entity that provides a function or functions.
Asset Group	A set of physical entities that collectively provide a function or functions.
Audit, RCM	A checking process to ensure that an RCM analysis complies with the requirements of this NES.
Audit, Technical	A checking process to ensure that an RCM analysis has made sound engineering judgements compatible with the technical characteristics of the assets within the analysis boundary.
Black Boxing	A technique that encapsulates an asset or a group of assets and subjects them to RCM analysis at a high level for a single failure mode. The resultant maintenance task will invariably be of the default type.
Bullet Point JIC	A brief synopsis of the work involved in undertaking a preventive or remedial task. A Technical Author develops the synopsis into a finalised JIC.
'Can'	The level of performance that an asset is capable of providing. For a maintenance strategy to be viable, the 'can' must be greater than the 'want' (*qv*).
Catastrophic Failure	A failure mode which may cause death or loss of platform.
Combination Task	A grouping of tasks, each usually drawn from one of the On Condition, Scheduled Restoration or Scheduled Discard tasks, which together reduce the risk of failure to a tolerable level.
Complex Structure	Structure that has a number of concurrent functions. These could include global and local strength, screening, ballistic protection, etc. It follows that most, or all, of these functions are lost when a structural failure occurs.
Condition Based Maintenance	A Remedial Task undertaken as a result of a measurable deterioration in the performance of an asset or Functional Asset Group.
Condition Monitoring	Any technique used to monitor the condition of an asset.
Conditional Probability of Failure:	The probability that a failure will occur in a specific period provided that the item concerned has survived to the beginning of that period.
Consolidated Task	A single task that is representative of many similar tasks. Their similarity is based on content rather than periodicity.
Corrective Maintenance	Maintenance actions taken after a failure has occurred.
Cost Effective (Task)	The cost of carrying out the preventive task over a period of time is less than the cost of correcting both the failure and any secondary damage caused by it.
Critical Failure	A failure mode in which one or more of the mission aims may not be met and/or the safety of the whole vessel is at risk until the failure is rectified.
Criticality Assessment	A means of establishing the risk to platform and personnel arising from the occurrence of a failure mode. It is based on a combination of the worst case consequences of the event coupled with the probability of its occurrence.

Term	Definition
Damage Tolerant Structure	Structure in which the Functional Failure of a single member will result in only a small reduction in the residual strength of the whole. The design allows for alternative load paths to mitigate the effect of failure.
Default Task	The alternative to a preventive task and will be either Failure Finding or No Scheduled Maintenance with the possibility of a Redesign.
Directed Survey	An explicit instruction for a ship surveyor to examine and report the condition of a defined SSI, irrespective of the condition in which it is found.
Effective Task	A task is said to be effective if it is worth doing, i.e. over a period of time it: a. reduces the probability of failure to a tolerable level, and b. is cost effective to undertake.
Engineering Judgement	That knowledge and experience possessed by individual members of a study team that Facilitators coalesce to form the basis of the decision making process employed by the team in the absence of quantitative data. The provenance of this knowledge should, wherever possible, be recorded in the study documentation.
Environmental Consequences	A failure mode or multiple failure has environmental consequences if it could breach any corporate, municipal, regional, national or international environmental standard or regulation that applies to the physical asset or system under consideration.
Environmental Deterioration Propagation Life (Assessed) ($EDPL_{assd}$)	The time from the onset of deterioration to an asset caused by exposure to the operating atmosphere until it can no longer deliver the required function(s) due to that deterioration, including any retarding effect due to protective coatings, etc. It takes into account the variation of the threat from the standard.
Environmental Deterioration Propagation Life (EDPL)	The elapsed time from the onset of deterioration to an asset caused by the operating atmosphere in which the asset resides to the time when it can no longer deliver the required function(s) due to that deterioration.
Environmental Deterioration Propagation Life (Maximum) ($EDPL_{max}$)	The time from the onset of deterioration to an asset caused by exposure to the least aggressive atmosphere (usually conditioned air) until it can no longer deliver the required function(s) due to that deterioration, ignoring any retarding effect due to protective coatings, etc.
Environmental Deterioration Propagation Life (Minimum) ($EDPL_{min}$)	The time from the onset of deterioration to an asset caused by exposure to the most aggressive atmosphere (usually associated with dirty oil storage) until it can no longer deliver the required function(s) due to that deterioration, ignoring any retarding effect due to protective coatings, etc.
Environmental Deterioration Propagation Life (Nominal) ($EDPL_{nom}$)	The nominal time from the onset of deterioration to an asset caused by exposure to a standard atmosphere (usually complete immersion in salt water) until it can no longer deliver the required function(s) due to that deterioration, ignoring any retarding effect due to protective coatings, etc.
Evident Failure	A failure mode whose effects will on their own eventually and inevitably become evident to the operating crew under normal circumstances.
Facilitator (RCM/Study)	A person trained in the application of the RCM technique and is, hence, qualified to lead a RCM study team.
Fail Safe	A design intent by which the inability of a mechanism to provide the required function is protected by one of two means; either the process will shut-down or a standby device will (automatically) subsume the function until a repair is undertaken on the duty mechanism.
Failure	The inability of an item to meet a desired standard of performance. See also Failure Mode.
Failure - Environmental	A failure mode whose consequence may breach any known environmental standard or regulation.
Failure - Non Operational	A failure mode which does not incur safety, environmental or operational consequences.
Failure - Operational	A failure mode that will affect mission profiles or the quality of customer service.

Term	Definition
Failure - Safety	A failure mode which may result in injury or loss of life.
Failure Effect	What happens when a failure mode occurs.
Failure Effects	The physical manifestations, if any, that result from the occurrence of a failure mode.
Failure Finding Task	A scheduled task used to determine whether a specific hidden failure has occurred.
Failure Finding Task Interval	The frequency at which a failure finding task is to be undertaken, based on the MTBFs of the protected and protective devices coupled with the acceptable frequency of multiple failure.
Failure Mode	A single event that causes a functional failure.
Failure Modes and Effects Analysis	A procedure by which each plausible failure mode in a system is analysed to determine the effects of failure of an asset.
Failure Modes, Effects and Criticality Analysis	An extension of the Failure Modes and Effects Analysis in which account is taken of the severity criteria and the probability of failure.
Failure Patterns	The various curves of conditional probability of failure as they relate to failures that occur due to infant mortality, random occurrences and "life". 6 different failure patterns are used for RCM purposes.
Failure Consequences	The way(s) in which the effects of a failure mode or a multiple failure matter (evidence of failure, impact on safety, the environment, operational capability, direct and indirect repair costs)
Fatigue Damage	The loss of structural function occasioned by the application of cyclic, tensile stresses to an element of structure. The nature of the material, imposed stress levels and the method of fabrication determine the number of cycles required to induce such damage.
Fatigue Damage (Deferred) (FD_{def})	The time that a first inspection for Fatigue Damage can be deferred. It takes account of the estimate of Life to Detectable Crack and ensures that an inspection task is in place at the time a crack is expected to occur.
Finalised JIC	The definitive work instruction for carrying out a maintenance activity, including safety precautions, spare gear, naval stores, tools and test equipment, etc., required together with a step-by-step procedure. The setting to work requirements, if any, will also be defined.
Function	What the "owner" or user of a physical asset or system wants it to do, expressed with numerical performance standards whenever possible.
Functional Asset Group	An asset, or group of assets, whose failure will have consequences in terms of safety, environment, mission or cost.
Functional Block	A logical block of functionality showing a common purpose or characteristic which is suitable for FMECA.
Functional Failure	A state in which a physical asset or system is unable to perform a specific function to a desired level of performance.
Functional Partitioning	A process by which the functions of a platform are identified and recorded. This process leads to the identification of Functional Blocks that may be suitable for RCM analysis.
Hard Time Task	A collective term for the Scheduled Restoration and Scheduled Discard Tasks.
Hidden Failure	A failure mode that will not, on its own, become evident to the operating crew under normal circumstances.
Hidden Function	A function whose unavailability will not be evident to the operating crew under normal circumstances.
Level of Detail	The level of physical decomposition at which failure modes are identified.
Level of Indenture	The level at which Failure Mode, Effects and Criticality Analysis is applied.
Level Of Repair Analysis	A process to determine the optimum level of indenture at which the various maintenance activities and types of maintenance and, hence, spare part strategy, should be applied.
Life	A limit imposed on an item that exhibits one or more age related failure modes. A life limit may be based on Safety, Operational or Economic criteria.

Term	Definition
Mean Active Repair Time	The time taken to restore a function, assuming all spares, tools, test equipment, etc., are available.
Minor Structure	See Other Structure.
Mission Degradation	The defined Mission Profile can be undertaken and completed, but with required platform functions either totally unavailable or operating at less than required performance.
Mission Loss	The inability to commence, or complete, a defined Mission Profile.
Mission Profile	Statement of the principal role(s) of the platform as defined by Staff Targets, Staff Requirements and other related documents.
Mu/Md	The ratio of the ultimate strength (Mu) of a ship's hull, usually in longitudinal bending, to the design strength (Md). Values are normally in the range of 1.1 - 1.5. This figure is not directly comparable to a Factor of Safety.
Multiple Failure	An event that occurs if a protected function fails while its protective device or protective system is in the failed state.
Natural Task Interval	The interval at which a schedulable task should be undertaken, determined solely by the physics of deterioration of the asset that is host to the failure mode under consideration.
Naval Stores	The tools, test equipment, materials (excluding spare gear (*qv*)) and consumables required to undertake and complete a maintenance task.
No Scheduled Maintenance	A conscious decision to do no maintenance whilst the Functional Asset Group is delivering the required function. The asset group will be run to functional failure and then repaired or replaced.
Non-operational consequences	A category of failure consequences that do not adversely affect safety, the environment, or operations, but only require repair or replacement of any item(s) that may be affected by the failure.
Normal Circumstances	In RCM terms the word "Normal" means: a. Nothing is being done to prevent a failure. b. No specific task is being done to detect that a failure has already occurred. c. The asset is being operated in accordance with standard operating procedures as defined in the operating context.
On Condition Task	A scheduled task used to detect potential failures. The detection of potential failures will call forward a "Remedial Task".
One-Time Change	Any action taken to change the physical configuration of an asset or system (redesign or modification), to change the method used by an operator or maintainer to perform a specific task, to change the operating context of the system, or to change the capability of an operator or maintainer (training)
Operating Age	The ageing of an asset based on a timescale related to its exposure to stress and, hence, usage, e.g. rounds fired, fatigue cycles, number of starts, etc. In cases where exposure to stress is continuous, it will relate to calendar time.
Operating Context	The circumstances in which a physical asset or system is required to operate.
Operating Crew	Anyone who has occasion to observe the asset or what it is doing at any time in the course of their normal duties and who can be relied on to identify and report any abnormal conditions. This could be, but is not necessarily restricted to, any member of the ship/submarine's staff.
Operational Availability	A measure of a system's preparedness to meet its operational requirement, usually measured by the ratio of Uptime to Total Time.
Operational Consequences	A category of failure consequences that adversely affect the operational capability of a physical asset, system or platform (output, product quality, customer service, military capability, or operating costs in addition to the cost of repair)..

Term	Definition
Opportunity Survey	A requirement for a ship surveyor to report either: a. a structural failure mode occurring on an SSI other than that which has generated the inspection interval, or b. a structural defect of any type that has occurred beyond the boundary of any specified SSI at the time a directed survey is being undertaken.
Owner	A person or organisation that may either suffer or be held accountable for the consequences of a failure mode by virtue of ownership of the asset or system.
Part of Ship Plan (PSP)	A process for the visual monitoring of the condition of all equipment and structure either within a compartment or designated areas of weatherdecks and external shell plating.
Period of Time	The time base against which the cost-effectiveness of undertaking a task is to be considered. Normally this is the remaining service life of the asset at the time it is being analysed.
Personnel Risk Safety Assessment	A method of determining the risk to personnel undertaking maintenance tasks derived from a RCM analysis.
P-F Interval	The interval between the point at which a potential failure becomes detectable and the point at which it degrades into a functional failure (also known as “failure development period” and “lead time to failure”).
Potential Failure	An identifiable condition that indicates that a functional failure is either about to occur or is in the process of occurring.
Practitioner (RCM)	An exponent of RCM authorised to train, facilitate, support and consult on the subject.
Predictive Task	See On Condition Task.
Preventive Maintenance (PM)	Maintenance actions intended to pre-empt failures. Preventive maintenance tasks include On Condition, Scheduled Restoration, Discard and Combination tasks.
Primary Function(s)	The function(s) which constitute the main reason(s) why a physical asset or system is acquired by its owner or user.
Primary Structure	Structure which contributes to the main structural strength of the vessel. Such structure will all become Significant Structure for an RCM analysis.
Probability of Occurrence	A measure of the frequency at which a specific failure mode might be expected to occur.
Project Leader (PE or SSA)	For the purposes of this NES, the MoD person responsible for the initiation, management and acceptance of an RCM analysis undertaken by in-house or contracted resources.
Protective Device or Protective System	A device or system which is intended to avoid, eliminate or minimise the consequences of failure of some other system.
Rationalised Task Interval	The task interval that results from adjustments made to the Natural Task Interval when a logical grouping of tasks is scrutinised to identify practicable maintenance strategies.
Redesign	The modification of physical hardware, its installation or location, operating procedures and/or its associated operator/ maintainer training.
Reliability centred Maintenance	A structured method of deriving the maintenance strategy for an asset in its operating context.
Reliability centred Stockholding	A method of defining the spare part requirements to support the RCM derived maintenance of an asset in its operating context.
Remedial Task	A task to ensure the continued function of an asset called forward once an On Condition Task has detected a Potential Failure condition. It is also called forward as a result of a failure mode for which NSM or Failure Finding has been chosen as an appropriate maintenance strategy.
Resnikoff’s Conundrum	In essence: a maintenance policy must be designed without using experiential data which will arise from the failures which the maintenance policy is designed to avoid. This is because existing maintenance will have suppressed those failures with really serious consequences such that the bulk of failure data will represent inconsequential failures.

Term	Definition
Risk Value$_{max}$	The maximum numerical value that can be awarded to a SSI as an Assessed Risk Value. This figure is non-dimensional.
Run-to-Failure	A failure management policy that permits a specific failure mode to occur without any attempt to anticipate or prevent it.
Safe Life	An age limit imposed to ensure that all age-related failures are avoided.
Safe Life Structure	Structure designed such that its inherent strength and the imposition of a Safe Life limit prevent failure from fatigue during its service life.
Safe Maintenance Strategy	A maintenance strategy that reconciles the requirements of safe operation with those of operational availability and cost effectiveness.
Safety Consequences	A failure mode or multiple failure has safety consequences if it could kill or injure someone.
Scheduled	Performed at fixed, predetermined intervals, including "continuous monitoring" (where the interval is effectively zero).
Scheduled Discard	Scheduled task that entails discarding an item at or before a specified age limit regardless of its condition at the time.
Scheduled Restoration	A scheduled task that restores the capability of an item at or before a specified interval (age limit), regardless of its condition at the time, to a level that provides a tolerable probability of survival to the end of another specified interval.
Secondary Damage	Damage to, or failures induced on, other systems, structures or personnel, as a direct effect of the occurrence of a failure mode.
Secondary Functions	Functions which a physical asset or system has to fulfil apart from its primary function(s), such as those needed to fulfil regulatory requirements and those which concern issues such as protection, control, containment, comfort, appearance, energy efficiency and structural integrity.
Secondary Structure	Structure that does not contribute to the main structural strength of the vessel but does provide watertight integrity or support local loading. Some of this structure will be categorised as Significant Structure for RCM analysis.
Ship Manager	Collective term for directing personnel involved in the operational or maintenance planning of a platform.
Significant Structure	Structure that carries major loads or has another function such as containment and whose failure will result in a reduction in residual strength, loss of load bearing function or flooding/leakage to sea. Such failures have safety or operational consequences.
Spare Gear	Those items (other than Naval Stores (*qv*)) intended to replace existing items during the course of a maintenance task.
Structural Risk Factor	A numerical value calculated for a SSI as a measure of its relative susceptibility to Accidental Damage.
Structural Sampling	An Age Exploration process for structure that addresses highest risk items.
Structurally Significant Item	An area of significant structure chosen to represent the overall condition of the whole of the significant structure and subjected to RCM analysis. See Asset.
Structure - Other	Structure that does not contribute to either structural strength or containment but serves some other non-vital purpose such as screening.
Superfluous Function	An element of functionality of an asset that has been rendered redundant and is no longer required. This usually occurs due to changes in the operating pattern of equipment.
System Operator Checks	A process undertaken by an equipment operator with the aims of: a. confirming availability of the equipment's function(s), and b. providing operator continuation training.
Technical Secretary	A person employed to populate the RCM RN and/or RCS RN databases during the course of an RCM analysis. They should not take an active role in the decision-making process. Their use during an analysis is optional.

Term	Definition
Tertiary Structure	See Structure - Other.
Tolerable Probability of Failure	The threshold, defined by the authority responsible for the operation and maintenance of an organisation's assets, above which the frequency at which a specific failure mode might be expected to occur cannot be endured by that organisation. The grounds for establishing the threshold can be based on the safety, operational or economic penalties that such failures accrue.
User	A person or organisation that operates an asset or system and may either suffer or be held accountable for the consequences of a failure mode of that system.
'Want'	The performance that a user requires from an asset. For a maintenance strategy to be viable, the 'want' must be less than the 'can' (*qv*).
Zero-based	In RCM terms, nothing is being done to either prevent a failure from occurring, or mitigating the effects of failure once the failure process has initiated.

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ANNEX C

THE RCM RN TOOLKIT

C.1. General

a. RCM analyses generate a large amount of data. To interpret the data requires significant processing, which lends itself to software support if the task is to be done efficiently. Therefore recording and subsequent interrogation of RCM analyses is to be by means of the RCM RN Toolkit. The application has been specifically developed to support the NES 45 RCM process.

C.2. Application Functionality

a. The RCM RN Toolkit provides extensive functionality to support the NES 45 RCM process. Table C.1 summarises the major areas catered for in the application.

RCM Process Area	Functionality Provided
Operating Context	Entry of platform and analysis level operating contexts.
FMECA	Entry of functions, functional failures, failure modes, failure effects and criticality assessments.
Failure Mode	Entry of failure mode details, including:- failure data, RCM justification, redesign justification, age exploration justification and spare gear requirements.
Task Selection	Entry of NES 45 algorithm decisions, task descriptions, task intervals and task skills.
Analysis Auditing	The application allows progressive auditing of RCM review group results throughout the RCM analysis.
Maintenance Packages	Collation and printing of tasks grouped by maintenance interval, skill level required or machinery state.
Job Information Cards	Collation and printing of task details in job information card format.
Asset Hierarchy	Failure Mode to Asset linking, and storage of the set up by the user to allow the tasks derived by RCM to be applied to specific assets within the class hierarchy.
Functional Hierarchy	Storage of the Functional Hierarchy derived for the ship class.
Fault Finding Guide	Linking and storage of Key Words from the FMECA to produce a Ships Staff fault finding guide.

Table C.1 – RCM RN Toolkit Summary of Application Functionality

C.3. Importing and Exporting Analyses / Templates

a. The RCM RN Toolkit contains synchronisation facilities and is to be used to exchange analyses/templates between RCM databases. This will enable facilitators and project managers to build up consistent libraries of templates which can be applied to common assets.

C.4. Failure Finding Intervals (FFIs) (RN Version Only)

a. A failure finding interval module, which automates calculation of FFIs, is a composite part of the software and is to be used to calculate FFIs for hidden failures. FFIs can be based upon:

- Acceptable risk of the multiple failure.
- Desired availability of the protective function.
- Economic criteria.

Note: This functionality is not available in the contractors' version of the RCM RN Toolkit.

C.5. Single and Multi-user Versions

a. The RCM RN Toolkit can be used in both stand alone and networked multi-user environments.

Note: The Contractors’ version supplied by the MOD is only available in the stand-alone version.

C.6. User Guide

a. A user guide covering the application in detail is available from the sponsor of this NES.

C.7. Software Availability

a. Applications for use of the current version of the RCM RN Toolkit should be referred to the sponsor of this NES.

ANNEX D

THE RCS RN TOOLKIT

D.1. Background

a. In order to achieve the full benefit of an RCM based maintenance strategy it is essential to ensure that the spares required to complete each maintenance task can be made available in the appropriate timescale. If this is not achieved then operational availability may be endangered. Traditional initial provisioning procedures, even those based on reliability or availability studies, will often err on the cautious in order to be certain of achieving the specified operational availability. Following a rigorous RCM analysis, it is possible to define and justify the optimum spares stockholding to provide the necessary insurance against the effects of specific failure modes. The MOD has developed the Reliability Centred Stockholding (RCS) toolkit in order to calculate the spares allowances as an integral part of the RCM review process.

D.2. Introduction

a. The RCS RN Toolkit consists of a number of windows and calculation tools that:

1) Link part requirements to RCM originating and remedial tasks.

2) Record the effects of a stockout on operational equipment.

3) Enable the user to identify the appropriate stocking policy for each part.

4) Record the stocking policy in the RCM database.

b. Achieving the correct stocking decision depends on a number of factors and provision of specific data. This includes:

1) The proposed RCM Maintenance policy.

2) The severity of effects if a part is not available.

3) Equipment configuration, including any standby capacity available.

4) The reliability of the equipment supported (MTBF data).

5) Mission length.

6) Details of the inventory re-supply chain.

D.3. System Data and Functionality

D.3.1. Analysis Data

a. Detailed data is required to calculate an individual part’s stock holding. This data includes:

1) Demand or MTBF.

2) The number of parts used in the Functional Group supported by the parts.

3) Equipment configuration details such as standby capacity.

4) Part purchase and repair costs.

5) Mission length.

6) Financial discount rate.

D.3.2. Parts List

a. A detailed list of the parts is also required, to include spares numbers, description and price.

D.3.3. Part Group Optimisation Calculations

a. Support Organisation. The support organisation that is modelled in RCS RN represents the re-supply chain from the UK mainland to the operational platform. It is used to model to any operational scenario, in order to provide information on the following stores requirements:

1) Base Stores

2) Forward Stores

3) On-board stores

4) "Hot" stores

b. RCS RN Calculation. Using the appropriate analysis data the RCS RN optimisation software calculates the optimum number of spare parts and their location in the supply chain for the required equipment availability. An illustrative calculation worksheet is shown in Figure D.1.

Mission Group Optimisation

Part Analysis Group To Indicate Water Depth

Mission Length 10 Days

0906-995318228	Cable assembly	003129	0	*On-board store*	**1**
0906-995330823	Recording Paper roll	003130	0	*On-board store*	**4**
0906-995333028	Range & Phase Coding PEC	003131	0	*Base Store*	**1**
0906-995333030	Power Supply Unit	003132	0	*Base Store*	**1**
0906-995432798	Pre-Amplifier PEC	003133	0	*Base Store*	**1**
0906-995432800	Motor Drive & Paper Drive PEC	003135	0	*Base Store*	**1**
0906-995432801	Transmitter Unit	003136	0	*Base Store*	**1**
0906-995439782	Trigger Suppression & Draught Delay PEC	003137	0	*Forward Support Unit*	**1**
0906-997872609	Transducer	003138	0	*Base Store*	**1**
0906-997872610	Transducer	003139	0	*Base Store*	**1**

Availability Achieved **99.51%**

Cost per platform supported **£2,964.11**

Figure D.1 - RCS Mission Group Optimisation Results Sheet

c. The RCS Toolkit enables the user to choose the best allocation of budget between a group of parts by specifying a *part group.* For example, if 99% system availability is needed, but a stockout of any of 20 parts could result in system downtime, the toolkit will choose the stock levels of individual parts that achieve 99% availability at minimum cost. The RCS RN part group optimisation attempts to achieve a target equipment availability at minimum cost.

d. Up to 5 different sensitivity analyses may be carried out to show the effects of varying lead times and demand rates. Several separate analyses may also be carried out for each part, so it is possible to determine the effects of changing supplier, or alternative strategies such as vendor stocking Analyses may include several hundred parts, depending on available computer memory and processor speed.

e. Availability Requirement. The user may also select one or more availability targets in order to compare the relative costs of different equipment availability levels.

D.3.4. RCS Decision Worksheet

a. The RCS Decision Worksheet summarises the RCS analysis of the stockout consequences recorded on the RCS Information Worksheet and documents the recommended stocking policy. An illustration worksheet is at Figure D.2.

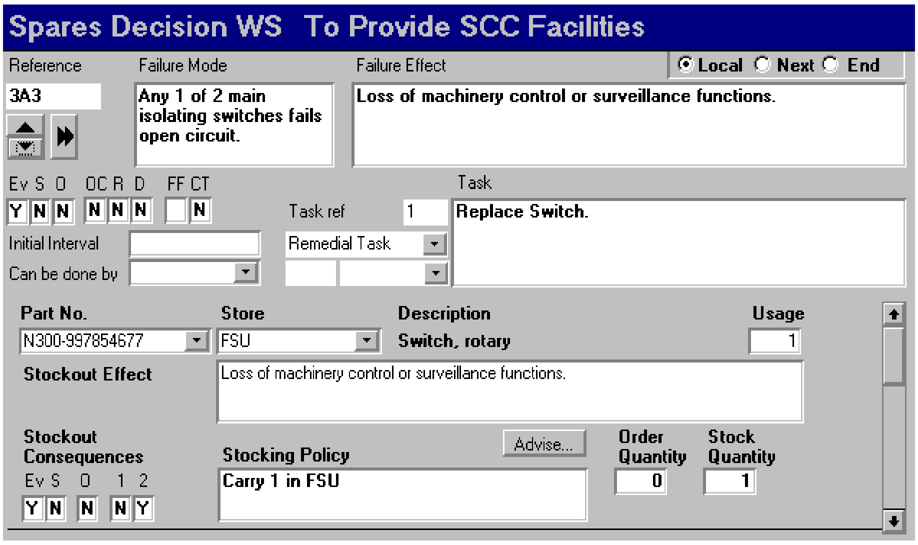

Figure D.2 - RCS Spare Parts Decision Worksheet

b. Graphical Presentation. All results for individual spare parts may be presented in a graphical form suitable for inclusion in final reports. Graphs may show up to five different cases to assist sensitivity analysis.

D.4 Analysing Individual Items

a. Part Optimisation. The RCS RN Toolkit optimises part holdings based on a number of targets, including discounted through-life costs, stockout rate and system availability. The most applicable in a military environment are:

1) Mission Success. Selects a stock level that achieves at least the probability of mission success specified by the user. This target is generally chosen if the assets are operated for a fixed mission length and it is difficult, or impossible, to obtain parts during the mission.

2) Mission Length. Selects a stock level that achieves at a minimum mission length. This target is often used where part re-supply during the mission is difficult or impossible.

b. Sensitivity Analysis. The information needed to undertake a stock analysis is seldom known with absolute certainty. Only approximate demand rates, lead times and financial information are available in the real world. RCS RN provides sensitivity analysis tools that show the impact of uncertain data. In most cases this enables the Commodity Manager (Logistician) to make a firm decision without having to spend time in compiling absolutely precise information.

D.5 Integration with RCM RN

a. RCS RN shares the same main database as RCM RN and so has direct access to all RCM analysis data. This enables the database engine to enforce referential integrity constraints between RCM and RCS tables.

b. RCS RN task part lists, stockout effects and RCS decision worksheet information is imported and exported with the associated RCM RN analysis.

D.6 Single and Multi-user Versions

a. The RCS RN Toolkit can be used in both stand alone and networked multi-user environments.

D.7 User Guide

a. A user guide covering the application in detail is available from the sponsor of this NES.

D.8 Software Availability

a. Applications for use of the current version of the RCS RN Toolkit should be referred to the sponsor of this NES.

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ANNEX E

RCM IN REGULATED AND /OR CONTROLLED ENVIRONMENTS

E.1. Introduction

a. With the implementation of modern safety standards and management methods it is essential that the RCM process liases fully with the relevant safety management team. Many decisions made within the RCM process have the ability to undermine the safety justification and mitigation agreed as part of the control mechanism for a given hazard in the safety management system. *It is therefore essential that the RCM team provide feedback to the safety management team to enable an evaluation to be made on the merits of updating existing safety cases or safety assessments.*

b. Application of the algorithm may potentially undermine the safety justification and mitigation agreed as part of the control mechanism for a given hazard in the safety management system. The study team must ensure that all decisions are justified and recorded within the relevant section of the software. Although analysis results must feedback to the safety management team to assess the implications on the existing safety case or safety assessments, it is not necessary for the regulating authority to be present at analysis meetings (but see E.5.a. below). This will ensure that an analysis has not been subjugated to conform to current regulatory thinking.

E.2. External Regulatory Authorities

a. Safety regulators typically include Safety and Reliability Directorate (SRD) for nuclear installations, and Lloyd's Register for RFA and other auxiliary vessels. International safety and environmental regulations are also laid down by the International Maritime Organisation (IMO) and, in the UK, by the Marine and Coastguard Agency (MCA).

b. Where NES 45 is to be used for reviewing the maintenance of similarly regulated shore installations, the appropriate authorities will include the Health and Safety Executive (HSE). The policy directorate of the HSE will provide guidelines on safety requirements for assets operated within the defence sector.

c. Where environmental considerations predominate, guidance can be sought from the Environmental Agency (EA) or, in Scotland, from the Scottish Environmental Protection Agency (SEPA), both of whom are responsible for setting environmental controls.

d. Other professional bodies will provide codes of practice, (i.e. the Institute of Electrical Engineers whose regulations cover such issues as earth bonding).

E.3. RCM within a Safety Justification

a. Where a safety justification already exists for the asset to be analysed, experience shows that it is better to de-couple the RCM analysis from such justification assumptions. A zero-based RCM analysis should be carried out, and then any anomalies between safety justification and RCM analysis identified and resolved later. Where RCM suggests a relaxation from the regulation, current rules should continue until the regulator endorses such a relaxation.

E.4. Regulatory RCM Data Requirements

a. In all cases license to operate assets with a significant safety dimension will be based on the review of risk, such as PRA, and on the maintenance tasks to be put in place to manage that risk. In all such cases, the audit trail will be vital. RCM will establish that data trail, providing the following information is captured during the analysis:

1) Plausible failure modes affecting safety for the operating context in question. For example, the risk associated with an aircraft crashing on a nuclear power station would be considered in an RCM context, whereas a similar event happening to remote, unmanned pumping station would probably not be addressed.

2) The value of tolerable risk used to assess hidden/evident safety failure modes, and the source of such data for auditability.

3) Values for component reliability and protective system demand rates for hidden failures for which failure finding is an appropriate default action. The source of such data should be included for auditability

4) Values for the life of equipments showing an age-related failure pattern, including the source information. Such data should show an unequivocal age/failure relationship. If this is not available it is more conservative to base a failure management strategy on a random failure pattern.

E.5. RCM Analysis Team Regulatory Responsibilities

a. Responsibility for ensuring that adequate safety and environmental guidance is provided to the analysis group is the responsibility of the Facilitator. It is recommended that an RCM analysis having significant regulatory implications should include within the group a representative from the regulating authority itself but only part-time in the specialist role. Where this cannot be achieved, it may be sufficient for a manufacturer to provide his own specialists in this field.

b. For RCM analysis of the NSRP or any of its associated systems, reference should be made to RR&A document RRA18453 – “Safety Justification for the Application of NES 45 RCM Methodology on NSRP”.

ANNEX F

FUNCTIONAL PARTITIONING AND OBSCURE FUNCTIONS

F.1. The following represents the typical functions that may be identified for a surface ship or submarine platform. As a rule, the functions for these types of platform are similar, but the engineering required to achieve the required performance levels for each platform will, of course, differ considerably.

Platform Function	Typical Asset Groups
Vessel Integrity:	
To Ensure a Sound Vessel	Primary and Secondary Structure
To Prevent Cathodic Corrosion	ICCP, Zinc Protectors
To Enable Making Fast of the Hull	Anchors & Cables, etc.
Marine Engineering:	
To Generate Electrical Power	Generators and Prime Movers Emergency Generators
To Pressurise HP & LP Air	Air Compressors and Pipework to end users
To Store and Distribute Fuels	Fuel Tank Coatings, Pumps and Distribution Pipework
To Produce, Chlorinate & Store Fresh Water	RO Plant, Distillers and FW Tank Coatings
To Store, Refrigerate, Prepare, Cook & Distribute Food	Storage, inc. Cold & Cool Rooms, Food Processing Equip., Cookers, Display Cabinets, Food Lifts
To Distribute Main Power Supply	Switchboards, Rectifiers/Transformers, Junction Boxes
To Enable Directional Control & Stable Platform	Rudders, Steering Gear & Telemotor Controls, Autopilot; Stabiliser Gear; Hydroplanes, Bow Thrusters
To Distribute Hot & Cold Fresh Water	User Pipework and flow control mechanisms, calorifiers
To Enable Personal Hygiene	Washing Machines, Tumble Dryers and Pressing Boards
To Ensure Personnel Safety and Rescue & Medical Treatment	Personal Safety Equip., Neil Robertson Stretchers, Sick Bay and Associated Equip.
To Propel the Ship	Gas & Steam Turbines, NSRP; Electric Motors, Gearboxes, Shafting & Propellers
To Collect, Treat & Store Human and other Waste	Urinals, WCs, Soil Pipes, CHT Equip.; Garbage Compactors
To Control & Monitor the Ship	Machinery Control and Surveillance
To Pressurise HP & LP SW	Pumps and Distribution Pipework, Hydrants
To Treat Air & Ventilate Machinery Spaces	HWTCs, ACUs, Vent Trunking, Air Diffusion Equip.; Electrolysers, CO_2 Absorption Unit
To Enable Systems Testing & Maintenance	Workshops, Maintenance Room Equipment.
To Allow for Storage, Access and Collective Protection	Store Rooms, WT/NWT Doors, Anti-slip Deck Coverings. Citadel, AFUs, Gas and Chemical Detectors, Radiac Monitoring
To Actively Protect from Flooding & Fire	Bilge Alarms, Fixed F/F Installations, Smoke Detectors and Fire Alarms
To Produce and Distribute Chilled Water	Chilled Water Plant and Distribution Pipework
To Permit Aviation Support	Flight Deck Lighting and Coating; Air Workshop Equip.; Helo Earthing Equip.
To Store Munitions	Magazine Safety Equip.; Munitions Handling

Platform Function	Typical Asset Groups
Combat System:	
To Compile Air Picture	Long Range Surveillance Radar
To Enable Point Defence AAW of the Ship	Phalanx; Goalkeeper; SeaWolf
To Facilitate Internal Communications	Main Broadcast
To Facilitate External Communications	ICS; SCOT, INMARSAT; ULF; LF; MF; HF; VHF; UHF; Visual Signalling; Acoustic Signalling; UWT
To Permit Management of the Combat System	Combat System Highway, Central and Distributed processing
To Prosecute Medium Range SuW	Medium Calibre Gun; NGS
To Enable Passive ASW Surveillance	Passive Hull Arrays; Towed Array
To Enable ECM & Above Water Countermeasures	Active EW; Chaff & IR Dispensers
To Collect and Display Platform Data	Windspeed & Direction, Log; Bathythermograph; Meteorological Equip., Depth Indication, Echo Sounder
To Enable Electronic Surveillance	Passive EW
To Prosecute Long Range SuW	Exocet; Harpoon; SubHarpoon; SLAM
To Deploy a Ship Launched ASW Capability	STWS launcher; Heavyweight Torpedo Discharge
To Deploy Sub Surface Countermeasures.	Towed and Projected decoys
To Manage Ships Signature	HVME; RASH; IR management equip. Acoustic Tiles
To Enable Active ASW Surveillance	Active Sonar
To Enable Above Water Surveillance	Surveillance Radars
To Prosecute Close Range AAW & SuW	Cannon, Small Arms
To Deploy and Control a Remote Prosecution	Helo Control Facilities; Special Communications, Personnel Launch and Recovery Systems

Table F.1 – Generic Platform Functional Blocks

F.2. Typical obscure functions may include the following:

Function/Equipment Keywords	Purpose	Equipments Affected
Protective coatings	To prevent surface deterioration.	Paint coatings, plasticising, enamelling, galvanising or metal spray coatings.
Decorative finish	Aesthetic appearance.	Paint coatings, decorative panels, floor coverings, deckhead panels (metal or MPL)
Fixed lifting points	On ship structure and equipment for removal purposes	Eyebolts, eyeplates and bolted lifting plates
Portable equipment removal arrangements	Ancillary Support Equipment (ASE)	Lifting beams, hoists, chain blocks, strops, slings and shackles
Structural support arrangements	Equipment support	Structural seats c/w welds, bolts
Access and safety arrangements	Safe working conditions. Access for maintenance purposes.	Deck coatings e.g. Non Slip Gangways and walkways, stanchions, guardrails, fitted guards, access panels and doors
Personal protection equipment	Protection to personnel prior to, during and after specific operations	Environmental testing gear, gloves, goggles , facemasks, respirators, aprons etc. Barrier creams, eyewash bottles, first aid kits
Warning and information signs	To be mounted/painted on equipment and/or adjacent structure to give specific information on safety and/or operation.	Safety boundaries, weapon swept arcs. Safety and warning signs (RADHAZ, hazardous contents, and radiation, pipe markings). Operating hazards, start and stop routines, equipment kill cards. Personnel escape routes.
Preparation for a period of extended readiness	Preparations for dormant or dead ship, P x O arrangements; leave periods, or on initial fit	Anti-condensation heaters, desiccants, additional lubrication, protective envelopes.
Bringing assets forward from extended readiness	Activation measures, failure finding and condition monitoring when active	Initial measurements when activated prior to settling into "active" maintenance regime.

Table F.2 – Obscure Functions

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ANNEX G

OPERATING CONTEXT FORMATS

G.1. Operating Contexts are to be written hierarchically, such that lower level statements can be directly related to the next higher level. The levels of Operating Context are illustrated in Figure G.1. Levels 0 and 1 are outside the scope of this NES.

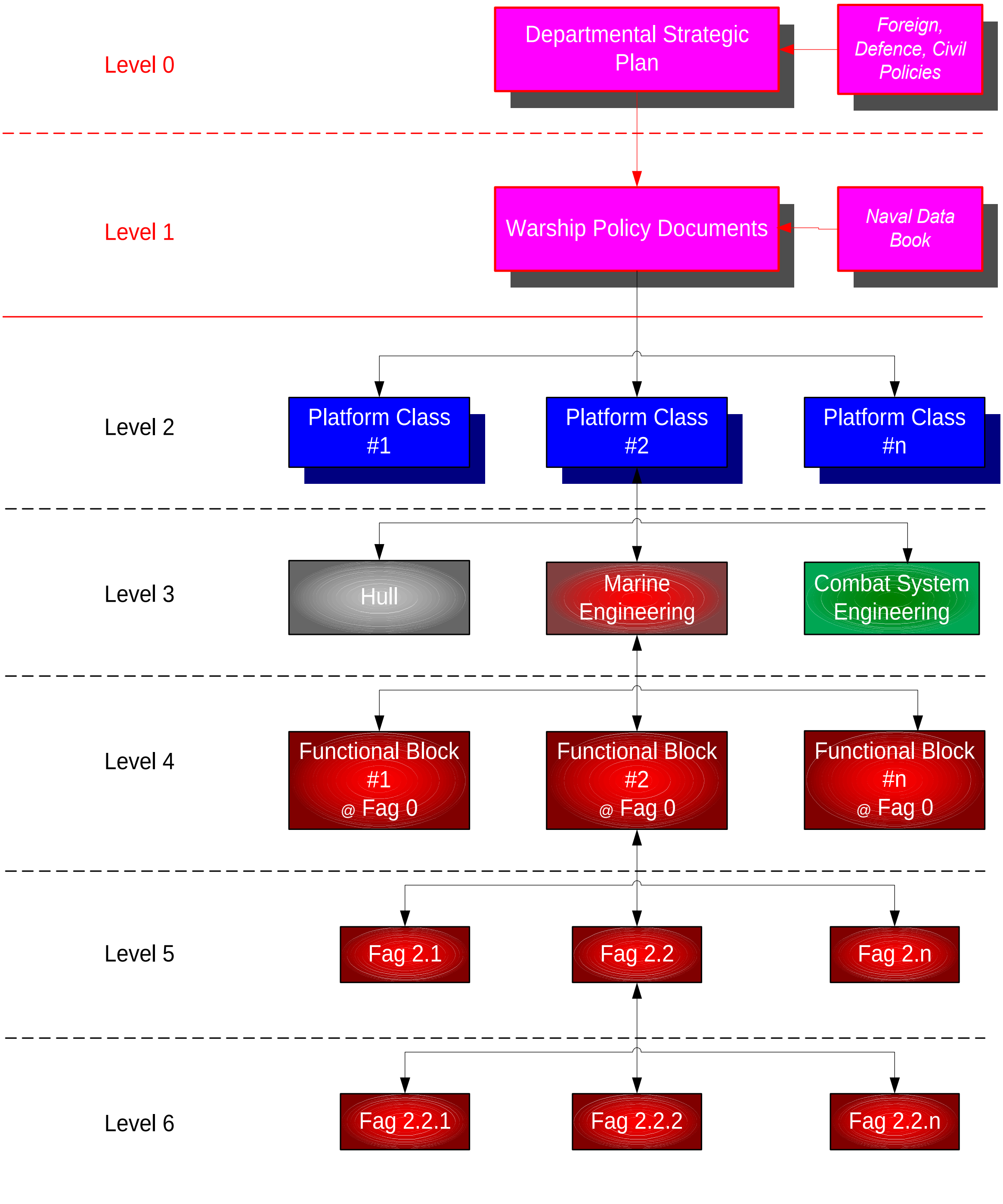

Figure G.1 – The Levels of Operating Context

PLATFORM OPERATING CONTEXTS (LEVEL 2)

G.2. Platform Operating Context formats are given below. Note that the highest level for a platform is Level 2, since Level 2 concerns the maritime doctrine enshrined in BR1806, which is itself developed from Governmental Policy Directives at Level 0.

LEVEL 2 OPERATING CONTEXT - GENERIC

ROLE

1. OPERATIONAL TASKS

1.1 In support of Defence Role 1:

1.2 In support of Defence Role 2:

1.3 In support of Defence Role 3:

2. AVAILABILITY

2.1 Availability for sea:

2.2 Ship Task Intrinsic Availability:

2.3 Weapon System Availability:

3. PRIMARY OPERATING PARAMETERS

3.1 Normal Operations:

3.2 Environmental Range:

3.3 Speed:

3.4 Endurance:

3.5 Manoeuvrability:

3.6 Flight arrangements:

3.7 Signature Reduction:

a. Noise:

b. Radar Cross Section:

c. Magnetic:

d. Others:

3.8 Protection:

3.9 RADHAZ and Mutual Radio Interference:

3.10 Hardening against Nuclear Weapon effects:

4. COMPLEMENT

4.1 Complement:

4.2 Current Scheme of Complement:

			Officer	CPO	PO	JR
Complement (xmt Flight)						
Flight						
NAAFI						
Margins:	*Training*	*Advancement Board*				
		Flexibility				
	Sub-total	*(inc. CO, exc Margins)*				
	Total:	*(inc. Margins)*				

4.3 Aircraft Complement:

5. ONBOARD TRAINING

5.1 Ship Board Training Equipment:

6. LOGISTICS

6.1 "n"-day endurance:

6.2 Fuel:

6.3 Naval Stores:

6.4 Victualling Stores:

6.5 Armament Stores:

6.6 Replenishment:

6.7 OASIS:

7. UPKEEP POLICY

7.1 Ship Life:

7.2 Upkeep Cycle (Historical Note, for information):

7.3 Upkeep Cycle Extension (Historical Note, for information):

7.4 AR&M:

7.5 Maintenance Policy (Historical Note, for information):

7.6 Design for Upkeep:

7.7 Onboard Upkeep Facilities:

- *a. Test equipment procedures:*
- *b. Workshops:*
- *c. Spares:*
- *d. Documentation:*

7.8 Base Upkeep Facilities:

8. UPDATE

8.1 Update Policy:

8.2 A&A Fitting Opportunities:

8.3 Planned Type A A&As:

8.4 Type B A&As Functional Groups:

8.5 Ship Equipment Fit:

9. THROUGH LIFE MANAGEMENT

9.1 Acceptance Policy:

9.2 Combat System Certificate of Clearance for Use:

9.3 Ship Safety:

LEVEL 3 OPERATING CONTEXT - GENERIC

G.3. Level 3 operating contexts are divided into Hull, Marine Engineering and Weapon Engineering departments. The Hull statements are formatted as follows:

Level 3 - Engineering Discipline Level – Structure (Generic)

1. *Key Hull Dimensions:*
   Length overall
   Length at waterline
   Beam at deck
   Beam at Waterline
   Deep displacement
   Draught at HO Sonar Dome
   at props (lowest sweep)
   amidships
   Deck heights above baseline 4-Deck
   3-Deck
   2-Deck
   1-Deck at ship centreline
2. *Key Constructional Features:*
3. *Underwater Appendages:*
4. *Machinery Removal Routes:*
5. *Citadel subdivision:*
6. *NBC Arrangements for Machinery Spaces:*
7. *Prewet:*
8. *Accommodation:*
9. *Domestic and Hotel Services Arrangements:*
10. *Medical and Dental Arrangements:*
11. *Educational and Recreational*
12. *Upper Deck Arrangements:*
    a. *Boats:*
    b. *Replenishment Facilities:*
    c. *Recovery of Man Overboard:*
    d. *Life Saving Equipment:*
    e. *Awnings:*
    f. *Anchors and Capstans:*
    g. *Guardwires and Safety Nets:*
13. *Structural Allowance for Towed Array Improvements:*
14. *Aviation Arrangements:*
15. *Design Disclosure Information:*
    Build Standard - Military or Commercial Standard to which the structure is designed.
    Design weight distribution -Full weight breakdown (NES 163) with supporting evidence.
    Primary design loading - Hogging, sagging (vertical and horizontal) and still water bending moments. Justifications to be based on intended areas of operation.
    Naturally induced loads - Sea (transverse, green seas, notional number of wave encounters, etc.), wind, ice, thermally induced.
    Artificially induced loads - Berthing, docking and grounding, helicopter and vehicles, shock.
    Ultimate strength - Panel collapse curves, sectional properties - sectional area, 2nd moment of area and location of the neutral axis, supporting assumptions - material properties, rolling and corrosion allowances, superstructure effect.
    Fatigue - Full fatigue calculations for each selected SSI.
    Vibration analysis - Design method used, main machinery excitation frequencies, main modes and frequencies of hull vibration, modes for masts and other structural features.
    Environmental protection - Coating schemes, extent of cathodic protection.
    Statutory requirements - For systems that include items subject to statutory tests, analysis of supporting structure for concentrated loads, lifting appliances, ramps, RAS points, etc.
    Build quality - List of all major and minor build concessions.

G.4. The Marine Engineering statements are formatted as follows:

Level 3 - Engineering Discipline Level – Marine Engineering (Generic)

MAIN MACHINERY

1. Main Machinery Configuration

2. Propulsion Machinery Control:

3. Steering Arrangements:

4. Stabilisers.

ELECTRICAL POWER GENERATION AND DISTRIBUTION

5. Normal Supply:

6. Emergency Supply:

7. Floodlighting:

AUXILIARY MACHINERY

8. Fuel Filling, Storage and Transfer System.

9. Fuel Supply System.

10. Fuel Stripping System.

11. Main Engine Lubricating Oil System.

12. Diesel Generator Lubricating Oil System.

13. Low Pressure Seawater System.

14. High Pressure Seawater (HPSW) System.
a. Machinery Cooling.
b. Firemain.

15. Individual Cooling Water Systems.

16. Chilled water system.

17. Fresh water system.

18. Domestic fresh water system.

19. High Pressure Air System.

20. Low pressure air system.

21. Essential LP air system.

22. Special Service Air (Masker) system.

23. Remotely operated valve facility.

24. Bilge and sullage system.

25. Refrigeration.

26. Domestic Ventilation.

27. Machinery Space Ventilation

DAMAGE CONTROL

28. NBCD Control Arrangements.

29. Magazine spraying arrangements
a. Manual systems.
b. Automatic systems
c. Rapid Reaction Spray Systems (RRSS).

30. Machinery Space Firefighting Arrangements

31. Smoke Detection.

32. Firefighting Equipment.

MARPOL AND ENVIRONMENT

33. Anti-Pollution:
a. Solid waste.
b. Liquid waste
c. Black water (Sewage).
d. Grey water.

AVIATION SUPPORT SERVICES

34. Fuel supply.

G.5. The Weapon Engineering statements are formatted as follows:

Level 3 - Engineering Discipline Level – Weapon Engineering (Combat Systems) (Generic)

GENERAL

1. AR&M Deficiencies:

COMMAND SYSTEM

2. The Command System:

3. Data Links:

NAVIGATION AND IDENTIFICATION

4. Navigation Arrangements:

5. Navigation Facilities:

6. Navigation Radar:

7. Navaids:

8. Conning Arrangements:

9. Identification Friend or Foe (IFF):

10. Meteorological Arrangements:

ASW

11. ASW Sensors - Long Range:

12. ASW Sensors - Medium Range:

13. ASW Sensors - Short Range:

14. ASW Sensors - Environmental:

15. ASW Weapons - Long Range:

16. ASW Weapons - Air Launched:

17. Sonobuoys:

18. ASW Weapons - Short Range:

19. ASW Decoys

20. ASW Training Arrangements:

21. Demolition Outfit:

22. Diving:

ASuW

23. ASUW Sensors:

24. ASUW Weapons - Long Range:

25. ASUW Weapons - Short/Medium Range:

AAW

26. AAW Sensors - medium range:

27. AAW weapons - short range

28. Additional Radar Displays and Operational CCTV requirements:

ELECTRONIC WARFARE

29. EW Sensors:

30. EW Decoys:

MINOR WEAPONS

31. Small Arms:

32. Saluting Guns:

EXTERNAL COMMUNICATIONS

33. LF/MF/HF Transmission and Reception:

34. VHF/UHF Transmission and Reception:

35. Satellite Communications:

36. Cryptography and Speech Secrecy:

37. Underwater Communications Equipment:

38. Main Communications Office (MCO) Arrangements:

39. CCR (LP) Arrangements:

40. CCR (HP) Arrangements:

41 Operations Room Arrangements:

42. Operations Room Annex Arrangements:

43. Bridge Comms Arrangements:

44. Broadcast Reception:

45. Message Handling:

46. Communications Control Outfit:

47. Maritime Distress:

48. Emergency Arrangements:

49. Visual Signalling:

50. Miscellaneous:

INTERNAL COMMUNICATIONS

51. Internal Communications:

RECREATIONAL AND MISCELLANEOUS COMMUNICATIONS

52. Facilities include:

53. Miscellaneous:

54. Administration:

TRAINING FACILITIES

55. These comprise:

NBC FACILITIES

56. NBC Arrangements:

G.6. Note that when necessary, each section can be annotated with the following notes as required:

Shortfalls:

Endorsed Enhancements:

Endorsed Savings Measures:

G.7. The continuous updating of these documents will ensure that all participants in the upkeep process are aware of the required, actual and modified performance parameters of the platform.

G.8. Operating contexts for Level 4 and lower levels (FAG 0 and lower order FAGs) should normally be formatted as follows. This list is not to be considered exhaustive:

1. System Boundary. The physical boundary of the system considered by the FAG, to include where the system boundary interfaces with adjacent systems/RCM studies.

2. Physical Description. A summary of the assets within the system boundary, which should include the following:

a. Manufacturer
b. Type, including rated capacity, etc.
c. Notional/Designed/Performance
d. Method of control
e. Special considerations, e.g. shock/vibration, filtering/drying, ship's/shore supplies required and source, etc.
f. Downstream users of the functional output.

3. System Functions. The performance requirements of the primary system function where its outputs cross the system boundary are to be quantified. Secondary functions are also to be identified quantitatively wherever possible.

4. System Inputs. Assumptions made about inputs into the systems from adjacent systems/RCM studies. These are to be stated quantitatively wherever possible. The usual assumption is that the system is available.

5. Mode of Operation. How the system is operated and the critical phases of system availability (ie. when the system availability is most crucial) are to be identified.

6. Standby Plant/Mitigation Actions. The availability of standby plant is to be identified, together with any other known procedures for dealing with loss of the primary or secondary functions, eg. whether systems can be reconfigured.

7. Existing Monitoring Facilities. What instrumentation or other devices are currently available to monitor the condition/performance of the assets within the system boundary.

8. Assumptions. Any assumptions made prior to the commencement of the RCM study, plus any further relevant assumptions drawn during the study itself, are to be recorded.

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ANNEX H

REQUIREMENTS FOR WRITING JOB INFORMATION CARDS (JICs)

H.1. Introduction.

a. This statement of requirements is to be used to produce detailed JICs from bullet point JICs derived as part of the RCM studies for use in RCM (RN) and RCMS. JICs are completed in soft copy in RCM (RN), which will be subject to audit by EPMs and/or Design Authorities.

H.2. IT Requirements.

a. The RCM (RN) software toolkit is to be used for the preparation of JICs and will create JICs as a standard output report from the analysis database.

H.3. General.

a. The software determines the basic format of the JIC, but certain sections are generated from free form memo fields within the data record, where the layout and content are the responsibility of the JIC author. These fields should be completed in accordance with the general requirements of BR1313 Chapter 2.

b. MoD staff will allocate Planned Maintenance Schedule (PMS) numbers and Maintenance Operations (MOP) numbers at a later date when the requirements of the revised Maintenance Schedule are known.

H.4. RCM RN JICs

a. The JIC will draw some of its text from data entered through its input form, although this is not the only screen that could be used for the purpose. To produce a JIC Report, the following fields should be completed:

1) **TASK TITLE**. The Task Title field must be manually populated from the analysis data record, using the same task identification number and text description as entered in Decision Worksheet 1.

2) **HANDBOOK NUMBER**. The relevant BR, Handbook or IPC number for the equipment is to be quoted here together with any other BR or handbook that may be referred to in the job description. A single line entry is to be used for each reference including the issue number, i.e. BR2798(11) Issue 13.

3) **SAFETY PRECAUTIONS**. All Safety Precautions that are required to be observed prior to the start of the work are to be included under this heading. This text is to be typed in **BOLD UPPER CASE** and should mirror that used in the appropriate technical handbook.

4) **TASK STEPS**. In writing the task steps or method the author should be guided by the following rules:

a) The JIC is to be considered the definitive directive concerning the method of carrying out the task together with all associated parameters to be achieved, unless these are classified greater than RESTRICTED. The JIC procedure should be safe for personnel working on the equipment or system, and should also not cause damage to the equipment/system components e.g. by specification of excessive test pressure or insulation test voltage.

b) Instructions are to be presented as a logical sequence of brief but unambiguous instructions in a step by step manner. Paragraph numbering and their sub-divisions are to be in accordance with JSP 182 Chapter X Para 27. Security classification is to be either UNCLASSIFIED or RESTRICTED, and content to be tailored accordingly by reference to higher classification documents where necessary.

c) Long or complex procedures involving detailed test figures or tolerances should not be included if they are to be found in the handbook, unless this is considered to be necessary e.g. where the working environment precludes the use of the handbook (which may be in hard copy or on fiche).

d) Where JICs call for measurements to be taken, instructions should be given as to where and in what manner they are to be recorded, e.g. "plot on the MMS1A, enter on DI. chart".

e) Tools to be used are to be described in the simplest possible manner consistent with clarity, specialist tools should be specified additionally by part number (NSN) where available.

f) A facility exists within the RCM RN software for including Technical Drawings within the JICs. Drawings included must be clear enough to understand at the required level of detail.

5) **WARNINGS**. Warnings are used to alert the reader to possible hazards that may cause loss of life, physical injury or ill health in any form. They must immediately precede the associated text to which they apply. Warnings are to be typed in **BOLD UPPER CASE. JICs should clearly differentiate between TAGOUT procedures, Man Aloft precautions and Diving isolations.**

6) **CAUTIONS**. Cautions are used to draw the attention to possible hazards that could cause damage to equipment or material and are to immediately precede the associated text. The captions of cautions are to be typed in **BOLD UPPER CASE** and the **text** in **bold lower case.**

7) **TASK INTERVAL**. The contractor should complete only the Natural Interval for the Task, deduced from the RCM analysis. The RCMIT will enter the rationalised interval once the platform maintenance consolidation and rationalising process has been completed.

8) **TASK DONE BY**. The Task Done By field should show who is required to undertake the task and is to use standard RN skill descriptors.

9) **TIME TO DO TASK**. The Time To Do Task field should show the predicted duration of the task, assuming all tools, spares etc are available.

10) **NAVAL STORES**. A listing of all the tools, test equipment, materials and consumables required to complete the task should be included under this heading. NATO Stock Numbers (NSNs) are to be inserted (where applicable) in the form IMC - Country of Origin followed by the final 7 digits for example O561-995216845. (O being alphabetical, not nought).

11) **SPARE GEAR LIST**. A list of all the parts that may need to be replaced during the execution of a task. NSNs (see Para 3.1j) and usage information are required and all efforts should be made to identify them using IPCs, PILs and other publications. Identifying NSNs can pose particular problems:

a) If a NSN cannot be easily identified then the RCMIT should be notified so that enquiries can be made in an effort to resolve the situation. All information about a part should be made available. If this is not possible then the manufacturer's reference number or similar identifier should be entered. As a last resort a temporary number should be used (see below). Descriptions should be as detailed as possible. It should be noted that the software overwrites previous details if a reference is duplicated, e.g. if an entry giving a reference of NP1 is made for a "generator" and later a duplicate entry of NP1 is made for a "washing machine", all previous descriptions will read "washing machine". Care is therefore required in carrying out this task and temporary number allocation will be controlled by the RCMIT.

b) If no spares are required for the task then "No spares required" should be entered in the Part Number box.

c) It is important to note that NSNs and Part Numbers entered in this specific field will be used elsewhere in other databases.

12) **DRAWINGS**. All relevant drawings should be listed including Drawing Number, Title and sheet number(s).

NOTE: AUTHORS SHOULD BE AWARE THAT WHERE THERE ARE DISCREPANCIES BETWEEN THE RECOGNISED EQUIPMENT HANDBOOK AND THE JOB INFORMATION CARD, THE LATTER WILL BE CONSIDERED TO BE THE DEFINITIVE DIRECTIVE.

H.5. JICs involving Isolation of systems.

a. Where necessary the operations in the JIC should cover the isolation and reconnection of the system invoking any **"TAG OUT"** working procedures in a generic manner. However, a

CAUTION should be inserted into the text of the JIC to make clear to the Operator that a final safety check be made prior to commencing the work, e.g. "Isolate the HPSW pump from the HPSW system and defuse power plus control circuits using Ship's approved TAGOUT procedures".

b. This is intended to avoid safety hazards being created by configuration changes between vessels. **Valve numbers and Fuseways are not to be identified in the text.**

H.6. Relating the content of the JIC to the originating study

a. It is imperative that the originating RCM study is consulted during preparation of JICs. Information is not to be extracted direct from BR's or other documentation without reference to the RCM analysis.

H.7. Risk Assessment

a. A full Personnel Safety Assessment in accordance with section 12.6 is to be conducted for each JIC and the results tabulated for each Functional Asset Group (FAG) and presented in a report.

H.8. Work in Way

a. It is imperative that all Work in Way associated with a JIC is fully defined in the text. This may involve the Author in ship familiarisation visits to ascertain the extent of the work.

H.9. Page Numbers

a. All JIC pages are to be numbered and must include the total number of pages within the JIC, e.g. 1 of 10, 2 of 10, etc.

H.10. Specialist Equipments

a. There are some equipments that are still under development, particularly in the weapons area, and a specialist contractor may be appointed as the delegated Design Authority. It will be necessary for the JIC Author to work in conjunction with the specialist contractor to ensure that the completed JIC is the optimal solution.

H.11. JIC Issue No

a. All new JICs are to be annotated at Issue 0.

H.12. Contents Detail

a. JICs should, where appropriate, incorporate the following sequential events:

1) Invoking of TAG out procedures

2) Application of Safety Checks

3) Working procedure

4) Restoration of services

5) Functional check(s)

6) Return of equipment/system to Operator/User

7) Instructions to debrief Section Head

b. Instructions to update the RCM database are to include the following:

1) Applicable S2022s.

2) Time taken to repair

3) Downtime

4) Spares used

5) Whether a new Failure Mode (with accompanying effects)

6) OPDEF Rectification (If appropriate)

7) Any difficulties experienced with RCMS

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FOLD-OUT

NES 45 RCM TASK ANALYSIS (Figure 10.5)

and

NES45 RCS STOCKHOLDING (Figure 14.1)

DECISION DIAGRAMS

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NES 45 RCM TASK ANALYSIS DECISION ALGORITHM

Figure 10.5

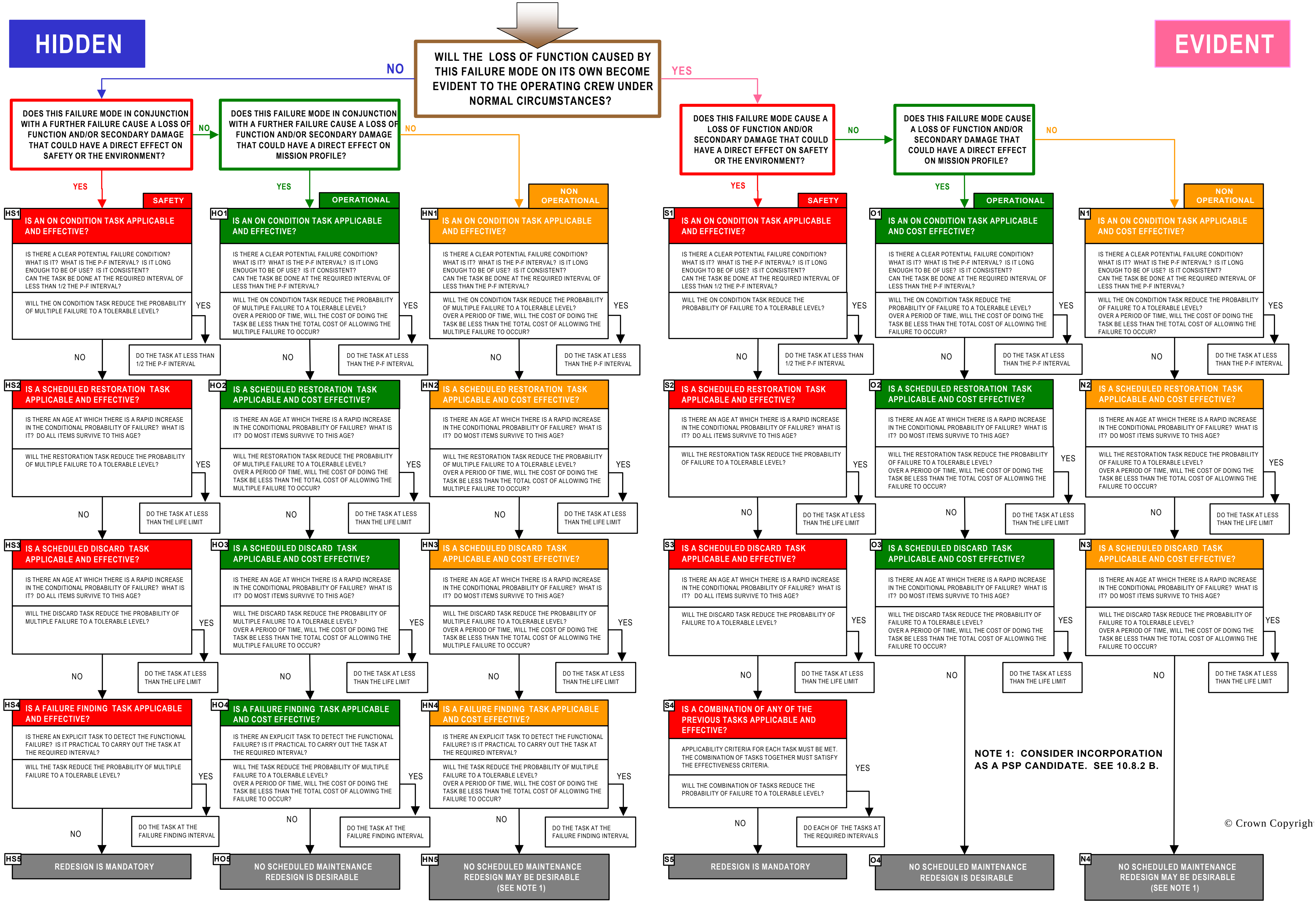
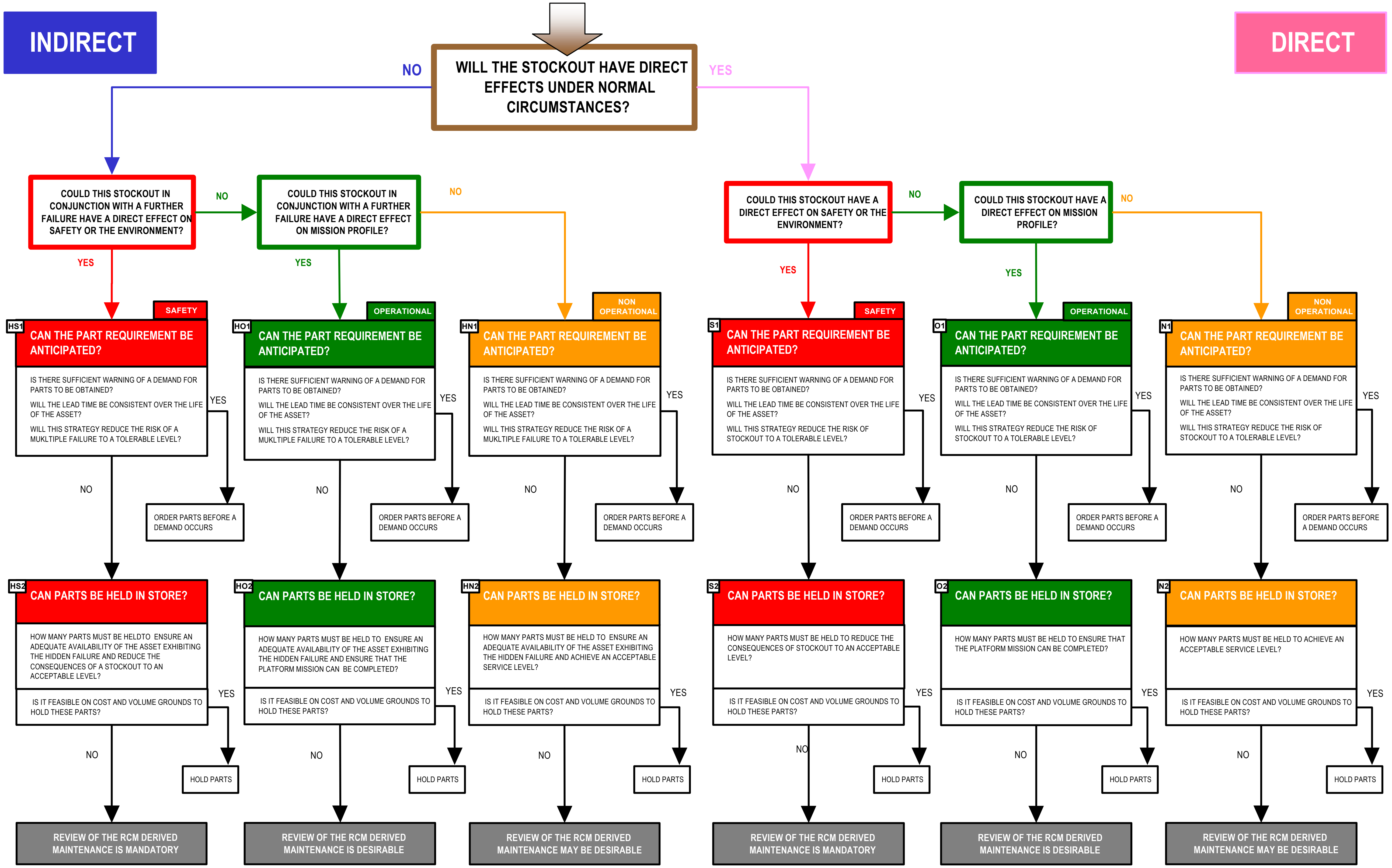


Figure 14.1

NES 45 RCS STOCKHOLDING DECISION ALGORITHM



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